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CD100 Series Servo Drive Manual



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Preface

We appreciated for choosing the CD100 Series Economical Servo drive!

CD100 series servo drive is an economic small and medium power AC servo unit developed by Simphoenix Electric. The power range of this series is 100W~7.5kW, with beautiful structure design, high integration, small volume, good heat dissipation, perfect protection and high reliability. Support MODBUS communication protocol, with the host computer can realize multi-steps servo drive operation. Advanced PID control algorithm is adopted to enhance the characteristics of fast response, small vibration, accurate positioning and high precision of the motion control system. A large number of advanced auxiliary functions are used to enhance the wide applicability of the product to the market, such as: adjustment-free rigidity coefficient setting, off-line rotation inertia projection, load disturbance observer, friction compensation, command feed forward, gain switching and other functions. The servo drive is easy to operate and widely used in CNC machine tools, packaging machinery, printing machinery, woodworking machinery, textile machinery, automatic production lines and many industrial control fields.

The CD100 series servo drives can support several different options of split frequency output, CANopen bus/full closed loop/dynamic braking function respectively.

This manual is an accessory sent with the drive. Please keep it in a safe place for future service and maintenance of the drive.

Due to the commitment to continuous product improvement, the information provided by our company is subject to change without notice. For the latest changes and more, please visit: www.simphoenix.com.cn.

Version: V102B00

Date of revision: 20230425

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Chapter 1 Product Introduction

1.1. Description of the servo drive and motor model

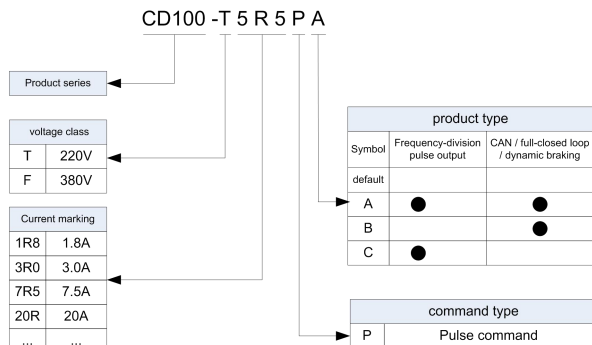


Figure 1-1 Servo drive model

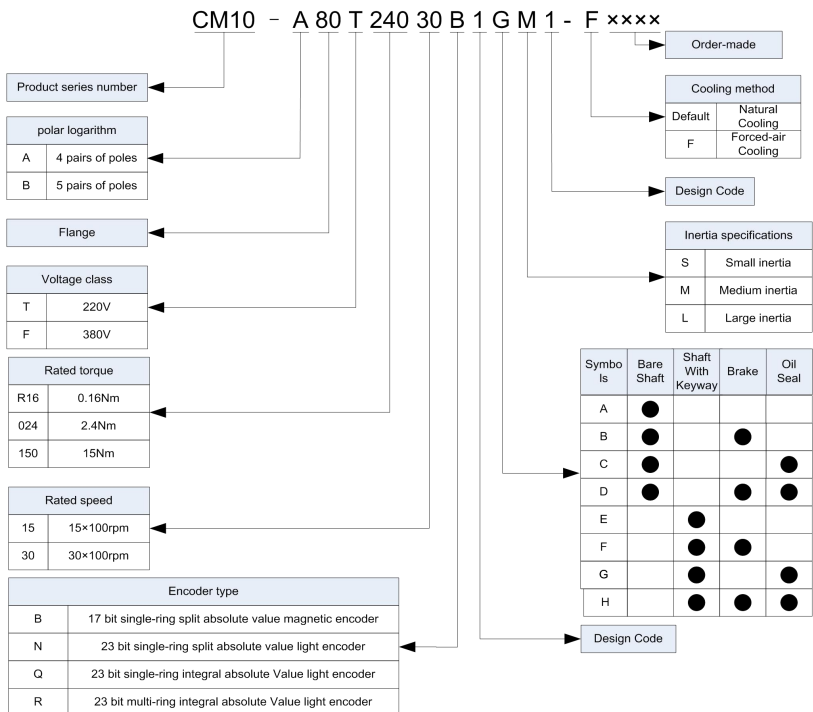


Figure 1-2 Servo motor model

1.2. Drive technical specification

Table 1-1 Drive technical specifications

Basic Specification	Drive model		CD100	T1R8/T3R0	Single Phase 220VAC-15%~+10%(50/60Hz)
				T5R5/T7R5	Three phase 220VAC-15%~+10%(50/60Hz)
	Control mode		Position,Speed, Torque,Position/Speed,Position/Torque, Speed/Torque		
	Feedback		Serial communication encoder : 17bit/23bit		
	Conditions of use	Temperature	Operating temperature : 0℃~+45℃(Ambient temperature +45℃ or more, please derating used).		
			Storage temperature : -20℃~+60℃.		
		Humidity	90%RH below, no condensation.		
		Ventilation	0.5g(4.9m/s²)		
		Protection	IP20		
Altitude	1000m below (>1000m, please derating used).				
	Others	1. No electrostatic interference, strong electric field, strong magnetic field, radiation, etc. 2. No corrosive gas, combustible gas, no water, oil, drug splash. 3. In an environment where there is little dust, dirt, salt and metal powder.			
Structure		Base installation type			
Position Control	Command source		External Pulse	Type: 1. pulse + direction 2. A, B orthogonal pulse 3. CCW+CW pulse. Maximum frequency: 500Kpps (differential) / 200Kpps (collector).	
			Internal Position	Relative and absolute position modes are configurable, and maximum speed and acceleration/deceleration times can be set.	
			Internal multi-steps Positions	A total of 16 segments, each with individually set maximum speed, acceleration and deceleration time and waiting time. Relative position and absolute position mode can be configured. Single operation and cycle operation can be configured, and any segment operation can be switched through DI terminal.	
	Electronic gear ratio		4 sets of 32bit electronic gear ratios.		
	Input control signal		Positive overtravel (POT), Negative overtravel (NOT), Gain switching (GAIN),Electronic gear ratio selection (GEAR), Inhibit pulse-in (INHP), Position error clear (PECLR), Position trigger (PTRG), Home position reference (REF), Get Home (GHOM).		
	Output control signal		Complete in position(COIN), Near in position (NEAR), Home completed (HOM).		

Speed Control	Command source	External analog	0~±10VDC/rated speed, maximum input voltage is ±12VDC.	
		Internal speed	Acceleration and deceleration time can be set up to 60S.	
		Internal multi-steps speeds	A total of 16 steps, each step can be set acceleration and deceleration time and running time individually. Single execution and loop execution can be configured. Exection of the 16 speeds is via Digital Input signals..	
	Speed Ratio	1 : 6000		
	Frequency Response	1.5kHz(Max)		
	Speed variation	Load variation	At 0 to 100% load: ±0. 2% or less (≤ rated speed)	
		Voltage variation	Rated voltage ± 10%:0% (≤ rated speed)	
		Ambient temperature variation	25 ±25°C: ± 0.1% or less (≤ rated speed)	
Input control signal	Zero speed clamp (ZCLMP), Gain switching (GAIN), Command inverse (CINV), External positive torque limiting (TCCW), External negative torque limiting (TCW), Internal multi-steps command switching (ICMD)			
Output control signal	Zero speed (ZSP), Speed achieve (SPA)			
Torque control	Command source	External analog	0~±10VDC/rated torque, maximum input voltage is ±12VDC.	
		Internal torque	Acceleration and deceleration time can be set up to 60S.	
		Internal multi-steps torques	A total of 4 steps, each step can be set build-up time of torque individually.Exection of the 4 torques is via Digital Input signals.	
	Current accuracy	±3%		
	Speed limit	3 limit modes.		
	Control inputs	External positive torque limiting (TCCW), external negative torque limiting (TCW), Internal multi-step command switching (ICMD), Positive jog (JOGP), Negative jog (JOGN), Command inverse (CINV).		
	Control outputs	Torque achieve (TQA), Speed limiting(SPL), Speed achieve (SPA).		
Input and output signals	Input signal	DI: 8 physical terminals and 8 virtual terminals witch can be defined as different functions and positive and negative logic switch. AI: Speed and Torque.		
	Output signal	DO: 5 physical terminals and 5 virtual terminals witch can be defined as different functions and positive and negative logic switch.		
	Pulse output	A, B, Z differential signal output, Z pulse open collector output.		
	Protection	Overcurrent, overvoltage, undervoltage, overload, abnormal main circuit detection, overheating, overspeed, abnormal encoder, abnormal CPU, abnormal parameters, etc.		
	Communication	RS-485 communication: support Modbus protocol.		
		CAN communication: support CANopen protocol.		
	Instructions	Charge (red),7-segment LED×6 (internal keyboard).		

1.3. The interface of the servo drive

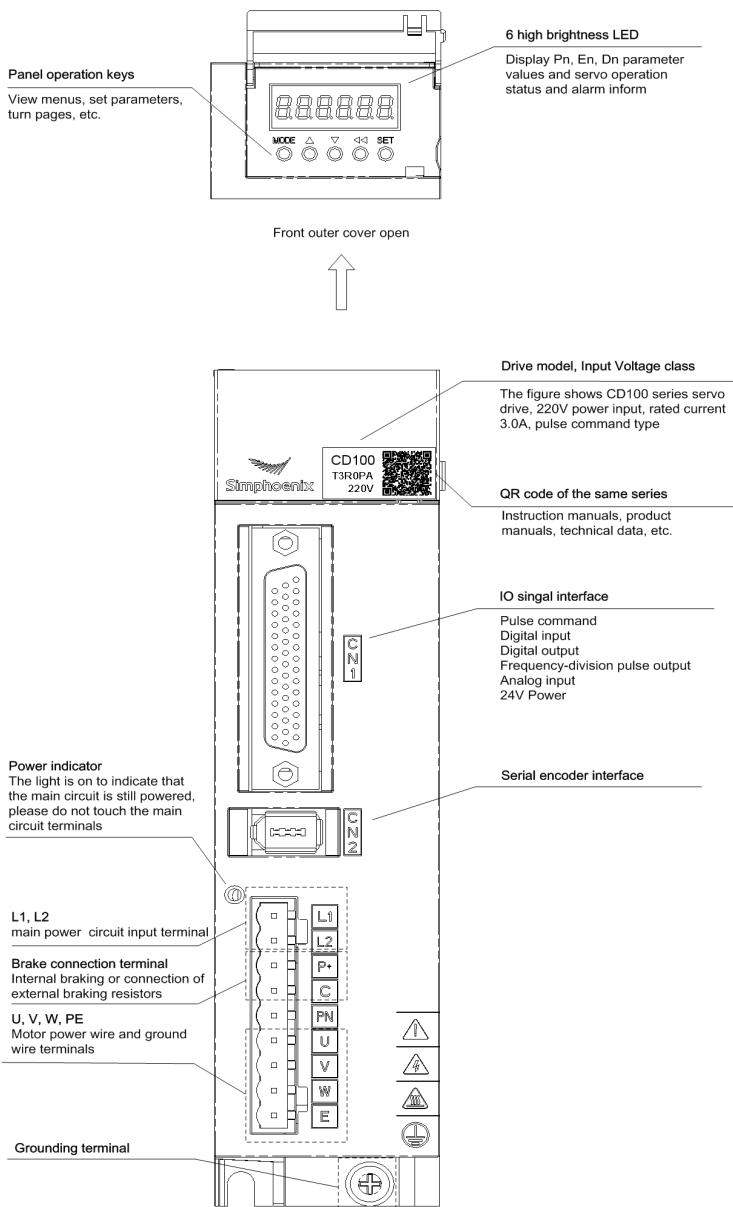


Figure 1-3 The interface of T1R8/T3R0 drive

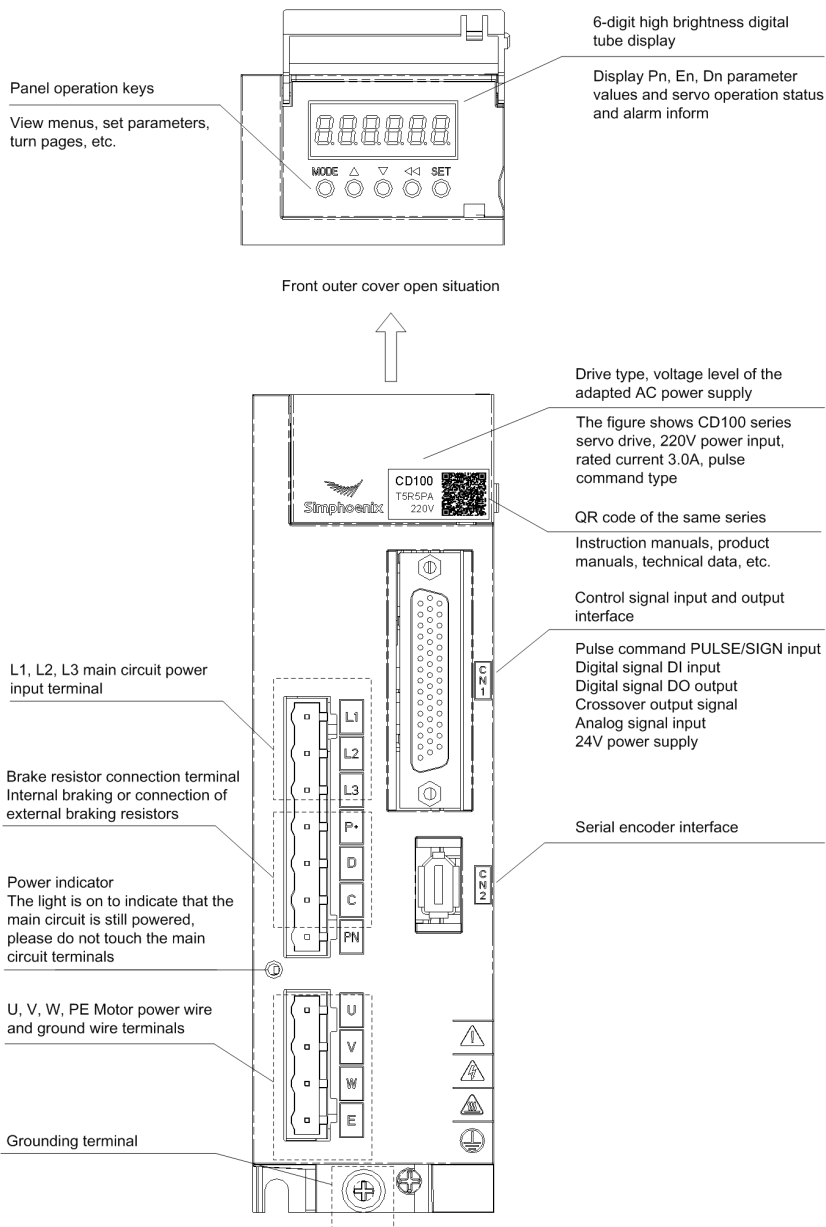


Figure 1-4 The interface of T5R5/T7R5 drive

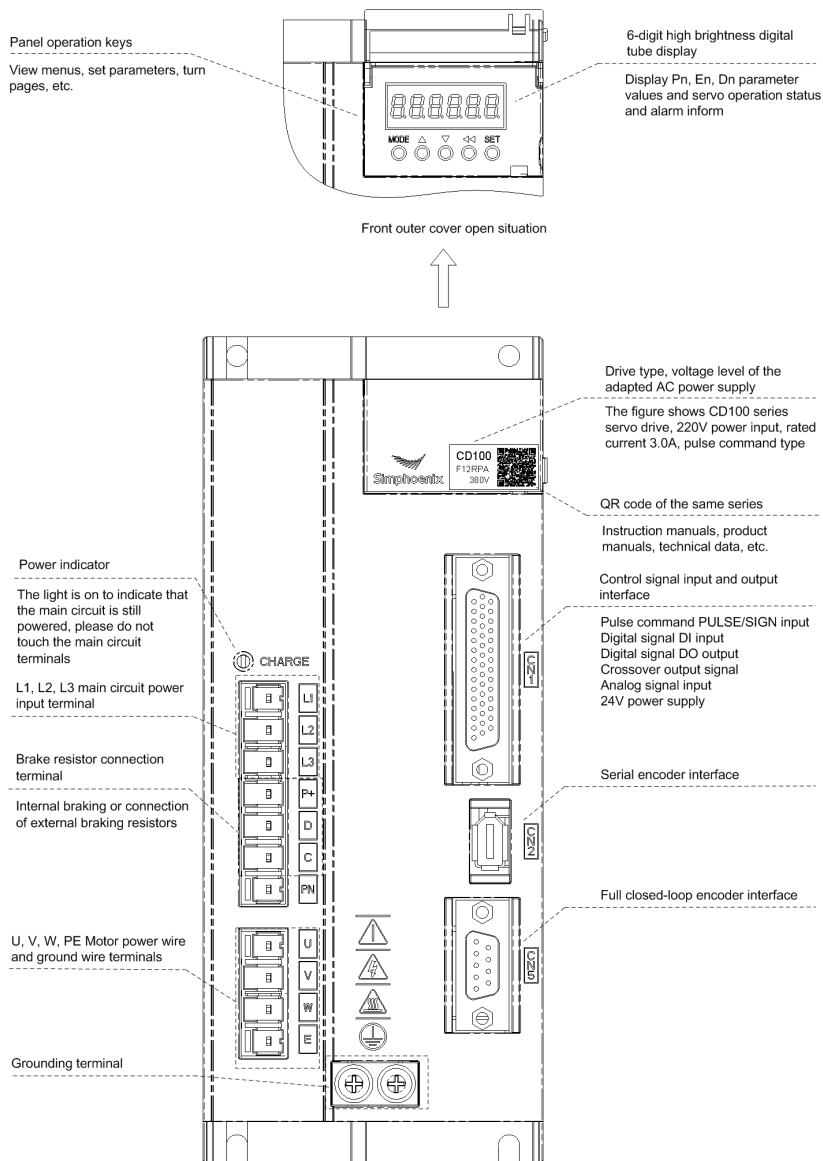


Figure 1-5 The interface of F4R0/F6R5/F8R5/F12R drive

1.4. Servo Driver Terminal Arrangement

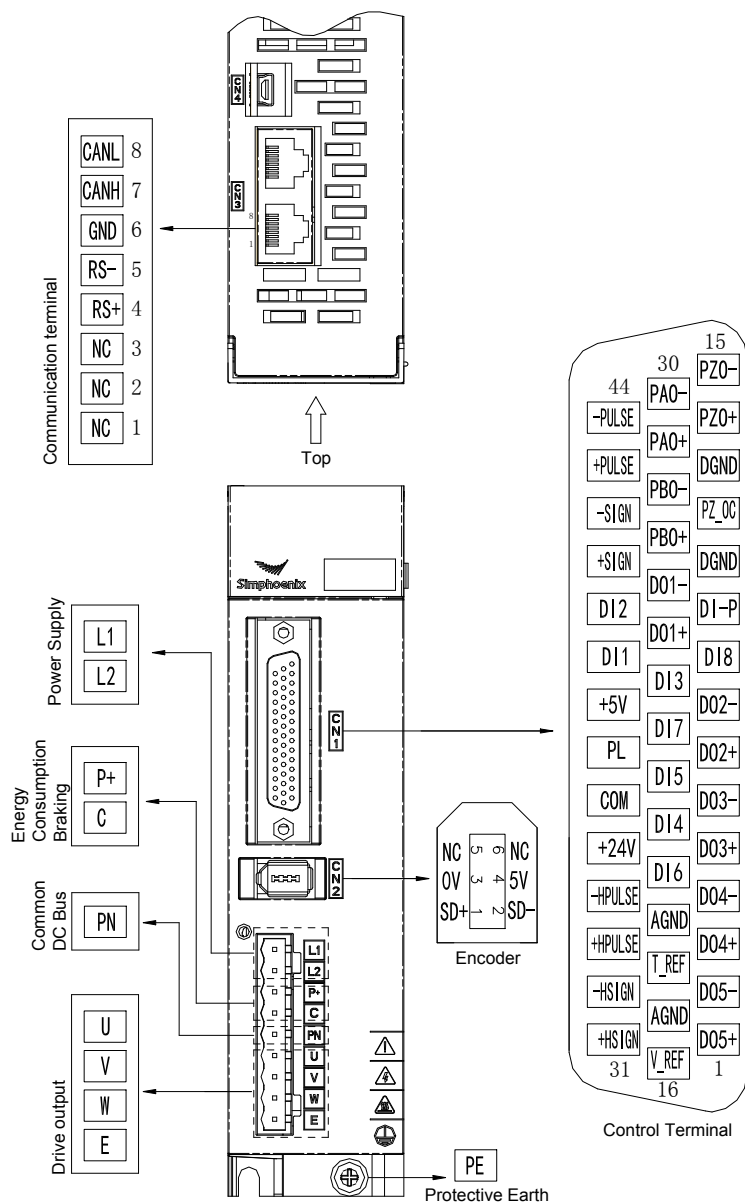


Figure 1-6 T1R8/T3R0 servo drive

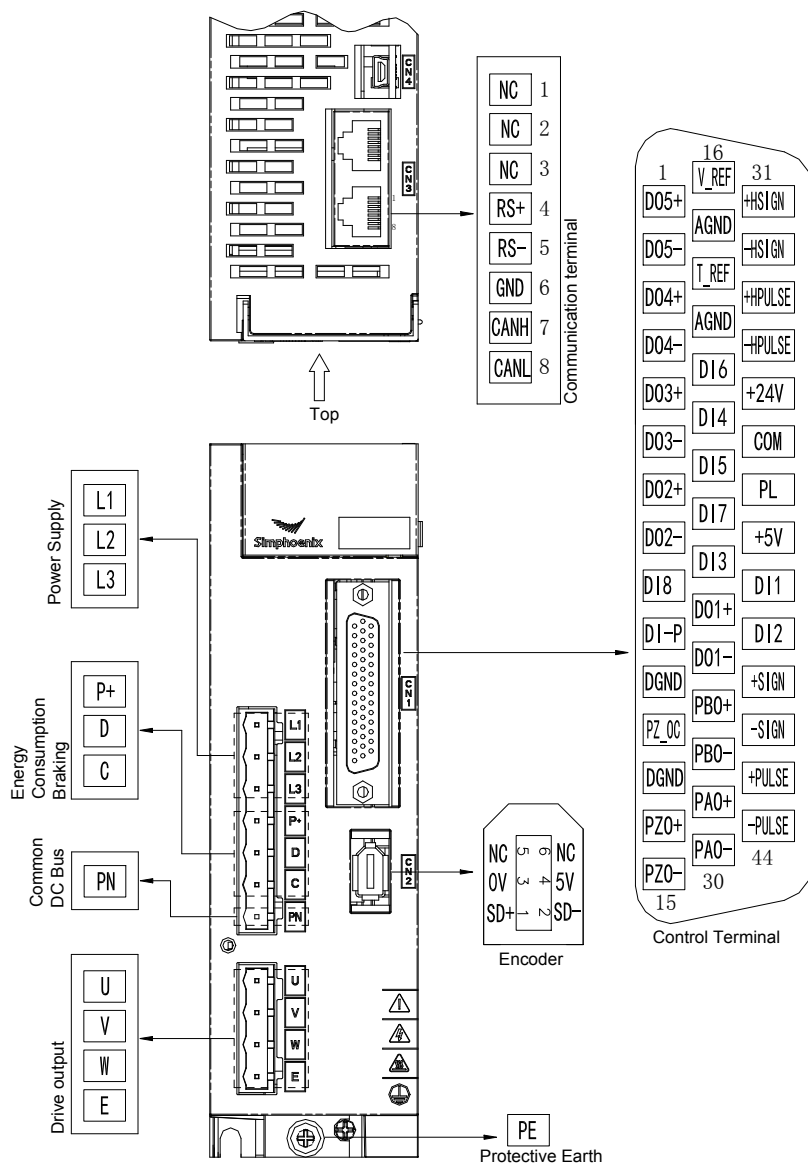


Figure 1-7 T5R5/T7R5 servo drive

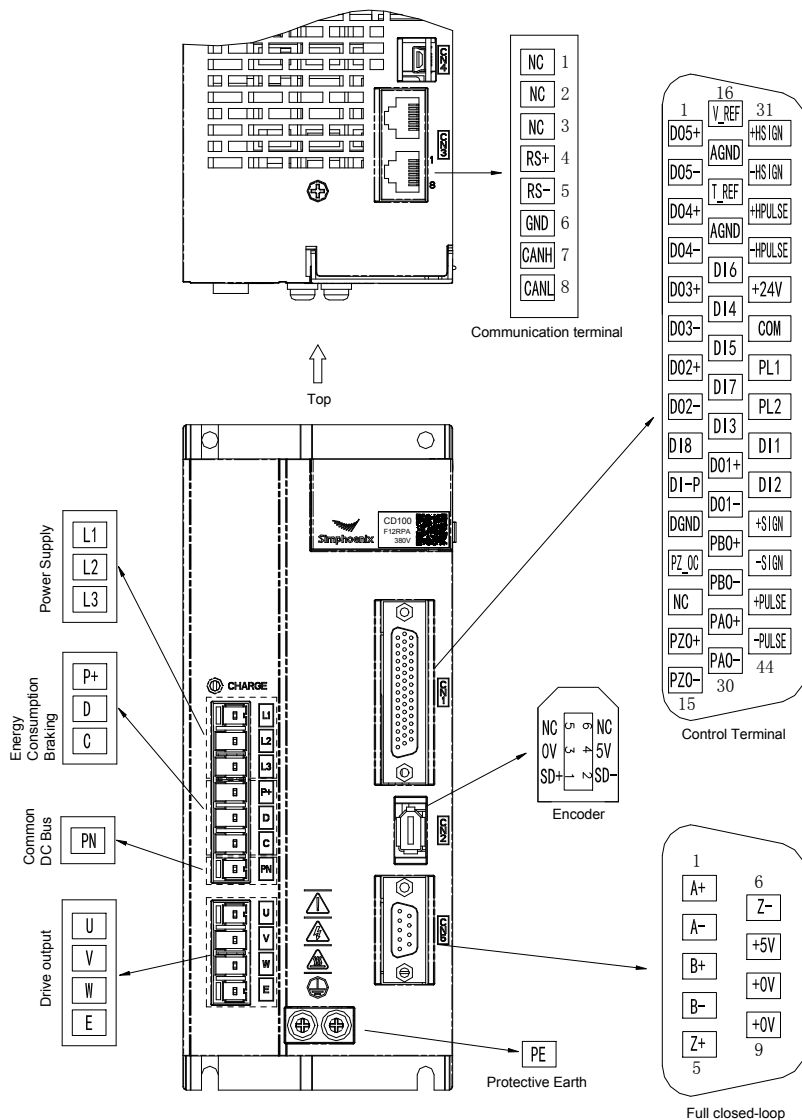


Figure1-8 F4R0/F6R5/F8R5/F12R servo drive

1.5. Servo driver and motor correspondence reference table

Refer to section 9.

Chapter 2 Installation Instruction

2.1. Notice

Please pay more attention to the followings.

- The servo motor can be installed in horizontal and vertical directions. Please using flexible couplings with mechanical connection.Ensure that the axis of the servo motor in a straight line with the axis of the mechanical load. Excessive axiality deviation will cause mechanical vibration and damage the servo motor bearings.

- It's forbidden to strike axially when installing, otherwise it is very easy to damage the encoder of the servo motor.

- when the distance between the drive and the motor exceeds 20 meters, please using the thicker cables of U/V/W connection and encoder connection...

2.2. Storage environmental conditions

In order for this product to be covered by our warranty and for future maintenance, it is important to note the following when storing it.

Type	Description
Storage temperature	-20℃~+65℃.
Storage humidity	Relative humidity 0%~95%, no condensation.
Other	Avoid storage in environments containing corrosive gases.

2.3. Installation ambient conditions

The following conditions need to be met for the installation and operation environment of this product.

Type	Description
Dust and gas	Must be installed in a dust-free, metal particle-free and oil-free place, avoid using in corrosive or flammable gas or liquid environment.
Temperature of working condition	0℃~+45℃, rated current operation, if the ambient temperature 45℃~50℃ must be used in derating, each increase of 1℃, current derating 2%, the highest use temperature should not exceed 50℃.
Vibration	< 4.9m/s ² .
Impact	< 49m/s ² .
Elevation	Less than 1000m; more than 1000m must be used to reduce the amount, each 1km higher, the current reduction of 10%.

- Do not use servo drives and motors in a closed environment

- If the product is installed in the electric cabinet that it's size and the ventilation conditions must be useful to prevent the internal electronic devices overheat..

- Notice that the vibration of the machine whether affect the other electronic device of the electric cabinet.

2.4. Installation space and dimensions

The installation direction must be in accordance with the regulations, otherwise it will cause failure.

In order to make the cooling work well, when installing the AC servo drive, its top, bottom, left and

right must keep enough space with adjacent items and baffles (walls), otherwise it will cause malfunction. Its suction and exhaust holes must not be sealed or placed upside down, otherwise it will cause malfunction.

Multi-steps Servo Drives are installed side-by-side in the electric cabinet. Users are requested to observe the installation spacing in the following diagram.

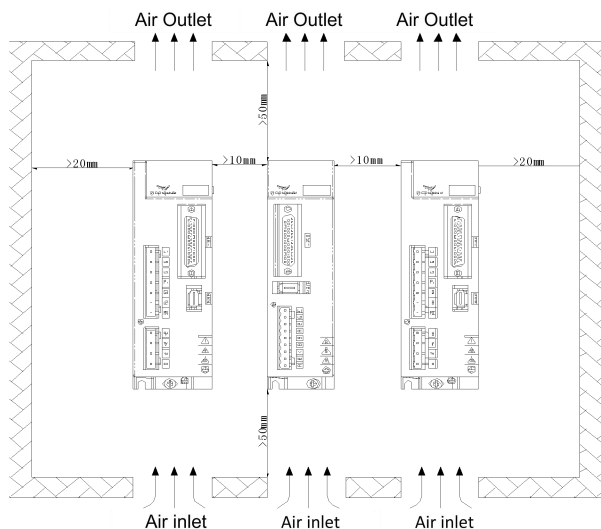
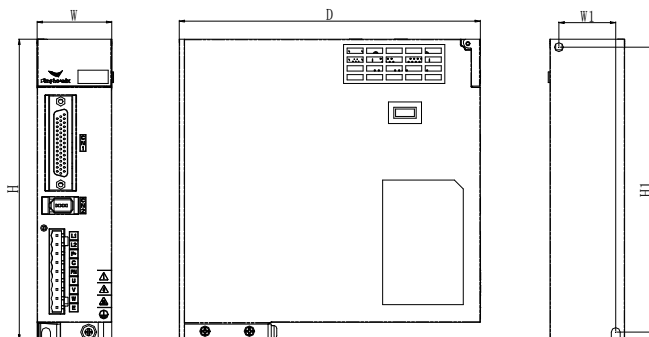


Figure 2-1 Servo drives installation method



Drive Model	W(mm)	W1(mm)	H1(mm)	H(mm)	D(mm)	Screw type
T1R8/T3R0	42	32	161	170	170	M4
T5R5/T7R5	50	40	161	170	170	M4
F4R0/F6R5/F8R5/F12R	80	64	186	195	182	M4

Figure 2-2 Dimensional drawing of drive installation

Chapter 3 Wiring

3.1. Servo peripheral devices and wiring

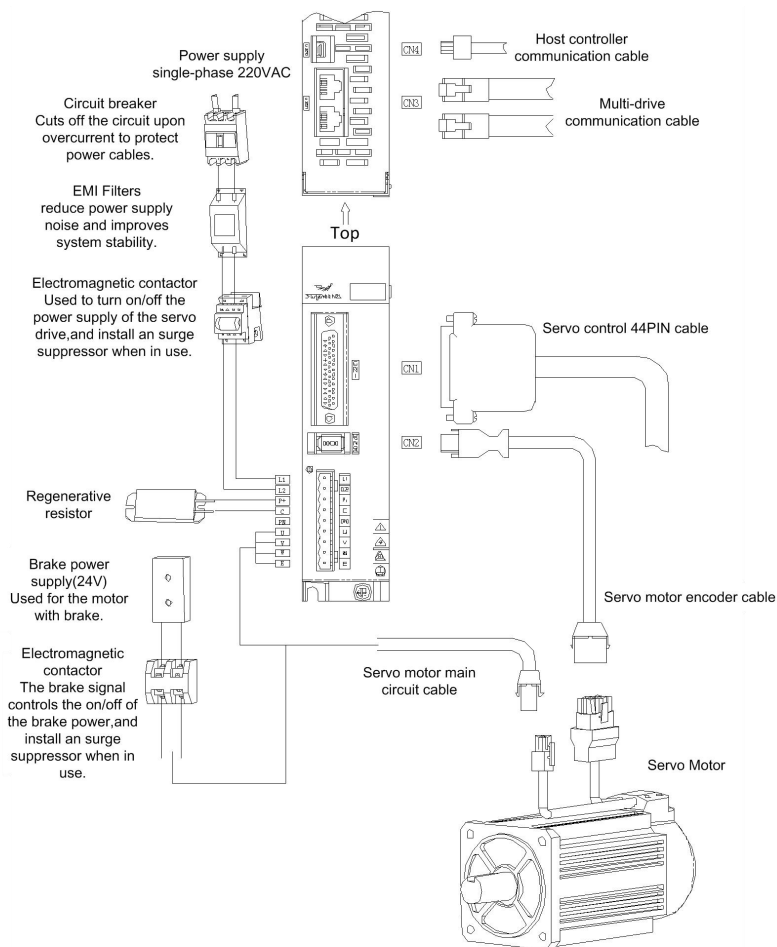


Figure 3-1 T1R8/T3R0 servo driver wiring diagram

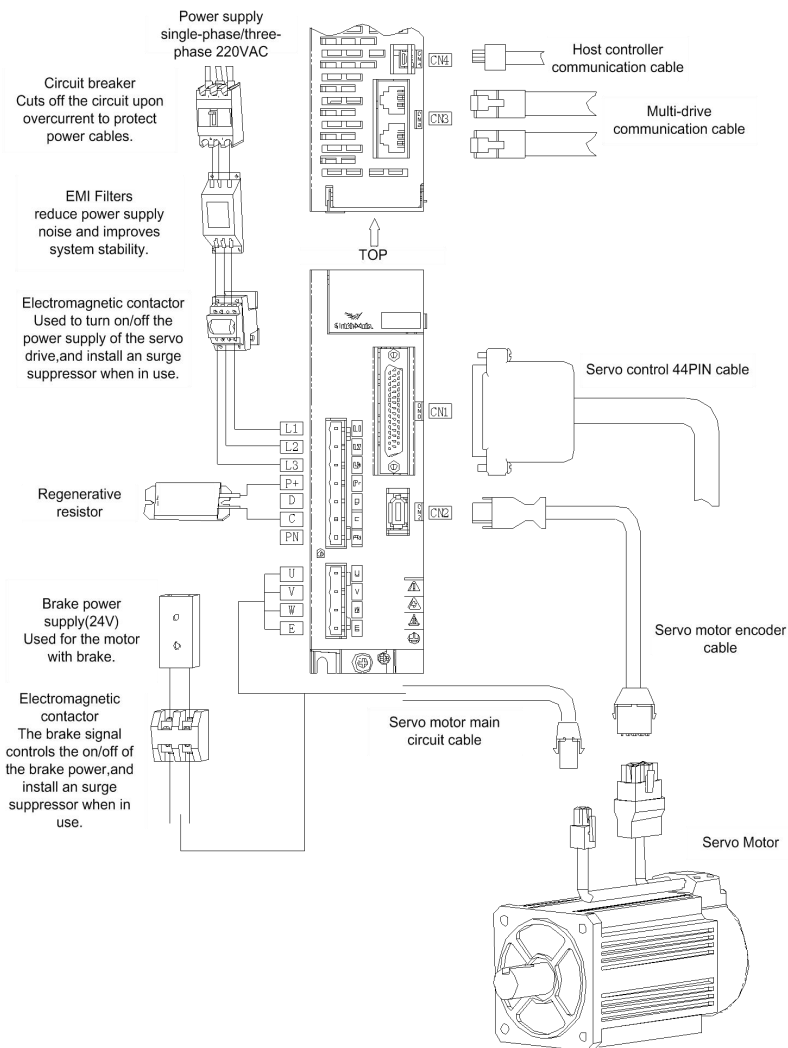


Figure 3-2 T5R5/T7R5 servo driver wiring diagram

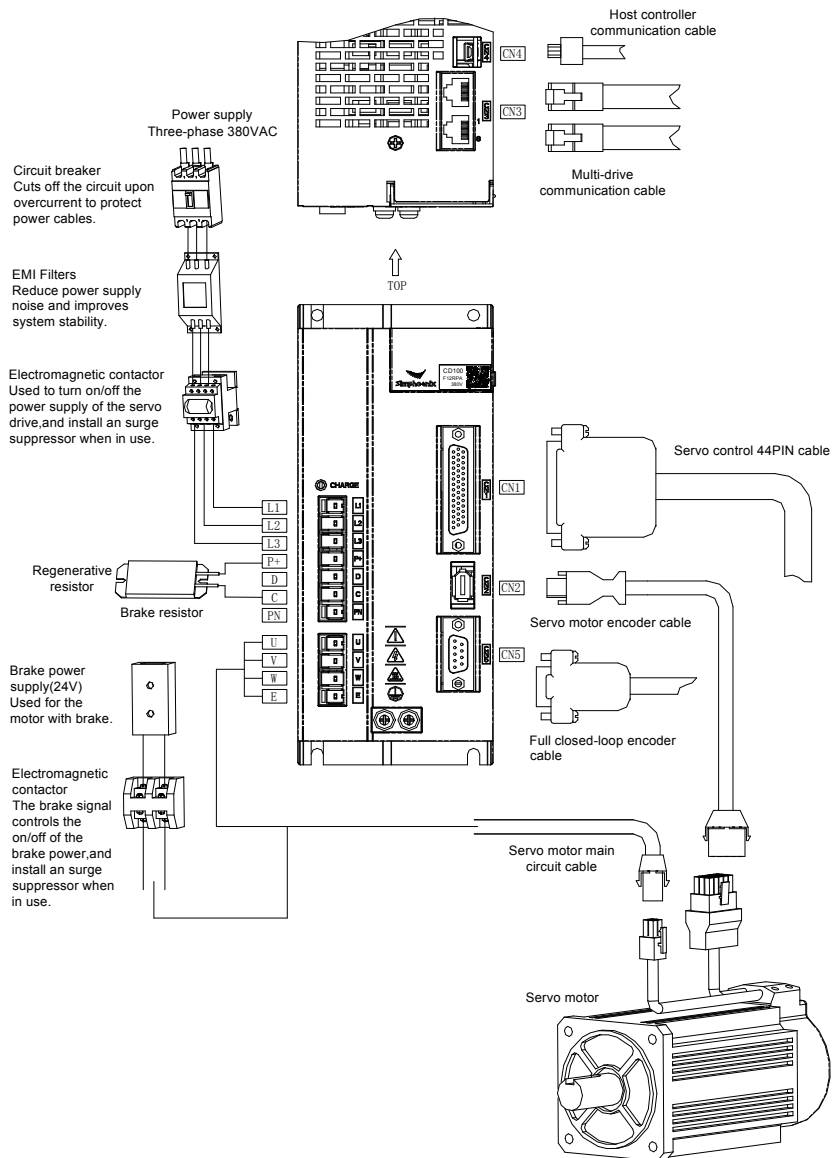


Figure 3-3 F4R0/F6R5/F8R5/F12R servo driver wiring diagram

3.2. Wiring precautions



•When the power is turned off, there may still be high voltage left in the servo drive. Do not contact the power terminal. After confirming that the CHARGE indicator light is off, the inspection operation can be performed.

•Confirm that the servo motor output U, V, W terminal phase sequence wiring is correct, wrong connection may cause the motor does not turn or runaway.

•Do not connect the input power cable to the outputs U, V, W, as this may cause damage to the servo driver.

•Please make sure the servo driver and motor are reliably grounded, otherwise there is a risk of electric shock.



•When using an external braking resistor, it should be connected to P+, C terminal (at this time P+ and D terminal shall be open), if using an internal braking resistor, then P+ and D terminal should be shorted (at this time P+ and C terminal shall be open).

•Do not pass power and signal wires through the same conduit or bundle them together. When wiring, the power line and signal line should be separated by at least 30cm, otherwise, it may lead to misoperation.

•Please do not turn on/off the power frequently. When it is necessary to turn on/off the power repeatedly and continuously, please keep it below 1 time per minute. Since there is a large capacitor in the power supply section of the Servo Drive, a large charging current (charging time of several hundred milliseconds) will flow when the power is turned on, so if the power is turned on/off frequently, the performance of the main circuit components inside the Servo Drive will be degraded.

•The maximum cable length for input and output signals is 3 m. The maximum cable length for main circuit and encoder cables is 30 m. Use twisted and isolated grounded signal cables.

•Please use thick wire (2.0 mm²) for grounding cable if possible.

•If the servo motor and machinery are insulated from each other, ground the servo motor directly.

3.3. Main circuit terminal name and function

Table 3-1 Main circuit terminals

Marking	Name	Drive Model	Function and connection
L1 L2 L3	Power input terminals	T1R8/T3R0	Input single-phase 220VAC power supply(no L3terminal).
		T5R5/T7R5	Input single-phase/three-phase 220VAC power supply.
P+ D C	Terminals for connecting external regenerative resistor	T1R8/T3R0	Internal regenerative resistor : None external regenerative resistor:connecting P+ and C terminal.
		T5R5/T7R5	Internal regenerative resistor: use a jumper bar to connect the P+and D terminals. External regenerative resistor:connecting P+ and C terminal.
U V W	terminals for connecting the servo motor	Connected to U, V and W phases of the servo motor.	
P+ PN	DC Bus Terminal	Used by the common DC bus for Multi-steps steps servo drives.	
PE	Grounding terminal	Connected to the ground terminal of the power supply and the ground terminal of the motor.	

3.4. Main circuit and power wiring

•KM is an electromagnetic contactor, Rly is a relay, D is a current-continuing diode, and SA is a surge suppressor.

3.4.1. Single-phase 220VAC wiring

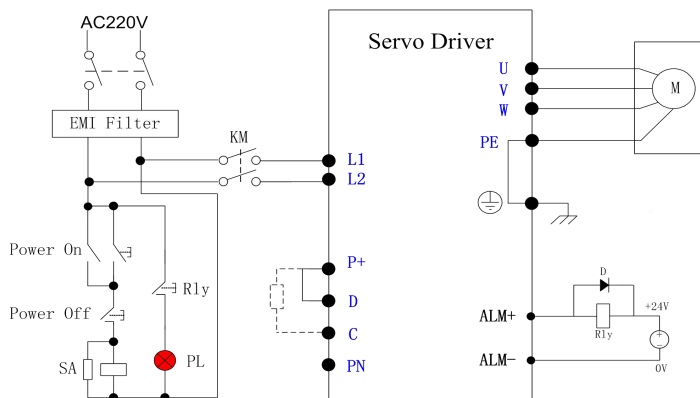


Figure 3-3 Single-phase 220VAC wiring diagram

3.4.2. Three-phase 220VAC wiring

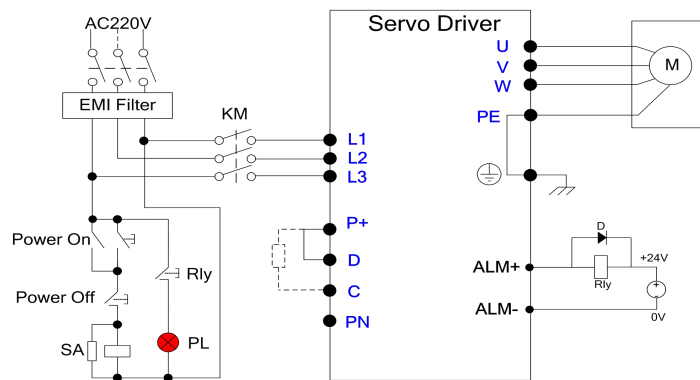


Figure 3-4 Three-phase 220VAC main circuit wiring diagram

3.4.3. Three-phase 380VAC wiring

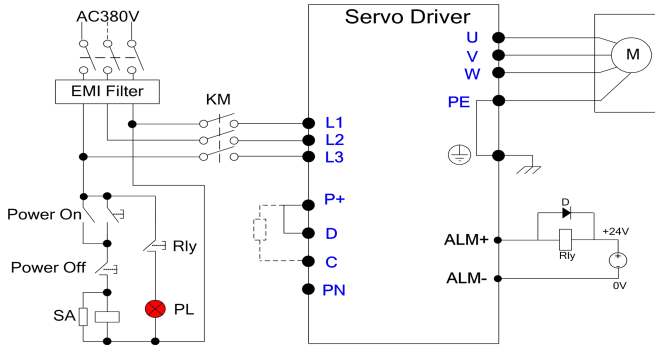


Figure 3-5 Three-phase 380VAC wiring diagram

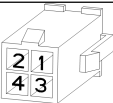
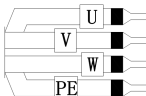
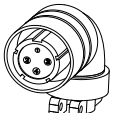
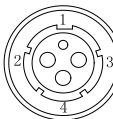
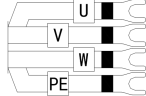
3.4.4. Main circuit cable diameter specifications

Table 3-2 Main circuit cable diameter specifications

Driver type	Input power cord	Motor connection wire	Brake resistance connection wire	Protective grounding wire
	L1/L2/L3	U/V/W	P+/D/C	PE
CD100-T1R8□□	0.50 mm ²	0.50 mm ²	0.50 mm ²	0.50 mm ²
CD100-T3R0□□	0.50 mm ²	0.50 mm ²	0.50 mm ²	0.50 mm ²
CD100-T4R5□□	0.75 mm ²	0.75 mm ²	0.75 mm ²	0.75 mm ²
CD100-T5R5□□	0.75 mm ²	0.75 mm ²	0.75 mm ²	0.75 mm ²
CD100-T7R5□□	0.75 mm ²	0.75 mm ²	0.75 mm ²	0.75 mm ²
CD100-F4R0□□	1.5 mm ²	1.5 mm ²	1.5 mm ²	1.5 mm ²
CD100-F6R5□□	1.5 mm ²	1.5 mm ²	1.5 mm ²	1.5 mm ²
CD100-F8R5□□	1.5 mm ²	1.5 mm ²	1.5 mm ²	1.5 mm ²
CD100-F12R□□	1.5 mm ²	1.5 mm ²	1.5 mm ²	1.5 mm ²

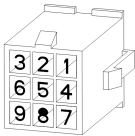
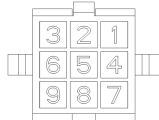
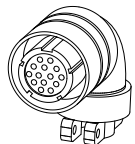
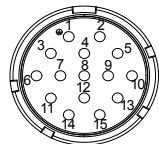
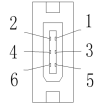
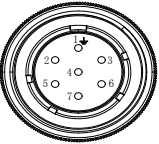
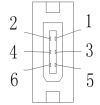
3.4.5. Motor power line terminal definition

Table 3-3 Motor power line terminals

Servo motor terminal wiring sequence					Drive side
Motor type	Terminal view	Terminal serial number	Pin	Definition	Terminal serial number
40/60/80 with amp plug			1	PE	
			2	U	
			3	V	
			4	W	
As follows,80 flanges replace amp plugs with small navigation plugs mainly for motor applications will occur reciprocating motion,high temperature and humidity and other harsh conditions					
80(Use a small pilot) 130/180 (Aviation plug)			1	PE	
			2	U	
			3	V	
			4	W	

3.4.6. Definition of encoder cable terminals

Table 3-4 Encoder wire terminals

Motor side					Drive side	
Motor type	Terminal profile	Terminal serial number	Pin	Definition	Pin	Terminal serial number
40/60/80 with amp plug			1	PE	PE	
			2	5V	4	
			3	0V	3	
			4	SD+	1	
			5	SD-	2	
			6	E+	NC	
			7	E-	NC	
			8	NC	NC	
			9	NC	NC	
As follows,80 flanges replace amp plugs with small navigation plugs mainly for motor applications will occur reciprocating motion,high temperature and humidity and other harsh conditions.						
80(Use a small pilot)			1	PE	NC	
			2	E-	NC	
			3	E+	NC	
			4	SD-	2	
			5	0V	3	
			6	SD+	1	
			7	5V	4	
130/180 (Aviation plug)			1	PE	NC	
			2	E-	NC	
			3	E+	NC	
			4	SD-	2	
			5	0V	3	
			6	SD+	1	
			7	5V	4	

Encoder cable precautions: Please ground the encoder shield mesh layer on the drive and motor side reliably, otherwise it will cause the drive false alarm.

3.4.7. Definition of brake terminals

Table 3-5 Brake terminals

Motor type	Brake terminal type	Motor side terminal	Pin	Definition
40/60	172233-1		1	24V
			2	0V
80/130	XS12K3P		1	24V
			2	0V
			3	NC
180	XS16K4TM		1	24V
			2	0V
			3	NC
			4	NC

3.4.8. Regenerative braking resistor selection

When the output torque of the motor is opposite to the direction of the operating speed, the motor will be in the power generation state, and this backfill energy will make the bus voltage rise, and its energy size depends on the inertia of the motor rotor and load. If the system inertia is small, the bus capacitor inside the drive can absorb the backfill energy, but the system inertia is large, the bus capacitor is not enough to absorb the backfill energy, must be consumed through the regenerative resistor, otherwise the bus voltage rises too high will lead to the drive reported overvoltage shutdown or even damage.

Rotor inertia regenerative energy calculation formula: $E_r = J \bullet V^2 / 182$, Unit J

J is the rotor inertia, Unit $\text{kg} \cdot \text{m}^2 \times 10^{-4}$, and V is the rated speed of the motor , Unit rpm.

Regenerative resistor capacity calculation formula: $P_r = 2 \times (nE_r - E_C) / T$, Unit W

nE_r is the total inertia of the load , E_C is the maximum braking energy that can be absorbed by the bus capacitor, and T is the action period.

Table 3-6 Regenerative braking resistor selection table

Drive model		Built-in regenerative resistor specifications		Allowable minimum external regenerative resistance value(Ω)	The capacitor can absorb the maximum braking energy $E_C(\text{J})$
		Resistance value(Ω)	Capacity(W)		
Single Phase 220VAC	CD100-T1R8	None	None	50	11
	CD100-T3R0	None	None	50	16
Single-phase / three-phase 220VAC	CD100-T4R5	50	40	25	19
	CD100-T5R5	50	40	25	29
	CD100-T7R5	25	100	20	34
Single-phase / three-phase 380VAC	CD100-F4R0	100	100	50	33
	CD100-F6R5	100	100	50	33
	CD100-F8R5	50	100	25	33
	CD100-F12R	50	100	25	48

- When using an external Regenerative resistor, the resistor is connected to the P+,C terminals and must be left open between P+ and D.
- The external Regenerative resistor must be greater than the resistance values listed in the table, otherwise it may cause damage to the drive.
- The Pn-119 Regenerative resistor load rate can be set according to the amount of regenerative energy, otherwise it will affect the use of this function.

3.5. Wiring of control circuit

3.5.1. Wiring diagram for three control modes

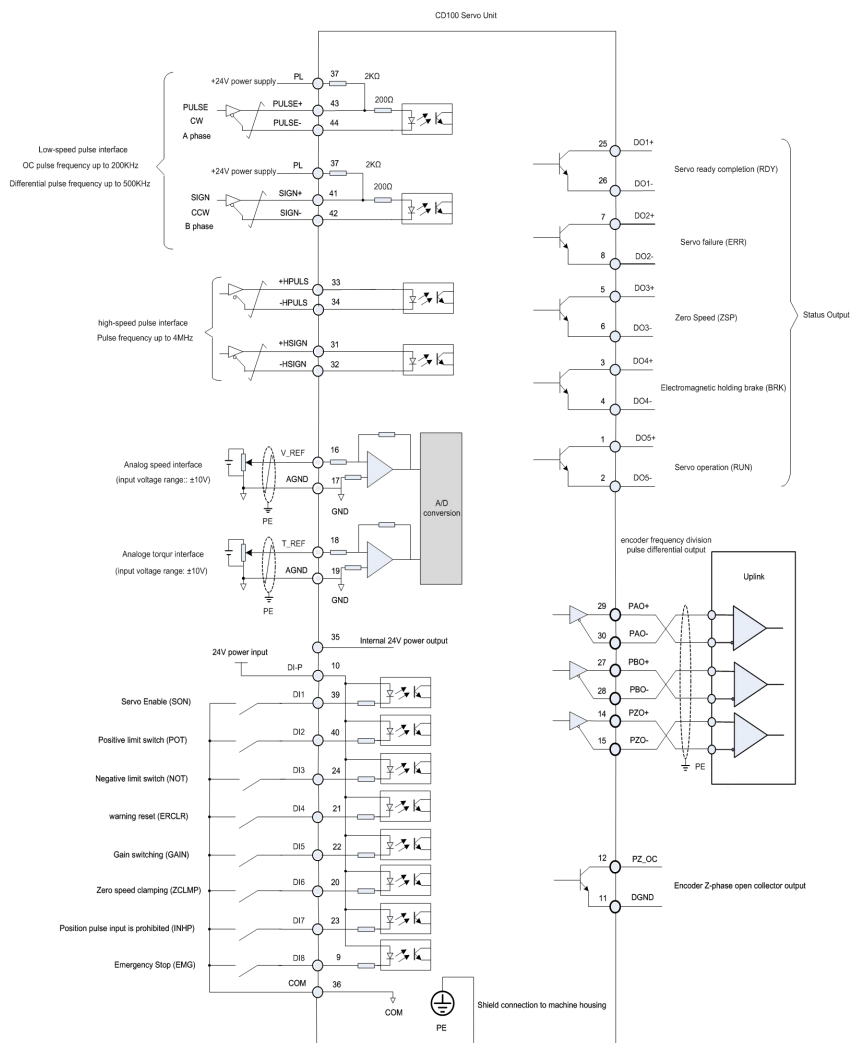


Figure 3-6 Wiring diagram for the position/speed/torque control mode

Note : The voltage range of the internal +24V power supply is 20V to 28V, with maximum operating current being 100mA.

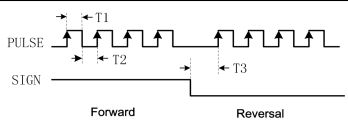
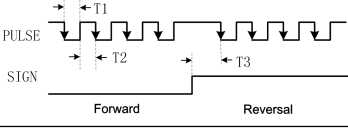
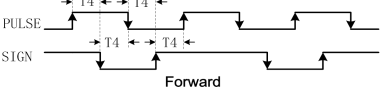
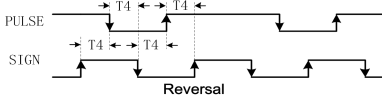
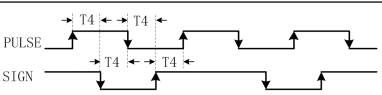
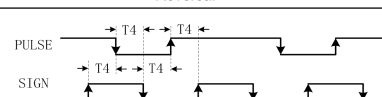
3.5.2. Pulse command input interface circuit

The command pulse input interface circuit is used to receive the command pulse signal sent from the upper device to the driver. The output of the upper device can be differential linear drive output or open collector output, the wiring of the two output methods is different, and the performance of the circuit is also different. In addition, plePulse coarse use twisted pair cable for the connection line.

Table 3-7 Description of position reference input signals

Signal name		Pin No	Function
Position reference	SIGN+	41	Position Direction Command Positive
	SIGN-	42	Position Direction Command Negative
	PULSE+	43	Position pulse positive
	PULSE-	44	Position pulse negative
	+24V	35	internal 24V power supply voltage range: 20V to 28V,maximum output current: 100mA
	COM	36	
	PL	37	Command pulse power input interface, the driver has internal 2KΩ resistor in series.

Table 3-8 Descriptions of the pulse form

Pulse form	Logical State	Pulse waveform
Pulse + Direction	Pn-101=0Positive Logic	
	Pn-101=1Negative Logic	
Two-phase quadrature pulses (2X frequency)	Pn-101=0 Positive Logic	
		
	Pn-101=1 Negative Logic	
		

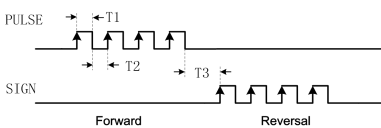
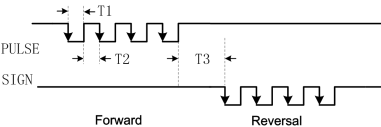
Pulse form	Logical State	Pulse waveform
CCW+CW	Pn-101=0 Positive Logic	
	Pn-101=1 Negative Logic	

Table 3-9 Correspondence between input frequency and pulse width

Pulse input	Maximum frequency	Minimum allowable time width				Voltage specification
		T1	T2	T3	T4	
Differential Inputs	500KHz	1us	1us	2us	0.5us	5V
Open collector input	200KHz	2.5us	2.5us	5us	1.25us	24V

•Differential pulse command input :

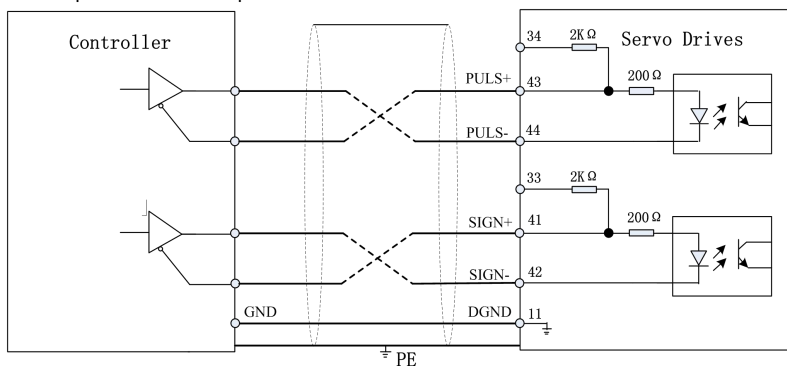
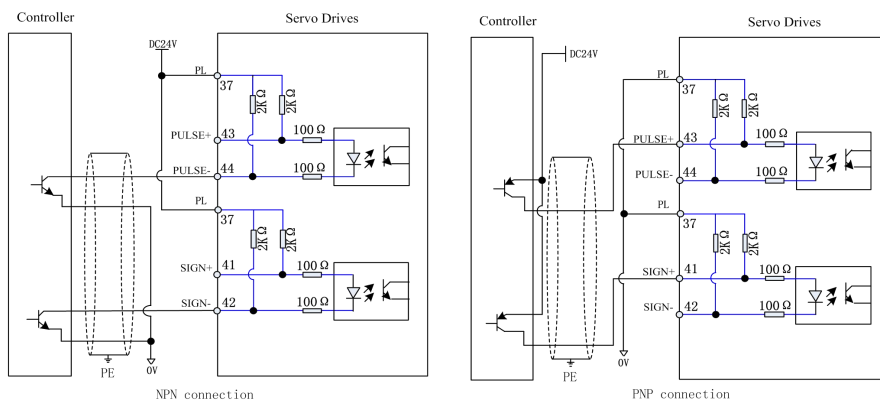


Figure 3-7 Differential pulse command input wiring diagram

-
- The image contains two wiring diagrams for Servo Drives, labeled 'NPN connection' and 'PNP connection'.
- NPN connection:** The Controller is connected to the Servo Drive. The +24V pin is connected to the +24V supply. The PL pin is connected to the Controller's ground. The PULSE+ pin is connected to the Controller's output line. The PULSE- pin is connected to the Controller's output line. The PL pin is connected to the Controller's ground. The SIGN+ pin is connected to the Controller's output line. The SIGN- pin is connected to the Controller's output line. The COM pin is connected to the Controller's ground. The PE pin is connected to the Controller's ground.
- PNP connection:** The Controller is connected to the Servo Drive. The +24V pin is connected to the +24V supply. The PL pin is connected to the Controller's ground. The PULSE- pin is connected to the Controller's output line. The PULSE+ pin is connected to the Controller's output line. The PL pin is connected to the Controller's ground. The SIGN- pin is connected to the Controller's output line. The SIGN+ pin is connected to the Controller's output line. The COM pin is connected to the Controller's ground. The PE pin is connected to the Controller's ground.

- Open collector pulse command input (external 24V power supply) :



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●Open collector pulse command input (external VCC + current limiting resistor) :

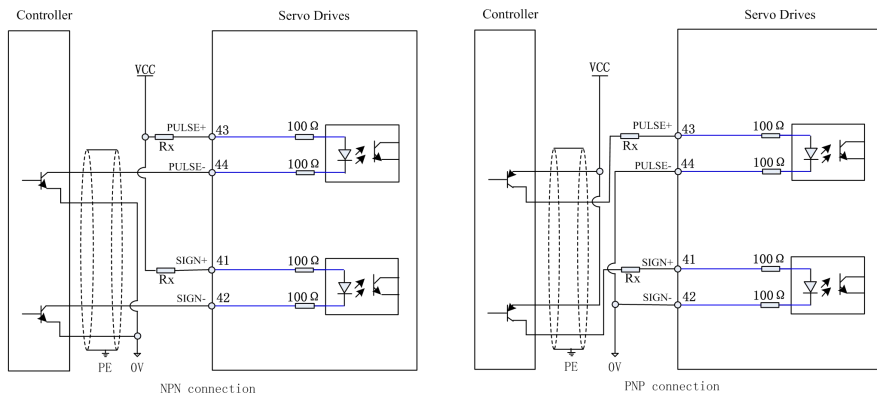


Figure 3-10 Open collector pulse command input wiring diagram (external VCC)

Calculation formula for current limiting resistor Rx : $R_x = (V_{cc} - 1.4 \text{ V}) / 10\text{mA}$

The recommended resistance values of Rx are as follows :

VCC voltage	Rx resistance value	Rx Power
24V	2K	0.5W
12V	800R	0.5W
5 V	0K	0W

3.5.3. Digital input interface circuit

The digital input interface of the servo driver is designed with bidirectional optocoupler, and the user can connect it by relay or transistor with open collector. When using a relay connection, please choose a relay for tiny current, otherwise it may cause poor contact or the circuit may not conduct. Note: 1. When using external power supply, disconnect internal 24V power supply from DI_P terminal.

2. Mixing of PNP and NPN inputs is not supported between Multi-steps steps DI terminals.

Table 3-10 DI signal description

Signal name		Defaultfunction	Pin number	Default function
DI input	DI1	SON	39	Servo enable
	DI2	POT	40	Positive limit switch
	DI3	NOT	24	Negative limit switch
	DI4	ERCLR	21	Warning reset
	DI5	GAIN	22	Gain switching
	DI6	ZCLMP	20	Zero speed clamping
	DI7	INHP	23	Position pulse input is prohibited
	DI8	EMG	9	Emergency stop
	DI_P	DI power supply	10	Common power supply of the DI signal
	+24V	+24V	35	Internal 24V power supply
COM	COM	36		

- The superior device is a relay output:

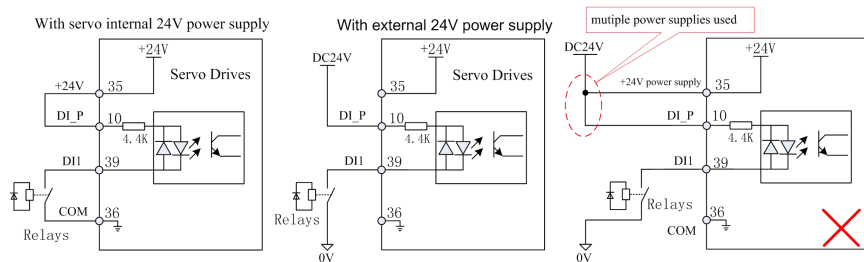


Figure 3-11 Digital input wiring diagram when the host controller is a relay output

Take DI1 as an example, the various wiring methods are as follows, and the DI1 to DI8 interface circuits are the same.

- The host controller is an NPN open collector output:

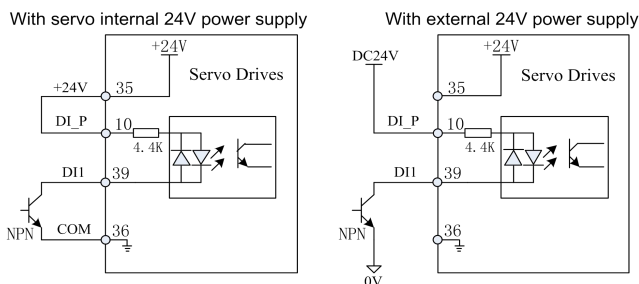


Figure 3-12 Digital input wiring diagram when the host controller is open collector (NPN) output

- The host controller is PNP open collector output:

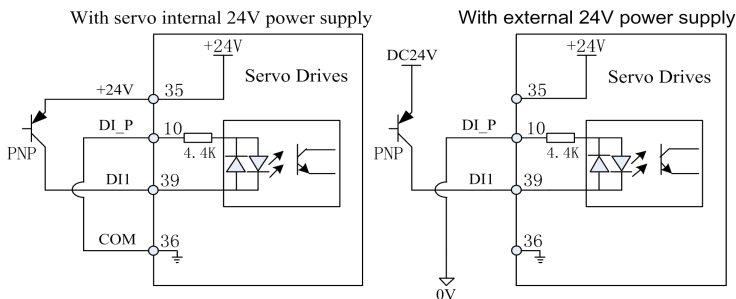


Figure 3-13 Digital input wiring diagram when the host controller is open collector (PNP) output

3.5.4. Digital output interface circuit

The digital output interface of the servo driver is designed with differential output.

Note:

1. The maximum allowable voltage of the servo driver digital output transistor is 30VDC.
2. The maximum allowable current of the servo driver digital output transistor is 50mA.

Table 3-11 DO signal description

Signal name		Default function	Pin number	Function
DO output	DO1+	RDY	25	Servo ready to complete
	DO1-		26	
	DO2+	ERR	7	Fault output
	DO2-		8	
	DO3+	ZSP	5	Zero speed
	DO3-		6	
	DO4+	BRK	3	Electromagnetic holding brake
	DO4-		4	
	DO5+	RUN	1	Servo enable
	DO5-		2	
	+24V	+24V	35	Internal 24V power supply
	COM	COM	36	

Taking DO1 as an example, the various wiring methods are as follows, and the DO1 to DO5 interface circuits are the same.

- The host controller is a relay input:

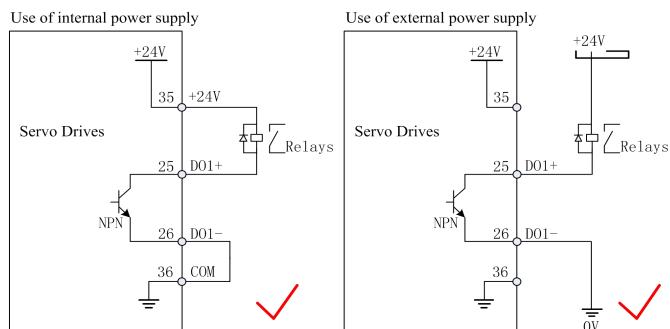


Figure 3-14 Correct wiring when connecting relays to digital output terminals

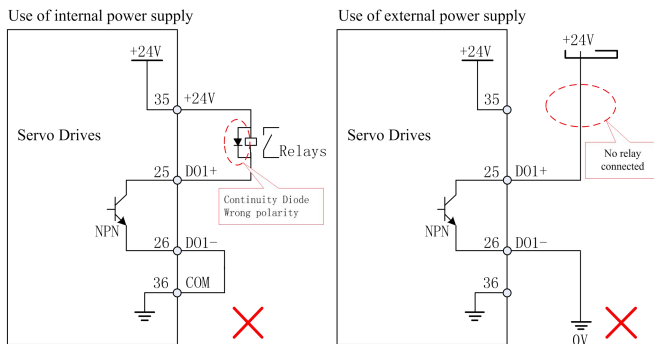


Figure 3-15 Wrong wiring when connecting relays to digital output terminals

- The host controller is an optocoupler input:

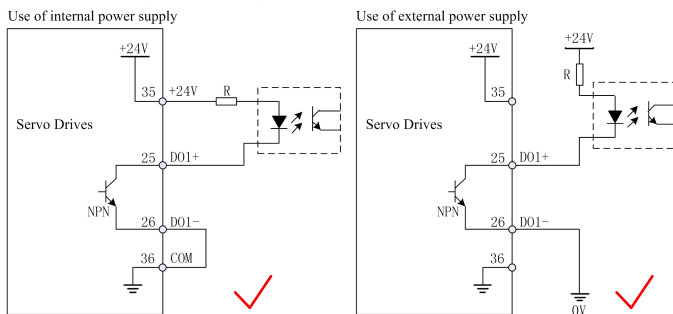


Figure 3-16 Correct wiring when digital output terminals are connected to optocouplers

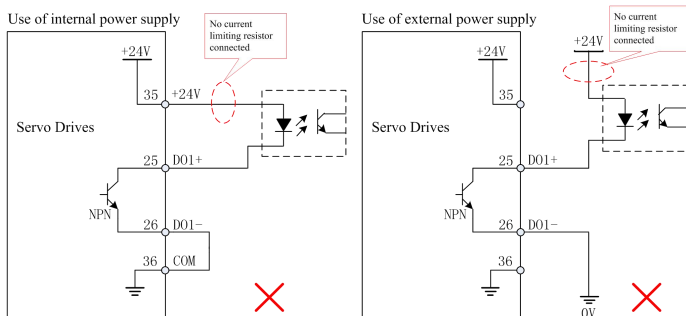


Figure 3-17 Wrong wiring when digital output terminal is connected to optocoupler

3.5.5. Encoder frequency division output signal interface circuit

The encoder divider output signal of the drive is a differential signal. It is normally used as a position feedback signal for position control by a superior device. On the superior unit side, please use the differential receiver circuit to receive it.

Note:

1. The maximum allowable voltage of the servo driver's internal output circuit is 30VDC.
2. The maximum allowable current of the servo driver's internal output circuit is 20mA.

Table 3-12 Encoder frequency division output signal description

Signal name	Pin number	Function
Crossover output	PAO+	29
	PAO-	30
	PBO+	27
	PBO-	28
	PZO+	14
	PZO-	15
	PZ_OC	12
	DGND	11
PE	Housing	Shielding layer

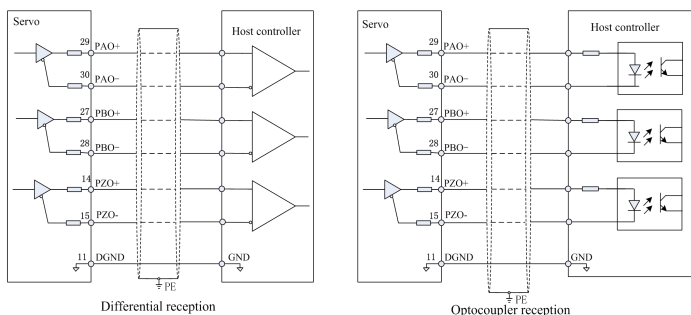


Figure 3-18 Encoder frequency division output connected to the host controller

The encoder phase Z frequency-division output circuit supports open-collector signal output. Generally, this circuit serves to provide feedback signals to the host controller in a position control system. Use an optocoupler circuit, relay circuit, or bus receiver circuit on the host controller side to receive feedback signals.

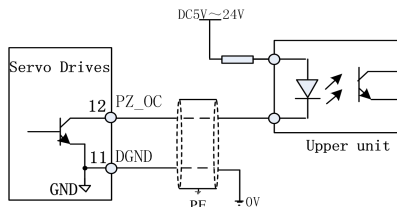


Figure 3-19 Encoder (open collector output) and upper unit connection

3.5.6. Analog input interface circuit

The external analog contains two signals of speed command or torque command. Please use twisted pair. The input standard is as follows :

Analog signal voltage range : $\pm 10V$, resolution of 12 bits

Maximum allowable voltage : $\pm 12V$

Input impedance : above 14k Ω

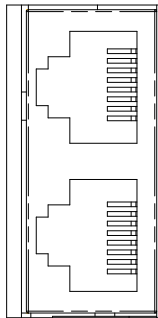
Table 3-13 Description of analog input signals

Signal name	Pin number	Function
V_REF	16	Speed analog input signal
T_REF	18	Torque analog input signal
AGND	17/19	Analog input signal ground

3.6. Communication and full closed-loop circuit wiring

3.6.1. CN3 terminal definition

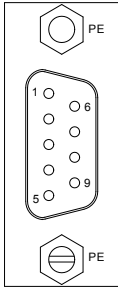
Table 3-14 CN3 terminal definition

Terminal position number		Pin	Definition		
	1	NC	1	NC	
	2	NC	2	NC	
	3	NC	3	NC	
	4	RS+	4	RS+	
	5	RS-	5	RS-	
	6	GND	6	GND	
	7	CAN+	7	CAN+	
	8	CAN-	8	CAN-	
	9	+5V	9	NC	+5V
	10	GND	10	NC	GND
	11	A+	11	NC	A+
	12	B+	12	RS+	B+
	13	B-	13	RS-	B-
	14	A-	14	GND	A-
	15	Z+	15	CAN+	Z+
	16	Z-	16	CAN-	Z-
		Housing	PE		

Note: The 220V driver fully closed-loop encoder input interface is multiplexed with the second port of the RJ45. When the customer selects a model with full closed-loop functionality, RS485 and CAN communication cannot be multi-steps machine parallel through the dual interface of RJ45. The 220V driver fully closed-loop encoder signal is connected to the CN5 terminal.

3.6.2. CN5 Terminal definition

Table 3-15 CN5 terminal definition

Terminal position number	Pin	Definition
	1	A+
	2	A-
	3	B+
	4	B-
	5	Z+
	6	Z-
	7	+5V
	8	GND
	9	GND

3.6.3. RS485 communication connection

Please use twisted pair for RS485 differential signal line, defined as follows:

Table 3-16 RS485 communication signal description

Communication type	Signal	Pin	Function
RS485	RS+	4/12	RS485 differential signal
	RS-	5/13	
	GND	6/14	Signal ground
	PE	Housing	

3.6.4. CAN communication connection

Please use twisted pair for CAN differential signal cable, defined as follows:

Table 3-17 CAN communication signal description

Communication type	Signal	Pin	Function
CAN	CAN+	7/15	CAN differential signal
	CAN-	8/16	
	GND	6/14	Signal Ground
	PE	Housing	

3.6.5. Fully closed-loop encoder connection

The 9~16 pin multiplexing of CN3(220V driver) provides pulse encoder signal input interface, which can constitute full closed loop control. CN5(3800V driver) is an independent full close-loop encoder terminal. The definitions are as follows:

Table 3-18 Description of fully closed-loop encoder signals

CN3 multiplexing pin	CN5 pin	Function
9	7	+5V power supply for encoder
10	8/9	
11	1	Encoder differential signal
12	3	
13	4	
14	2	
15	5	
16	6	
Housing	Housing	Shielding

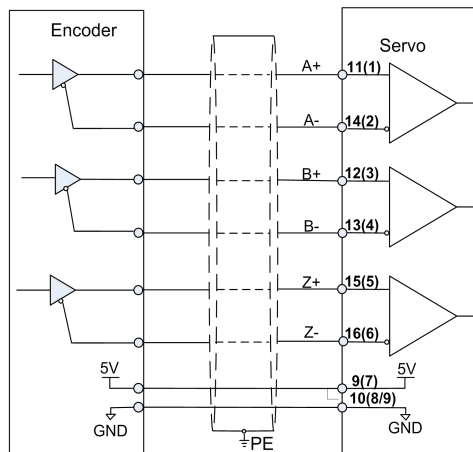


Figure 3-20 Fully closed-loop input wired to encoder unit

Encoder wiring matters:

- 1、Differential line signal lines two in the cable for twisted, such as A + / A- must be a group of twisted pair;
- 2、The length of twisted pair cable within 5m, please use the cross-sectional area of 0.2mm² cable (24AWG), cable length of each increase of 1m, the cross-sectional area of the wire needs to increase by 0.05mm², to ensure the normal power supply and normal transmission of signals;
- 3、Please make sure the shielding network layer is reliably grounded, otherwise it will cause the driver false alarm.

Chapter 4 Panel Display and Operation

4.1. Panel operator

The panel operator allows you to display the status, perform auxiliary functions, set parameters and monitor some of the parameters of the servo unit, as shown in Figure 4-1:

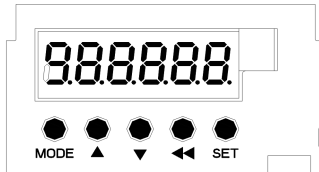


Figure 4-1 Panel Operator

The keys and corresponding functions are shown in the following table:

Number	Key name	Function	Effective form
1	MODE	Mode switch or return to previous menu level	Press to validate
2	▲ UP	Increase/increase the value	Press to validate/long press to validate
3	▼ DOWN	Decrease/decrease value	Press to validate/long press to validate
4	SHIFT	Shift to the left or view high values	Press to validate
5	SET	Store the set value or go to the next level menu	Press to validate

4.2. Panel display and switching

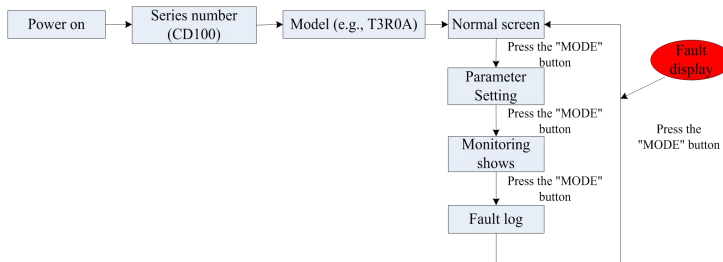


Figure 4-2 Panel display process

- After the power is turned on, the panel displays the series number and model number of the servo driver in turn, and then enters the normal screen.
- Press the "MODE" key to cycle through the different display modes.
- In any display mode, if the driver fails, immediately switch to the fault display mode, LED digital tube flashing synchronously, press the "MODE" key to switch to the normal surface.

- The normal screen is divided into 2 types: the status of the servo drive and the values of the monitoring parameters, set by the hundreds digit of Pn-003.

- No key action within 30S, if the drive has no warning and fault, the panel automatically switches to the normal screen; if the drive has fault or warning, the panel automatically switches to the warning or fault screen.

4.3. Normal screen

4.3.1. Status display of the servo drive

Display	Name	Description
	Power On	1. The three horizontal bars of the first digital tube from the left indicate the position/speed/torque control mode respectively. 2. The three horizontal bars of the second digital tube from the left indicate forward/zero speed/reverse rotation respectively. 3. The first decimal point from the left indicates the action of holding brake, lit for holding brake closed (loose gate), extinguished for holding brake open (holding brake). 4. The second decimal point from the left indicates the action of the braking resistor, lit for the braking resistor on, off for the braking resistor off. 5. The rightmost decimal point blinks to indicate that the servo is enabled and the motor is energized.
	Ready	
	Running	
	JOG	
	Homing	
	Alarm	
	Error	

4.3.2. Numerical display of normal monitoring parameters

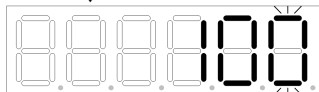
1. Only when the hundreds digit of Pn-003 is set to 1, the value of the normal monitoring parameter is displayed at the normal screen; otherwise, the status of the Servo Drive is displayed.

2. When the positive number is greater than 6 digits or the negative number is greater than 5 digits, the display is divided into two screens, press the "SHIFT" key from the low first screen (low) to the second screen (high) in order to cycle through the display.

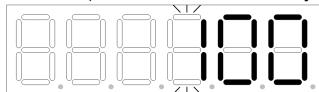
3. One of the leftmost decimal points is lit when the first screen is displayed, and the two leftmost decimal points are lit when the second screen is displayed.

4. If there is no key action within 30S, it will automatically switch to low display.

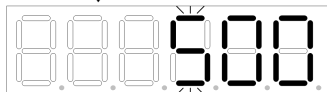
↓ Press "SET" button 1 time



↓ Press the "SHIFT" key 2 times



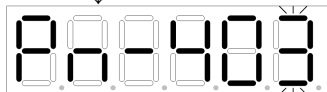
↓ Press the "UP" button 4 times



↓ Press "SET" button 1 time



↓



↓ Press the "MODE" key 1 time



↓ Press "MODE"



4.4. Monitoring parameters

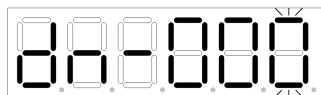
1. When the positive data to be modified is greater than 6 digits or negative data is greater than 5 digits, it will be displayed in two screens, press the "SHIFT" key to cycle from the first screen (low) to the second screen (high) in turn.

2. One leftmost decimal point lights up when the first screen is displayed, and the two leftmost decimal points light up when the second screen is displayed.

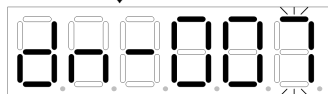
3. When the parameter is positioned, press and hold the "UP" key/"DOWN" key to quickly increase or decrease the value.

4. The value of monitoring parameters cannot be modified, only viewed.

Take the example of viewing the accumulated value of external pulse input, dn-007 = 123456789.



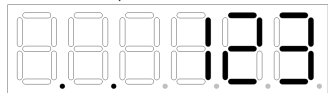
↓ Press the "UP" button 7 times



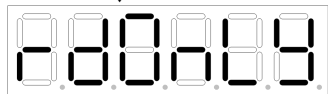
↓ Press "SET" button 1 time



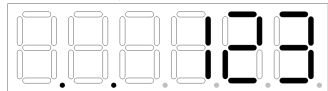
↓ Press the "SHIFT" key 1 time



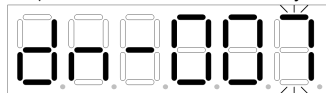
↓ Press the "UP"/"DOWN" key 1 time



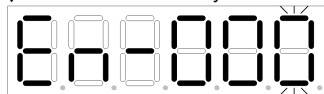
↓



↓ Press the "MODE"/"SET" key 1 time

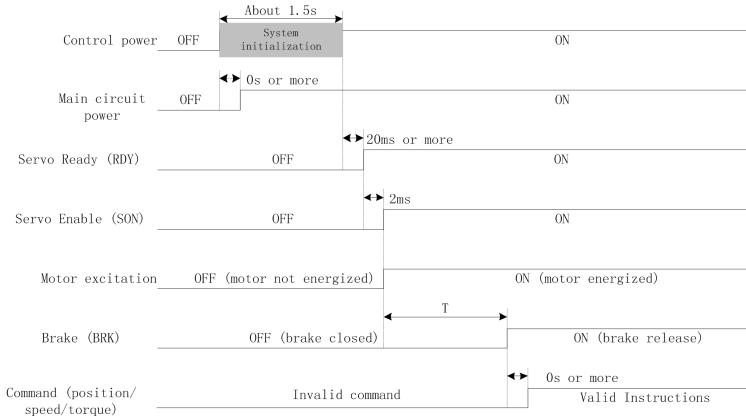


↓ Press the "MODE" key 1 time



4.5. Operation Timing Diagram

4.6.1. When the power is turned on

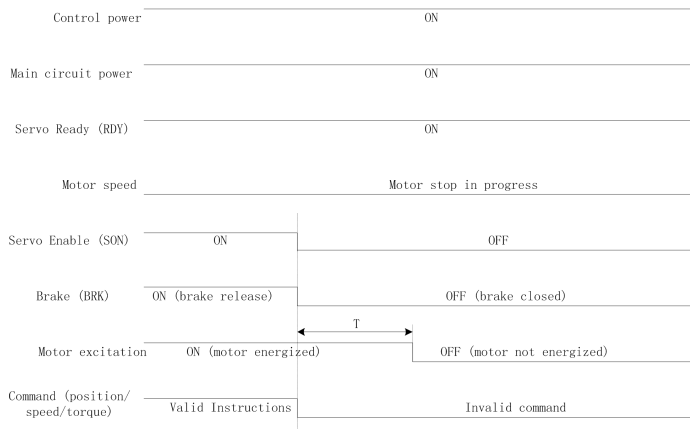


After the control power is established, the MCU initialization action takes 1.5s, after which the protection function starts to work. Please connect all input and output signals to the driver (especially those that may become the protection function trigger circuit, such as forward/reverse disable, etc.) Designed to be confirmed before the protection function work starts.

1. After the system initialization is completed, if the main circuit power is on and there is no other fault, the RDY signal output is valid after about 20ms.

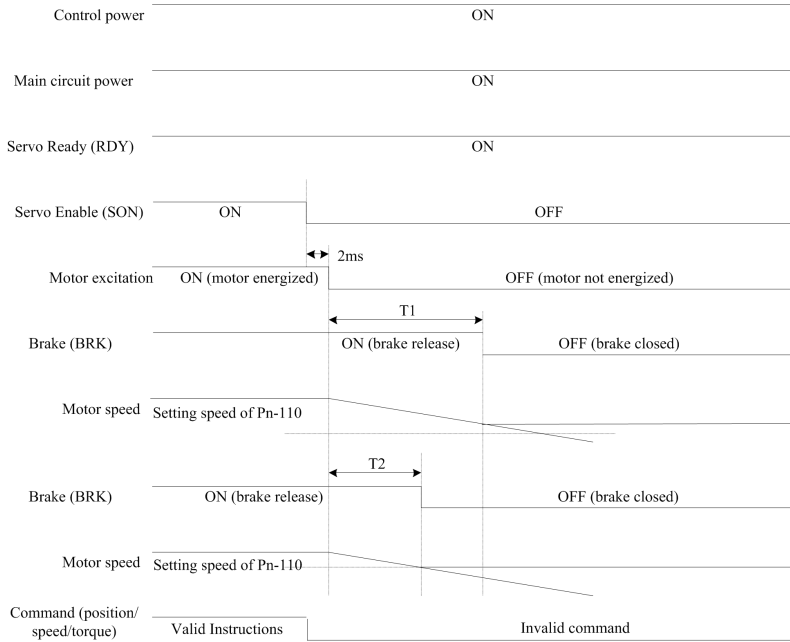
2. the time T between the servo enable signal (SON) is effective and the brake release of the holding brake is set by parameter Pn-108, the range is 0~500ms. the servo does not respond to the command (position/speed/torque) during this time.

4.6.2. Servo ON->OFF when the motor stops



1. The delay time T between the closing of the holding brake and the non-energizing of the motor is set by parameter Pn-109, and the range is 0~500ms.

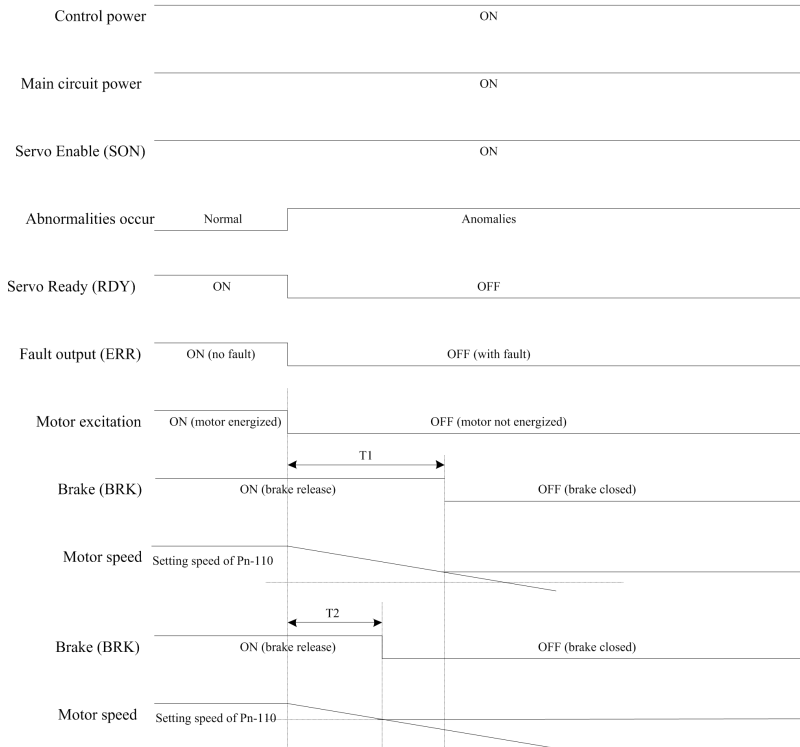
4.6.3. Servo ON->OFF when the motor rotates



1.If the setting time T1 of Pn-111 is shorter than the time when the motor speed drops to the setting speed of Pn-110, the fault occurs after T1 time when the holding brake is closed.

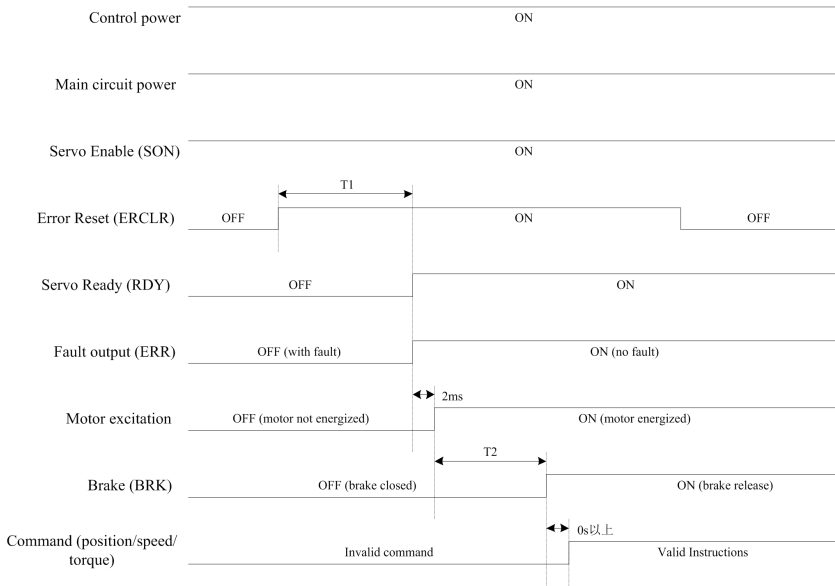
2. If the time T2 for the motor speed to drop to the set speed of Pn-110 is shorter than the set time of Pn-111, the holding brake closes after T2 time when the fault occurs.

4.6.4. In case of malfunction



1. If the setting time T1 of Pn-111 is shorter than the time when the motor speed drops to the setting speed of Pn-110, the holding brake closes after T1 time when the fault occurs.
2. If the time T2 for the motor speed to drop to the set speed of Pn-110 is shorter than the set time of Pn-111, the holding brake closes after T2 time when the fault occurs.

4.6.5. Fault reset

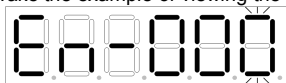


1. The fault reset input is valid on the rising edge, and the valid level of the input should be greater than the filtering time T_1 of DI terminal.

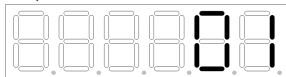
2. The time T_2 after the motor is energized until the holding brake is released, set by parameter Pn-108, the range is 0~500ms. this time the servo does not respond to the command (position/speed/torque).

4.6. Fault record

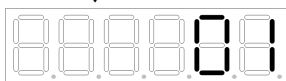
Take the example of viewing the last fault record, En-000 = 01.



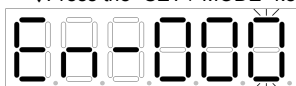
↓ Press "SET" button 1 time



↓ Press the "UP"/"DOWN" key 1 time



↓ Press the "SET"/"MODE" key 1 time



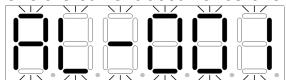
↓ Press the "MODE" key 1 time

Normal screen

4.7. Alarm and fault display

1. In any state, an alarm or fault occurs and immediately jumps to the alarm or fault screen.
2. After the current alarm is lifted, the alarm screen will no longer be displayed; after the current fault is reset, the fault screen will no longer be displayed.

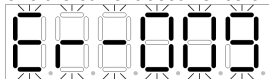
Take the current occurrence of a positive limit (POT) alarm as an example.



↓ Press the "MODE" key once or the alarm is automatically lifted

Normal screen

Take the current occurrence of DC bus undervoltage as an example.



↓ Press the "MODE" key 1 time

Normal screen

Chapter 5 Control Mode

5.1. Detection before operation

In order to ensure personal safety and avoid damage to the servo drive, before running, please carry out strict inspection and confirmation of the following items.

- Please remove the load connected to the servo motor, including connectors and related accessories. After the servo motor can operate normally under without loadcondition, the servo motor can be connected to the load. The axis of the servo motor must be well concentric with the axis rod of the equipment.
- Please implement insulation treatment at the connection part of the distribution terminal and check whether the driver is reliably grounded.
- Check whether the fastening part of the motor is loose, and all the four screws of the fixed motor must be tightened. Conductive objects such as screws or metal sheets and flammable objects cannot be stored in the servo drive.
- Check whether the power supply of the main circuit and control circuit of the servo driver is correct, and pay special attention to distinguishing the main circuit power input (L1, L2, L3 is single-phase / three-phase----AC220V / AC380V).
- Do not connect the power input terminal (L1, L2, L3) directly to the output terminal (U, V, W) of the servo drive, otherwise the servo drive will be damaged.
- Please check the servo drive, servo motor model is correct.
- Check whether the servo motor code is set correctly.(Through the parameter Pn-001 view and modify, motor model and code please refer to this manual 9.6 chapter).
- The definition of the motor rotation direction described in this manual : facing the motor shaft extension, the rotation axis counterclockwise rotation (CCW) is positive rotation, and the rotation axis clockwise rotation (CW) is reverse rotation.

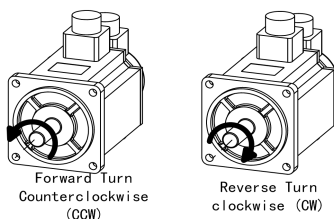


Figure 5-1 Rotation direction definition

5.2. JOG

The JOG operation method of the servo drive is used to test run the drive and motor, and the user does not need additional wiring. For safety reasons, it is recommended that the JOG operation be performed at low speed and without load.

Step1 : Set Pn-011 = 1 to run the servo drive in JOG mode

Step2 : Set Pn-211 = 0001, Pn-200 = 0001, Pn-202 = 0001, use DI1 auxiliary input to force the servo drive to enable.

Step 3 : Modify the value of Pn-007 / 008 / 009 / 010 to set the speed / acceleration time / deceleration time / maximum torque in JOG mode.

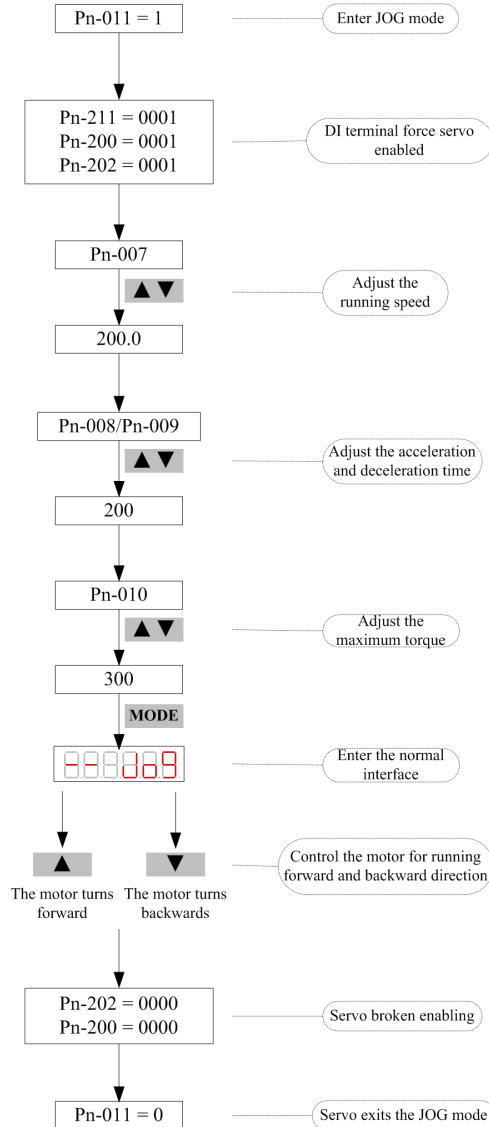


Figure. 5-2 Tap operation steps

Note 1 : If the motor is not running or not working properly, please check whether the UVW wiring and encoder wiring of the motor are correct.

Note 2 : The rotation direction of the motor can be set and changed by the parameter Pn-101.

5.3. Position control mode

Position control mode is applied to precision positioning occasions, such as industrial machinery. The device has three kinds of position command input mode : pulse, internal digital quantity (register) and internal multi-steps instruction. Provides 4 sets of 32-bit electronic gear ratios that can be selected and switched through DI terminals. The 16-segment internal position command is used for trajectory planning. The absolute position and relative position modes can be configured, and can also be switched between any segments through DI terminals.

The position control mode architecture is as follows :

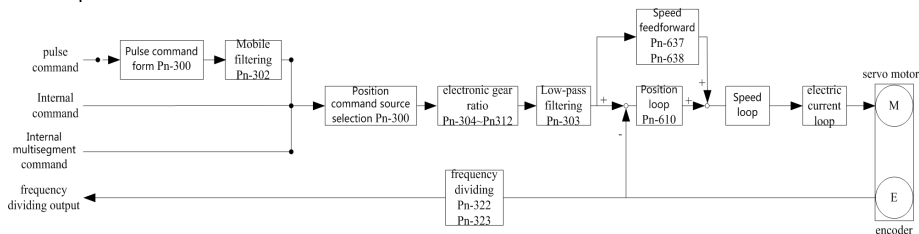


Figure 5-3 Position control

5.3.1. Pulse position command

When the parameter Pn-300 is set to 0, the command source of the position control mode is the pulse command. The pulse command is input to the servo driver by the upper device through the terminal. The pulse command is divided into low-speed pulse and high-speed pulse. Each pulse has three types, each type has positive and negative logic, set by the parameter Pn-101.

Function code	Name	Description	Factory value
Pn-300	Position command source and pulse pattern	units : Location command source 0 : pulse command 1 : Internal digital quantity given (set by Pn-314) 2 : Internal multi-steps position command Tens: Pulse command form 0 : pulse+direction 1 : Positive / reverse pulse 2 : orthogonal pulse Hundreds: Pulse Command Input Terminal Selection 0 : Low speed input terminal 1 : High speed input terminal Thousands: Minimum pulse width of external pulse Low speed terminal High speed terminal 0 : 600ns(0.83M) 0 : 150ns(3.33M) 1 : 2.4us(208K) 1 : 600ns(0.83M) 2 : 4.8us(104K) 2 : 1.2us(416K) 3 : 9.6us(52K) 3 : 2.4us(208K) 4 : filterless 4 : filterless	0000

For a detailed description of the logic pattern, minimum time width and connection mode of the pulse, please refer to chapter 3.5.2.

5.3.2. Internal digital position command

When the parameter Pn-300 is set to 1, the command source of the position control mode is the internal digital quantity. The driver sets up a position command buffer inside the parameters Pn-314 ~ Pn-317 and triggers the position command effectively with an external DI terminal (PTRG).

Function code	Name	Description	Factory value
Pn-314 Pn-315	Internal position command setting value	-2147483648~2147483647	131072
Pn-316	Internal position command runs at maximum speed	Unit is 0.1rpm	100.0
Pn-317	Internal position command acceleration and deceleration time	Unit is ms	100

The internal position command can be a relative position command or an absolute position command. After the PTRG trigger instruction, it can take effect immediately, or it can delay waiting for the entry into force. These are set by the parameter Pn-318.

Function code	Name	Description	Factory value
Pn-318	Internal position command pattern	units : command type 0 : relative displacement 1 : absolute displacement Tens: Continuous position trigger 0 : Receive position command after PTRG trigger. 1 : After the PTRG is triggered, the position command is not received before the end of the previous position. hundreds : filterless Thousands:filterless	0000

The relevant parameters of the internal position command can be modified in real time by communication, with special attention to the fact that the trigger PTRG of the position command is valid at the rising edge.

5.3.3. Internal multi-steps position command

When the bits of the parameter Pn-300 are set to 2, the command source for the position control mode is an internal multi-steps position command. In this control mode, the driver has a single-axis positioning function, and simple process control such as point control or trajectory planning can be realized without the host computer.

In the multi-steps position command mode, up to 16 segments of position control can be realized. Each segment can be set separately for displacement, running speed, acceleration and deceleration time and waiting time. It can run in a single operation, cycle operation or jump operation between segments by DI (ICMD1 ~ ICMD4) switching. PA-Group parameters are used to configure these functions.

The ten-bit setting of the parameter Pn-318 is also applicable to multi-steps position command, which can be configured to take effect immediately after the PTRG trigger instruction or wait for it to take effect.

Function code	Name	Description	Factory value
Pn-A00	Multi-steps displacement control mode	0 : DI switch runs 1 : Cycle running 2 : Stop after a single run	0
Pn-A01	Number of multi-steps displacement end points	1~16 , The number of segments used to select the run during a cycle or single run.	16
Pn-A02	multi-steps displacement instruction type	0 : Relative displacement command 1 : Absolute displacement command	0

Table 5-1 Multi-steps positions

Position segment	ICMD4	ICMD3	ICMD2	ICMD1	PTRG	Corresponding parameters
1	0	0	0	0	↑	Pn-A04 ~ Pn-A09
2	0	0	0	1	↑	Pn-A10 ~ Pn-A15
3	0	0	1	0	↑	Pn-A16 ~ Pn-A21
4	0	1	0	0	↑	Pn-A22 ~ Pn-A27
...	...	...	...	...	...	...
16	1	1	1	1	↑	Pn-A94 ~ Pn-A99

Note :ICMD1 ~ ICMD4 is 0 means DI turn off, is 1 means DI turn on. PTRG ↑ indicates the rising edge of DI.

5.3.4. Electronic gear ratio setting

The electronic gear function can set the working movement amount equivalent to one pulse of the input instruction of the upper device to any value. The one pulse of the command from the upper device, that is, the minimum unit, is called 'reference unit'.

The CD100 series contains four sets of 32-bit electronic gear ratios, and one of them can be selected by arbitrary switching of DI terminals (GEAR1 / GEAR2).

function code	name	description	Factory value
Pn-304 Pn-305	Electronic gear molecule 1	1~536870911	131072annotation ^①
Pn-306 Pn-307	Electronic gear molecule 2	1~536870911	131072 annotation ^①
Pn-308 Pn-309	Electronic gear molecule 3	1~536870911	131072 annotation ^①
Pn-310 Pn-311	Electronic gear molecule 4	1~536870911	131072 annotation ^①
Pn-312 Pn-313	Electronic gear denominator	1~2147483647	10000

Note : ① When the encoder bit is 17, the electronic gear molecule is 2¹⁷, and when the encoder bit is 23, the electronic gear molecule is 2²³

Electronic gear ratio calculation formula :

$$\frac{N \times P}{\Delta L} \times \frac{B}{A} = Pr \times M$$

$$\frac{B}{A} = \frac{Pr \times M \times \Delta L}{N \times P} = \frac{Pr}{\Delta L} \times \frac{M}{N}$$

ΔL : Command units ,such as 0.001 mm / pulse (ball screw or pulley), 0.1° / pulse turntable.

P_r : Encoder pulse, the number of pulses generated by the servo motor rotating 1 lap encoder.

P : Mechanical units, such as ball screw pitch, turntable rotation angle of 1 circle, pulley circumference, etc.

M/N : Reduction ratio, the servo motor rotates M turns when the load shaft rotates N turns, ball screw M/N is 1 / 1.

attention :

1.When the external pulse command is A, B orthogonal type, the servo driver inside the way to count in double frequency.

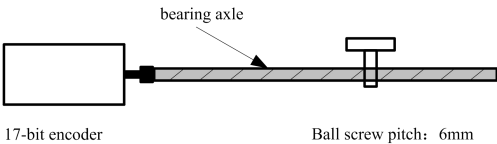
2. The range of electronic gear ratio is $0.02 \leq B/A \leq 2500$.

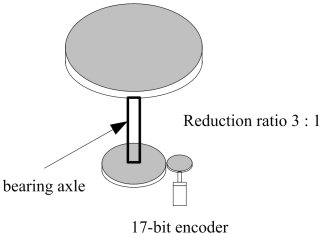
Electronic gear ratio setting steps are as follows :

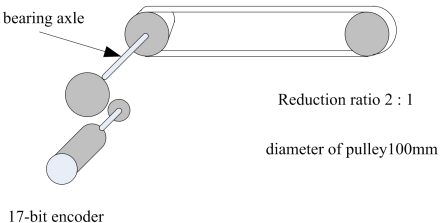
- Confirm mechanical specifications such as ball screw pitch, reduction ratio and pulley diameter.
- Confirm the servo motor encoder resolution, that is, 1 lap of the pulse number.
- On the basis of considering the mechanical specifications, positioning accuracy and other factors to determine the command unit.
- Calculate the amount of reference unit required for the load shaft to rotate 1 circle according to the reference unit.
- Electronic gear ratio (B/A) is calculated by the formula of electronic gear ratio.
- Set the calculated value as the electronic gear ratio.

The following table illustrates how to calculate the electronic gear ratio.

Table 5-2 Electronic gear ratio calculation

Ball screw		
Procedure	Command unit: 0.001mm  17-bit encoder Ball screw pitch: 6mm	
1	Confirm mechanical composition	Ball screw pitch : 6mm reduction ratio : 1/1
2	Encoder	131072pulse/rev
3	Determining the command unit	0.001mm/pulse
4	Calculate the command unit amount required for the load shaft to rotate one circle	$\frac{6\text{mm}}{0.001\text{mm/pulse}} = 6000\text{pulse}$
5	Calculation of electronic gear ratio	$\frac{B}{A} = \frac{131072}{6000} \times \frac{1}{1}$
6	Set-up parameter	32768(molecule)
		1500(denominator)

Rotary table		
Command unit: 0.1°		
Procedure		
1	Confirm mechanical composition	Turntable 1 circle rotation angle : 360° reduction ratio : 3/1
2	Encoder	131072 pulse/rev
3	Determining the command unit	0.1°/pulse
4	Calculate the command unit amount required for the load shaft to rotate one circle	$\frac{360^\circ}{0.1^\circ/\text{pulse}} = 3600\text{pulse}$
5	Calculation of electronic gear ratio	$\frac{B}{A} = \frac{131072}{3600} \times \frac{3}{1}$
6	Set-up parameter	32768(molecule) 300(denominator)

Belt + pulley		
Command unit: 0.01mm		
Procedure		
1	Confirm mechanical composition	Pulley diameter: 100mm/ perimeter 314mm Deceleration ratio : 2 / 1
2	Encoder	131072 pulse/rev
3	Determining the command unit	0.01mm/pulse
4	Calculate the command unit amount required for the load shaft to rotate one circle	$\frac{314\text{mm}}{0.01\text{mm}/\text{pulse}} = 31400\text{pulse}$
5	Calculation of electronic gear ratio	$\frac{B}{A} = \frac{131072}{31400} \times \frac{2}{1}$
6	Set-up parameter	32768(molecule) 3925(denominator)

5.3.5. Inhibit pulse-in function

The inhibit pulse-in function refers to the function of stopping the command pulse input count in the position control mode. To use this function, the DI terminal needs to be configured as an INHP function number. When the INHP input is ON, the pulse command signal stops calculating, so that the motor remains locked.

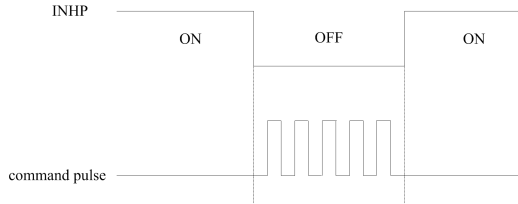


Figure 5-4 Pulse prohibition function

5.4. Speed control mode

Speed control mode is applied to precision speed control occasions, such as CNC machining. This device has three kinds of speed command input mode : analog, internal digital quantity (register) and internal multi-steps instruction. The voltage input range of the speed analog is $\pm 10V$, 12-bit resolution. The parameters of the internal digital instruction can be modified in real time by communication. 16-segment internal speed command Each segment can be individually set for speed, acceleration / deceleration time, and run time, and can be switched between any segments via DI terminals.

The speed control mode architecture is as follows :

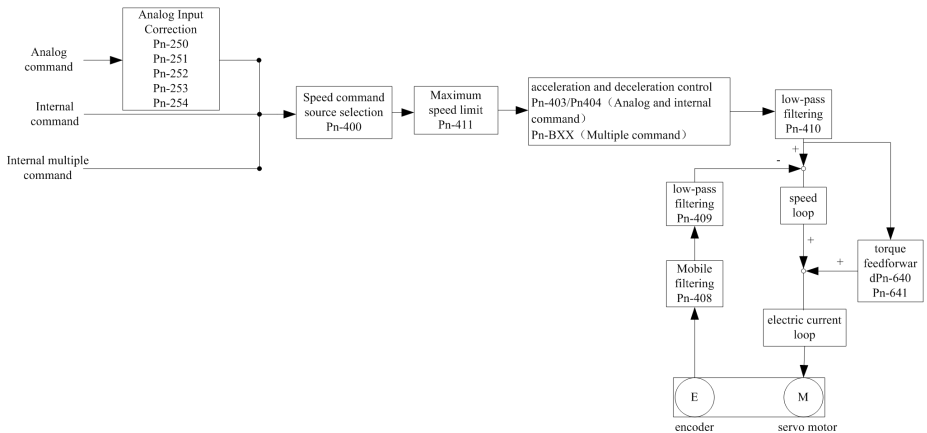


Figure.5-5 Speed control

5.4.1. Analog speed instruction

When the parameter Pn-400 is set to 0, the command source of the speed control mode is an analog command. The analog quantity command is input to the servo driver by the upper device through the terminal.

function code	name	description	Factory value
Pn-250	Analog channel function configuration	Units: Speed Simulation Command Polarity 0: bipolarity 1: Single positive polarity 2: Mono-negative polarity Tens: Torque Simulation Command Polarity 0: bipolarity 1: Single positive polarity 2: Mono-negative polarity Hundreds: reserved Thousands: reserved	0000
Pn-251	Speed analog command ratio	Rotational speed per volt of simulated voltage	300
Pn-252	Speed analog command filtering time	First-order smooth filtering	0
Pn-253	Speed analog command zero offset		0
Pn-254	Speed analog command dead zone		0

The speed analog command ratio specifies the proportional relationship between the voltage of the analog input channel and the speed, that is, the slope of the curve.

Actual speed = speed analog voltage × Pn-251

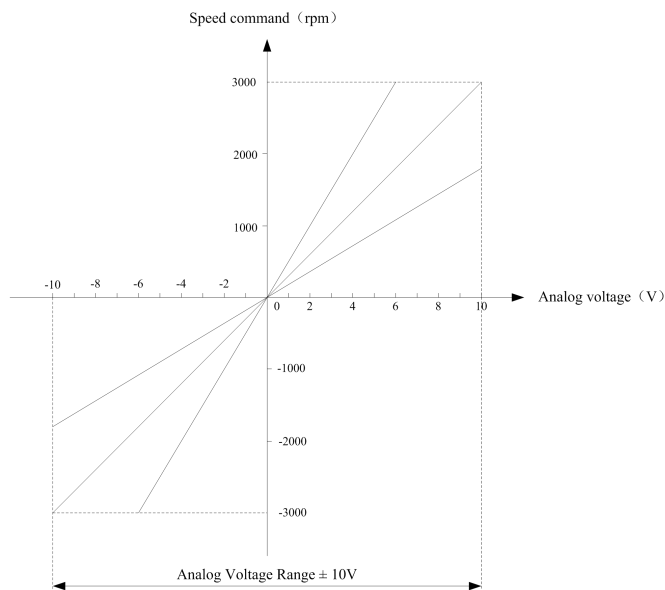


Figure.5-6(a) Analog command ratio

The speed analog command can be smoothed by the parameter Pn-252. The value of this parameter should not be set too large, otherwise it will affect the speed response of the drive.

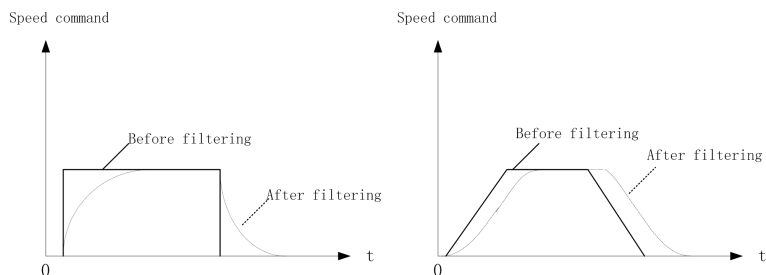


Figure 5-6 (b) Analog command filtering

When using analog to realize speed control, even if the analog command is 0V, the servo motor may still rotate at low speed. This is because the command voltage of the upper device and the external loop has a small deviation in mV, which is called bias. This situation requires the parameter Pn-253 to eliminate the offset.

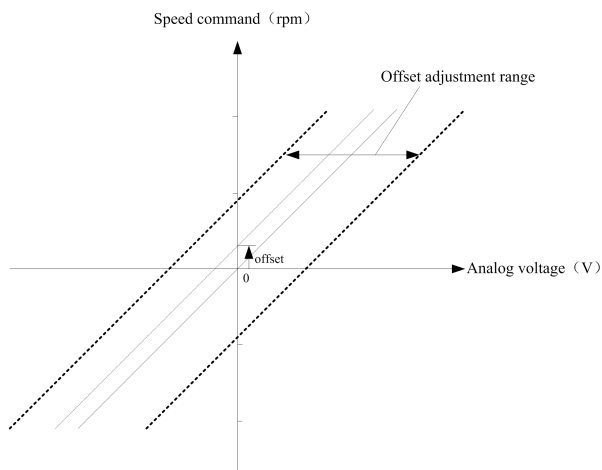


Figure 5-7 Analog command offset

In some applications, the speed command is 0 when the analog quantity is within a certain voltage range. This voltage range is called the dead zone and is set by the parameter Pn-254.

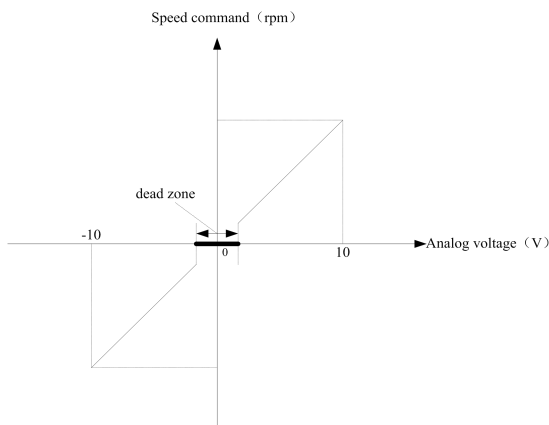


Figure5-8 Analog command dead zone

5.4.2. Internal digital speed command

When the bit of the parameter Pn-400 is set to 1, the command source for the speed control mode is an internal numeric quantity. The driver sets up a speed command buffer through the parameters Pn-401~Pn-404.

Function code	Name	Description	Factory value
Pn-401 Pn-402	Internal speed command setting value	-6000.0~6000.0	100.0
Pn-403	Speed command acceleration time	Both analog and internal speed commands are valid.The unit is ms.	100.0
Pn-404	Speed command deceleration time		100

5.4.3. Internal multi-steps speed commands

When the bit of the parameter Pn-400 is set to 2, the command source of the speed control mode is the internal multi-steps speed command. In this control mode, the driver can realize simple process control without the upper computer.

In multi-steps speed command mode, up to 16 speed segments can be controlled, with each segment individually setting the speed command value, acceleration and deceleration time, and running time. It can run in a single operation, cycle operation or jump operation between segments by DI (ICMD1~ICMD4) switching. PB-group parameters are used to configure these functions.

Function code	Name	Description	Factory value
Pn-B00	Multi-steps speed control	0 : DI switching operation 1 : cycle operation 2 : Single run stop	0
Pn-B01	Number of multi-steps speed end points	1~16 , The number of segments used to select the run during the cycle or single run.	16
Pn-B02	Multi-steps speed running time unit	0 : second 1 : minute	0

Table 5-3 Multi-stage speed

Velocity duration	ICMD4	ICMD3	ICMD2	ICMD1	Corresponding parameters
1	0	0	0	0	Pn-B06 ~ Pn-B10
2	0	0	0	1	Pn-B11 ~ Pn-B15
3	0	0	1	0	Pn-B16 ~ Pn-B20
4	0	1	0	0	Pn-B21 ~ Pn-B25
...	...	...	...	...	...
16	1	1	1	1	Pn-B81 ~ Pn-B95

Note : ICMD1~ICMD4 is 0 means DI turn off, is 1 means DI turn on.

5.4.4. Zero-speed clamping function

The zero-speed clamping function is used when the upper device is not configured with a position loop in the speed control mode.

When the zero-speed clamping (ZCLMP) signal is set to be effective, if the set value of Pn-416 is 0 and the speed command value reaches below the set value of Pn-417 (zero-speed clamping threshold), or the set value of Pn-416 is 1, the position loop is constructed inside the servo driver, ignoring the speed command and fixing the servo motor within ± 1 pulses of the zero-speed clamping effective position, even if it is rotated by external force, it will return to the zero-speed clamping position.

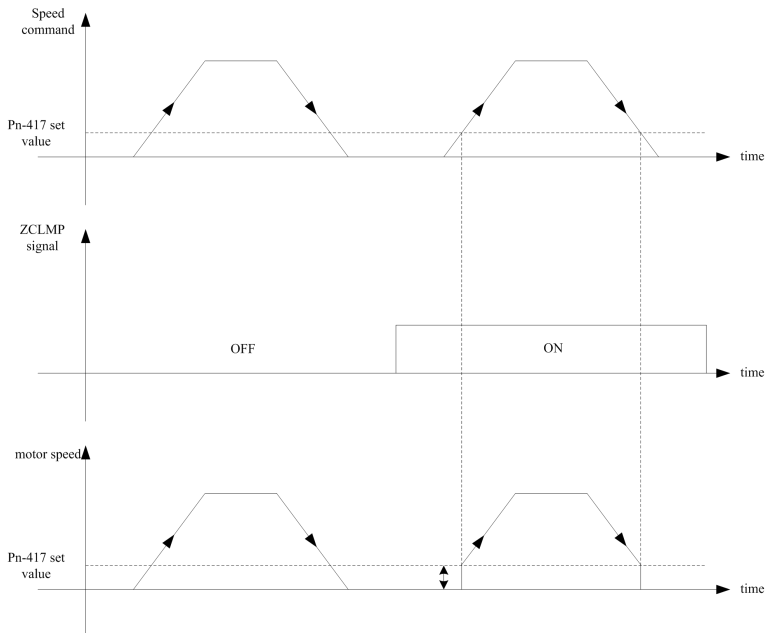


Figure 5-9 Zero-speed clamping

5.5. Torque control mode

The torque control mode is applied to occasions where torque control is required, such as printing machines, winding machinery, etc. This device has three kinds of torque command input mode : analog, internal digital (register) and internal multi-steps instruction. Torque analog voltage input range is $\pm 10V$, 12 bit resolution. The parameters of the internal digital instruction can be modified in real time by communication. 4-segment internal torque command Each segment sets the torque value and settling time individually and can be switched between any segments via DI terminals.

The torque control mode architecture is as follows :

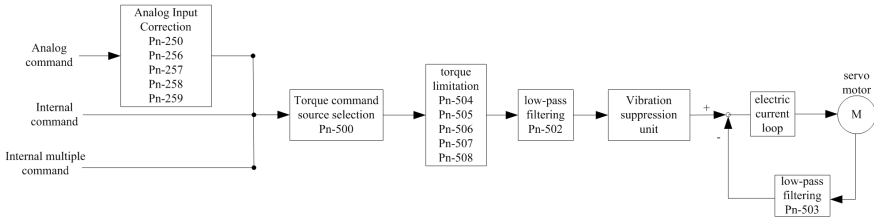


Figure 5-10 Torque control

5.5.1. Analog torque command

When the parameter Pn-500 is set to 0, the command source of the torque control mode is an analog command. The analog quantity command is input to the servo driver by the upper device through the terminal.

Function code	Name	Description	Factory value
Pn-250	Analog channel function configuration	Units : speed analog command polarity 0 : Bipolar 1 : Positive unipolar 2 : Negative unipolar Tens: torque analog command polarity 0 : Bipolar 1 : Positive unipolar 2 : Negative unipolar Hundreds:Reserved Thousand:Reserved	0000
Pn-256	Torque analog command ratio	Analog voltage per volt corresponding torque (percentage of rated torque)	30
Pn-257	Torque analog command filtering time	First-order smooth filtering	0
Pn-258	Torque analog command zero offset		0
Pn-259	Torque analog command dead zone		0

For the correction application of parameters such as the proportion of analog torque command, filtering time, offset and dead zone, please refer to chapter 5.4.1.

5.5.2. Internal digital torque command

When the parameter Pn-500 is set to 1, the command source of the torque control mode is the internal digital quantity. The driver passes the parameter Pn-501 torque command buffer.

Function code	Name	Description	Factory value
Pn-501	Internal torque command	-300%~300%	100%

5.5.3. Internal multi-steps torque command

When the parameter Pn-500 is set to 2, the command source of the torque control mode is the internal multi-steps torque command. In this control mode, the driver can realize simple process control without the upper computer.

Multi-steps torque command mode can achieve a maximum of 4 torque control, each speed can be set separately torque command value and torque settling time. It can be switched by DI (ICMD1~ICMD2) to realize jumping operation between segments. PB-group parameters are used to configure these functions.

Function code	Name	Description	Factory value
Pn-B88	The first torque value	-300~300	100
Pn-B89	The first torque value establishment time	0~65535	50
Pn-B90	The second torque value	-300~300	200
Pn-B91	The second torque value establishment time	0~65535	50
Pn-B92	The third torque value	-300~300	50
Pn-B93	The third torque value establishment time	0~65535	50
Pn-B94	The fourth torque value	-300~300	50
Pn-B95	The fourth torque value establishment time	0~65535	50

Table 5-4 Multi-steps torque

Torque segment	ICMD2	ICMD1	Corresponding parameters
1	0	0	Pn-B88~Pn-B89
2	0	1	Pn-B90~Pn-B91
3	1	0	Pn-B92~Pn-B93
4	1	1	Pn-B94~Pn-B95

Note : ICMD1 ~ ICMD2 is 0 means DI turn off, is 1 means DI turn on.

5.5.4. Torque limitation

To protect the servo drive and motor, torque limits are required. The torque limitation is effective for position control, speed control, torque control and hybrid control modes.

Function code	Name	Description	Factory value
Pn-504	Selection of Torque Limit Source	0 : Internal positive and negative torque limit 1 : External positive and negative torque limit (Through TCCW / TCW terminal selection) 2 : Analog input T _ REF torque limit 3 : Min [Analog Input T _ REF, External Positive and Negative Torque Limit (TCCW / TCW Terminal Selection)] 4 : Min [Analog Input T _ REF, Internal Positive and Negative Torque Limit]	0
Pn-505	Internal positive torque limitation	0~300%	300
Pn-506	Internal negative torque limitation	-300%~0	-300
Pn-507	External positive torque limitation	0~300%	300
Pn-508	External negative torque limitation	-300%~0	-300

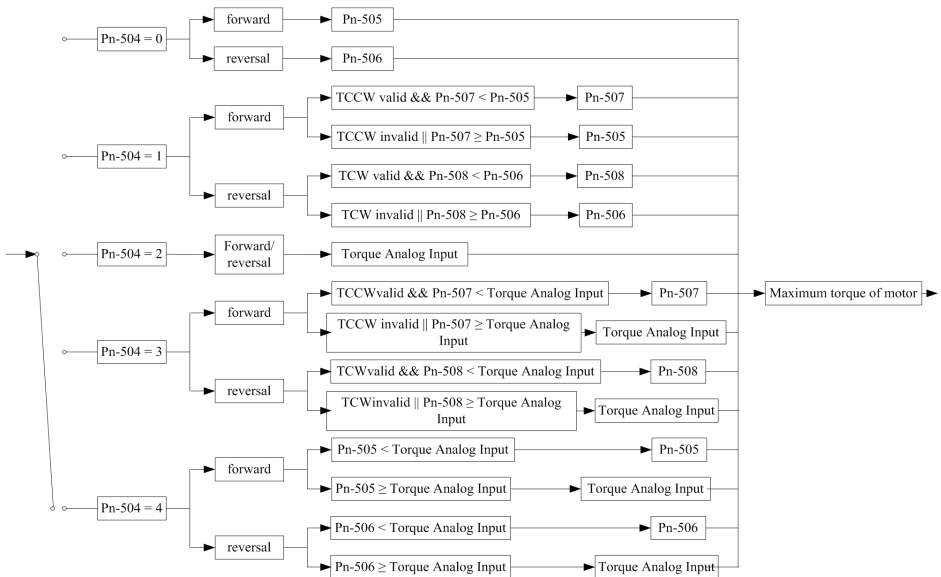


Figure 5-11 Torque limitation

5.5.5. Speed limit in torque mode

In the torque control mode, if the torque command is greater than the load torque on the mechanical side, the motor will continue to accelerate, resulting in overspeed and even damage to the mechanical equipment. Therefore, the speed of the motor must be limited.

Function code	Name	Description	Factory value
Pn-509	Source of Torque Mode Speed Limit	0 : Internal positive and negative speed limit 1 : Analog input for speed limit 2 : Select either 1st or 2nd by DI (ICMD4) The 2nd speed limit input DI Invalid, Pn511 as the positive and negative speed limit DI effective, Pn512 as the positive and negative speed limit	0
Pn-511	Torque Control Forward Speed Limit Value	0~6000.0	3000.0
Pn-512	Reverse speed limit value of torque control	0~6000.0	3000.0

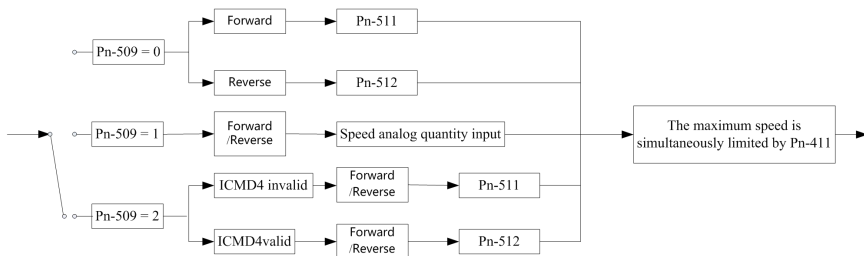


Figure 5-12 Speed limit in torque mode

5.6. Homing

The homing is to link the Z pulse position of the motor encoder to the coordinates inside the drive.

After receiving the get home signal, the servo drive actively locates the relative position of the motor shaft and the mechanical origin according to the pre-set mechanical origin, and first searches for the origin, because the deceleration must be stopped after finding the origin. Therefore, a small distance from the actual origin is exceeded according to the deceleration curve, and then the offset is moved to the actual origin position on this basis. Homing is usually applied to the first time to find the zero point.

Other position commands and software limit functions are blocked during homing. After the homing is complete, the servo driver outputs the DO signal (HOM) after the homing is complete, while the current absolute position of the motor (dn-015 is PUU unit, dn-020 is pulse unit) is consistent with the origin position offset.

5.6.1. Homing parameters

Using the homing function, it is necessary to set up the mechanical origin switch, limit switch and so on in advance. Ensure that the mechanical equipment is not damaged during the origin return process.

The relevant parameters are as follows :

Function code	Name	Description	Factory value
Pn-336	Homing control	Units:Reference point selection 0 : Forward looking for POT signal as reference point 1 : Find the NOT signal as the reference point 2 : Forward looking for REF rising edge as reference point 3 : Reverse finds the REF rising edge as the reference point 4 : Forward looking for REF falling edge as reference point 5 : Reverse looking for REF falling edge as a reference point 6 : Forward looking for Z pulse as reference point 7 : Reversal find Z pulse as reference point 8 : The current position is the reference point Tens:regression model 0 : Find the Z-pulse backward as the origin 1 : Forward the Z pulse as the origin 2 : Direct to the reference point as the origin 3 : Look back for the reference point as the origin 4 : Forward the reference point as the origin 5 : The current position is the origin Hundreds:The limit treatment method 0 : Warning 1 : Automatic reverse Thousand: starting mode 0 : Close the homing function 1 : First enable automatic homing after servo power on 2 : Homing triggered by GHOM function of DI terminals (function number 30)	0122
Pn-337	First high-speed homing speed	Speed of search for origin	100.0
Pn-338	Second low speed homing speed	Move a section of offset, return to the speed of the origin	20.0
Pn-339	homing acceleration and deceleration time		100
Pn-340 Pn-341	Origin position offset		0

5.6.2. Rajectomy of the homing

The homing function of CD100 series can support all 35 regression methods and trajectories specified by CiA402 protocol by properly setting parameter Pn-336.

Special Note :Set the filter time of the DI terminal to more than 10ms when the origin is regressed. Otherwise, it may cause the servo drive to fail to search for the origin or the homing trajectory error.

Table 5-5 Homing method

Serial number	Reference point and origin	Explanation	Parameter value
1	NOT , Zpulse	Reverse to find NOT, backward (forward) to find Z pulse.	0x2101
2	POT , Zpulse	Forward looking for POT, backward (reverse) looking for Z pulse.	0x2100
3	REF , Zpulse	Forward looking for REF rising edge, backward looking for Z pulse, infinite bit.	0x2002
4	REF , Zpulse	Forward looking for REF rising edge, forward looking for Z pulse, infinite.	0x2012
5	REF , Zpulse	Reverse to find REF rising edge, backward to find Z pulse, infinite bit.	0x2003
6	REF , Zpulse	Reverse to find REF rising edge, forward to find Z pulse, infinite bit.	0x2013
7	REF , Zpulse	Forward looking for REF rising edge, backward looking for Z pulse, POT limit.	0x2102
8	REF , Zpulse	Forward looking for REF rising edge, forward looking for Z pulse, POT limit.	0x2112
9	REF , Zpulse	Forward looking for REF falling edge, backward looking for Z pulse, POT limit.	0x2104
10	REF , Zpulse	Forward looking for REF falling edge, forward looking for Z pulse, POT limit.	0x2114
11	REF , Zpulse	Reverse to find REF rising edge, backward to find Z pulse, NOT limit.	0x2103
12	REF , Zpulse	Reverse to find REF rising edge, forward to find Z pulse, NOT limit.	0x2113
13	REF , Zpulse	Reverse to find REF falling edge, backward to find Z pulse, NOT limit.	0x2105
14	REF , Zpulse	Reverse to find REF falling edge, forward to find Z pulse, NOT limit.	0x2115
15	--	--	--
16	--	--	--
17	NOT , NOT	Reverse to find NOT, directly with NOT as the origin.	0x2121
18	POT , POT	Forward looking for POT, directly to POT as the origin.	0x2120
19	REF , REF	Forward looking for REF falling edge, directly to REF as the origin.	0x2024
20	REF , REF	Turn to find the rising edge of REF, directly with REF as the origin.	0x2022
21	REF , REF	Reverse to find the REF falling edge, directly with REF as the origin.	0x2025
22	REF , REF	Reverse to find REF rising edge, directly to REF as the origin.	0x2023
23	REF , REF	Forward looking for REF falling edge, directly with REF as the origin (backward), POT limit.	0x2134
24	REF , REF	Forward looking for REF rising edge, directly to REF as the origin (backward), POT limit.	0x2132

Serial number	Reference point and origin	Explanation	Parameter value
25	REF , REF	Forward to find the falling edge of REF, directly with REF as the origin (forward), POT limit.	0x2144
26	REF , REF	Turn forward to find the rising edge of REF, directly with REF as the origin (forward), POT limit.	0x2142
27	REF , REF	Reverse to find REF falling edge, directly to REF as the origin (backward), NOT limit.	0x2135
28	REF , REF	Reverse to find REF rising edge, directly to REF as the origin (backward), NOT limit.	0x2133
29	REF , REF	Reverse to find REF falling edge, directly to REF as the origin (forward), NOT limit.	0x2145
30	REF , REF	Reverse to find REF rising edge, directly to REF as the origin (forward), NOT limit.	0x2143
31	--	--	--
32	--	--	--
33	Zpulse , Zpulse	Reversal to find Z pulse	0x2027
34	Zpulse , Zpulse	Forward looking for Z pulse	0x2026
35		Taking the current position as the origin	0x2058

(1). Reference point : Reverse limit signal

Origin : Z pulse

Trajectory : high-speed reversal to find reverse limit NOT signal, in case of NOT rising edge after low-speed positive rotation, through the NOT falling edge, in case of the first Z pulse shutdown.

Parameter : Pn-336 = 0x2101

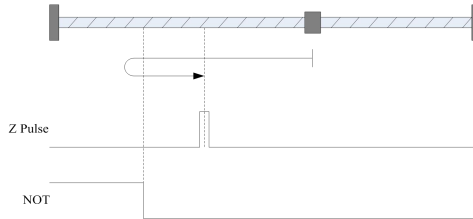


Figure 5-13 Homing method 1

(2). Reference point : positive limit signal

Origin : Z pulse

Trajectory : high-speed forward to find the positive limit POT signal, encounter POT rising edge after low-speed reversal, through the POT falling edge, encounter the first Z pulse stop.

Parameter : Pn-336 = 0x2100

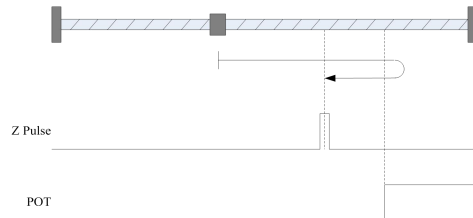


Figure 5-14 Homing method 2

(3). Reference point : homing reference signal

Origin : Z pulse

Trajectory : The reference signal REF is invalid when the homing starts, high-speed forward turn to find REF, low-speed reversal after the REF rising edge, after the REF falling edge, the first Z pulse stops.

The reference signal REF is valid when the homing starts, and reverses at a low speed. After the REF falling edge, the first Z pulse stops.

Parameter : Pn-336 = 0x2002

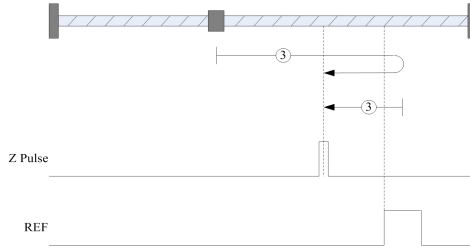


Figure 5-15 Homing method 3

(4). Reference point : homing reference signal

Origin : Z pulse

Trajectory : Homing reference signal REF is invalid when the homing starts, low-speed forward looking for REF, after encountering REF rising edge continue low-speed forward, encounter the first Z pulse stop.

The homing reference signal REF is valid when the homing starts, high-speed reversal, through the REF falling edge, low-speed forward to find REF, continue to low-speed forward after the REF rising edge, meet the first Z pulse stop.

Parameter : Pn-336 = 0x2012

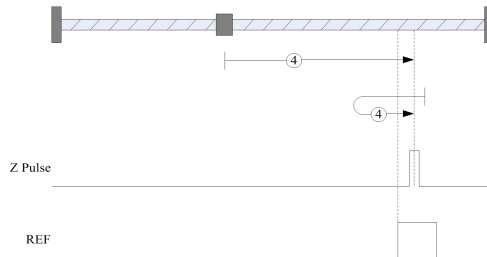


Figure 5-16 Homing method 4

(5). Reference point : homing reference signal

Origin : Z pulse

Trajectory : homing reference signal REF is invalid when the homing is started, high-speed reversal to find REF, encounter REF rising edge after low-speed forward, through the REF falling edge, encounter the first Z pulse stop.

The homing reference signal REF is valid when the homing starts, low speed forward, through the REF falling edge, encounter the first Z pulse stop.

Parameter : Pn-336 = 0x2003

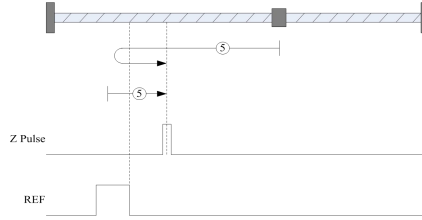


Figure 5-17 Homing method 5

(6). Reference point : homing reference signal

Origin : Z pulse

Trajectory : homing reference signal REF is invalid when the homing starts, low speed reversal to find REF, meet the REF rising edge to continue low speed reversal, meet the first Z pulse stop.

The reference signal REF of the homing is effective when the homing is started. The high-speed forward rotation, through the REF falling edge, the low-speed reversal to find the REF, through the REF rising edge, meets the first Z pulse shutdown.

Parameter : Pn-336 = 0x2013

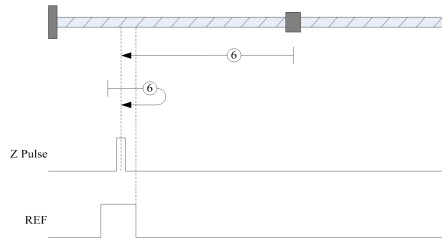


Figure 5-18 Homing method 6

(7). (8). (9). (10). The reference point is the homing reference signal, the origin is the Z pulse, and the positive limit signal is added. The search trajectory is as follows :

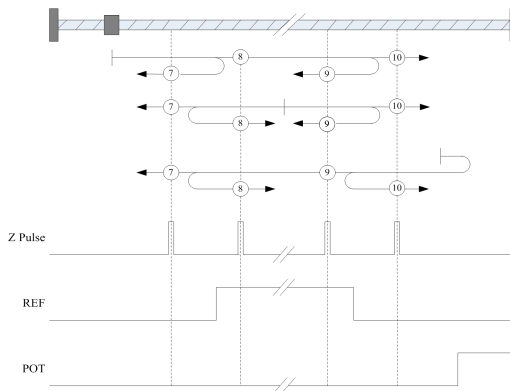


Figure 5-19 Homing method 7/8/9/10

(11). (12). (13). (14). The reference point is the homing reference signal, the origin is the Z pulse, and the reverse limit signal is added. The search trajectory is as follows :

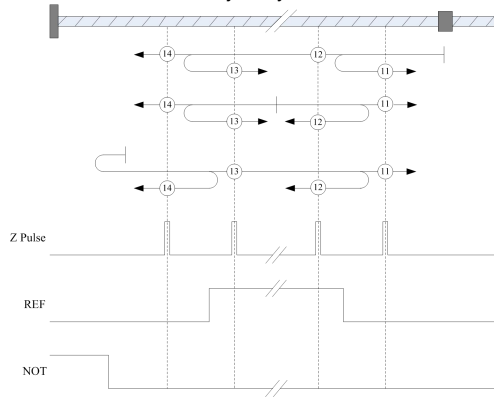


Figure 5-20 Homing method 11/12/13/14

(17).~(30).and (1).~(14).except that the reference point and the origin are the same. Taking (19). (20). as an example, the search trajectory is as follows :

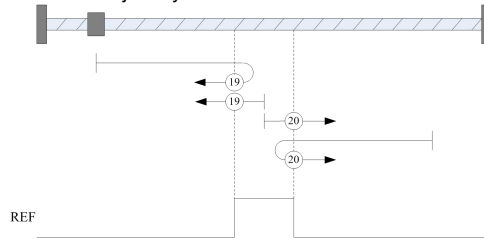


Figure 5-21 Homing method 19/20

(33). (34). The reference point and the origin are both Z pulses, and the search trajectory is as follows :

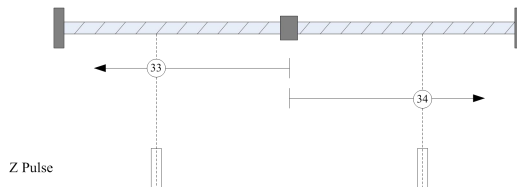


Figure 5-22 Homing method 33/34

(35). Take the current position as the origin.

5.7. The use of absolute position system

The absolute encoder not only detects the position of the servo motor in a single circle, but also counts the number of rotating circles of the servo motor. The single-turn resolution is 17 bits (131072), and the 16-bit multi-turn data can be recorded (-32768 ~ 32767).

When using a servo motor with an absolute encoder, an encoder connection cable with a battery box must be used at the same time. When the servo drive is powered off, the encoder backs up and records data via the battery, and when powered on, the drive calculates the absolute mechanical position from the absolute encoder position, eliminating the need for repeated homing operations.

When the CD100 Series Servo Drive matches an absolute encoder motor to build an absolute position system, parameter Pn-016 needs to be set to 1 (absolute position system). When power is first applied, the Servo Drive will report a warning of AL-010 (absolute position lost) because the coordinate system has not yet been established. This warning needs to be initialized for the absolute position system before it disappears.

In the absolute position system, the data size of its position has a certain limit. When the number of motor rotations exceeds the range of -32768~32767, the servo driver will report a warning of AL012 (absolute encoder multi-turn data overflow). From the user's point of view, the absolute position is a 32-bit data and must be between -2147483648~2147183647, otherwise a warning AL-020 (absolute position count overflow) will be generated.

Warnings AL-010, AL-012 and AL-020 can be detected by setting parameter Pn-810 to 1 mask

5.7.1. Absolute position system coordinate initialization

The CD100 series servo drives provide two methods for initializing the coordinates of the absolute position system : homing and parameter setting

homing : When the homing process is completed, the multi-turn data of the absolute encoder is reset, the system coordinates are defined to the position of a single circle, and the deviation is saved to the EEPROM.

Parameter setting: First set parameter Pn-815 to 0x0814 to allow absolute position zeroing operation, then set parameter Pn-814 to 1 to trigger the absolute position system zeroing action, after completion the multi-turn data of the absolute type encoder is reset, the system coordinates are defined to the single-turn position, and the deviation is saved to EEPROM.

5.7.2. Absolute Position Counting and Reading

The upper device can read the absolute position information of the absolute encoder in real time, and the absolute position is counted by two methods : encoder unit (pulse value) and user unit (PUU value).

$$\text{Pulse value} = \text{PUU value} \times \text{Electronic gear ratio}$$

When the motor rotates clockwise, the number of circles is defined as negative ; when rotating counterclockwise, the number of circles is defined as positive, and the maximum countable number of circles ranges from - 32768 to + 32767. When the number of circles exceeds this range, a warning of AL-012 will be generated. At this time, the coordinate initialization must be re-initialized to clear AL-012. If the system rotates in the counterclockwise direction and the value reaches 32767, the position of the next circle reaches, and the value will become - 32768. If the number of circles continues to increase, the value will increase in the direction of - 32768, - 32767, - 32766... If the system is clockwise, after reaching the maximum value of - 32768, it will become 32767, 32766..., and so on.

Number of pulses = Number of clusters × 131072 + Number of pulses (0-131072) + Bias

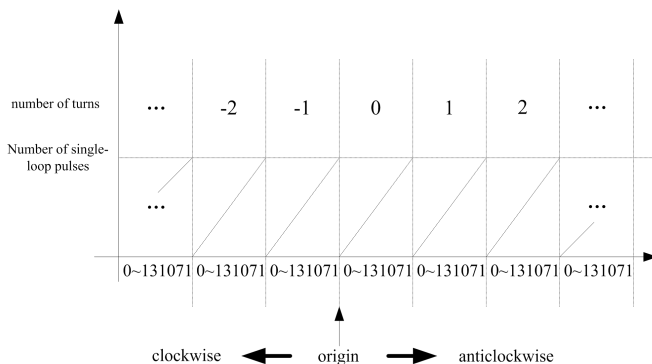


Figure 5-23 Pulse numerical counting method

The PUU value is a 32-bit absolute position data with positive and negative signs. When the motor rotates in the positive direction, the absolute position will increase; when the motor rotates in the negative direction, the absolute position will decrease. The positive rotation direction of the motor can be defined by Pn-101, not by the positive and counterclockwise direction to make a judgment, in short, make the encoder feedback value increases the direction of positive rotation.

If the motor continues to rotate in the same direction, when the PUU value exceeds the range limit of -2147483648~2147483647, the driver will report the absolute position counter overflow warning AL-020, and then the coordinate initialization must be done again to clear the warning. When the positive rotation exceeds the maximum value of the positive PUU, the value changes from 2147483647 back to -2147483648, -2147483647..., and when the negative rotation exceeds the maximum value of the negative PUU, the value changes from -2147483648 back to 2147483647, -2147483646... 2147483646....

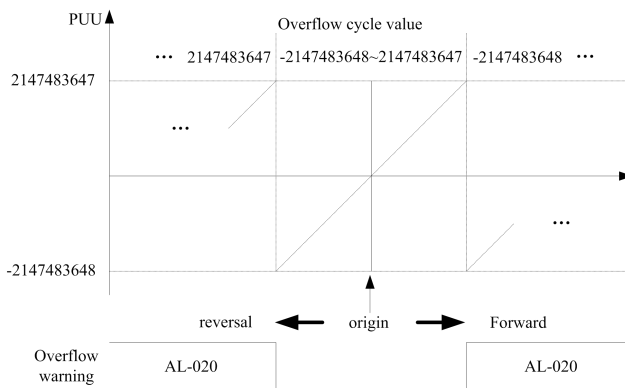


Figure 5-24 PUU value counting method

5.7.3. Absolute position system related parameters

Function code	Name	Description	Factory value
Pn-016	Absolute position system selection	0 : Incremental position system 1 : Absolute position system	0
Pn-810	Absolute position data overflow warning shielding selection	0 : Not shielding 1 : Shielding	0
Pn-814	Absolute position zeroing	0 : No operation 1 : Zeroing the absolute position of the current encoder Note: To ensure system safety, you need to set Pn-815 to 0x0814 first to allow the operation of absolute position to zero.	0
Pn-815	Absolute position zeroing enable	A setting of 0x0814 allows the absolute position to be zeroed, other values have no effect.	0x0000
dn-015 dn-016	System absolute position (PUU value)	-2147483648~2147483647	-
dn-020 dn-021	System absolute position (pulse value)	-2147483648~2147483647	-

Chapter 6 Adjustment

Servo drives need to drive the motor quickly and accurately to track commands from the host computer or internal settings. To achieve this requirement, the servo gain must be properly adjusted.

Servo gain contains several parameters, such as position loop gain, velocity loop gain, filter, load rotational inertia ratio, etc., which affect each other. Therefore, the setting of the servo gain must take into account the balance between the setting values of each parameter.

The general process of gain adjustment is as follows.

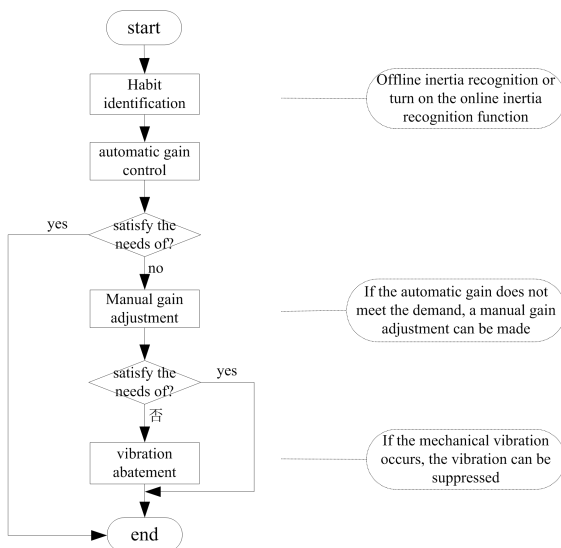


Figure 6-1 Gain adjustment process

6.1. Load inertia identification

Define the equivalent to the load inertia on the shaft as J_L and the motor body inertia as J_M , then the total inertia equivalent to the motor shaft is $J_L + J_M$.

The load inertia ratio β is.

$$\beta = \frac{J_L}{J_M}$$

The load inertia ratio (Pn-602) is an important parameter of the servo system. If the load inertia ratio is not accurately estimated, the best gain effect cannot be obtained. The load inertia ratio can be identified both offline and online.

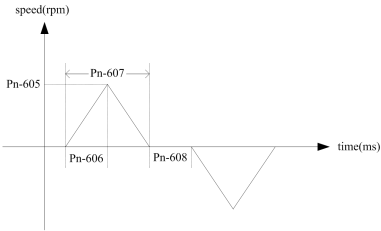
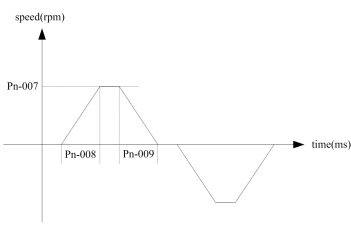
6.1.1. Offline inertia identification

Before off-line inertia identification, please confirm the following :

1. The actual load inertia ratio is below 120.0 times.
2. The time to accelerate to 2000RPM should be within 1S.
3. the positive and negative directions of the mechanical limit switch have more than 1 circle of movable stroke.
4. the motor can run to the maximum speed identified within the identified stroke range.
5. If the initial value of the inertia ratio is small, and the actual load inertia ratio is greater than 30.0, then the initial value is recommended to 5.0 as the initial value, and gradually increase the inertia ratio until the identification process panel display updates.

The off-line inertia identification is divided into two modes : positive and negative triangular wave and JOG point motion. The instruction forms of the two modes are different.

Table 6-1 Offline inertia identification

Projects	Positive and negative delta wave(Pn-604 = 0)	JOG point action(Pn-604 = 1)
Command form		
Identification steps	<pre> graph TD Start([start]) --> Pn003[Pn-003 = 0160] Pn003 --> SetSpeed[Set the speed / acceleration / deceleration time / travel / interval] SetSpeed --> ServoEnabling[Servo enabling] ServoEnabling --> Pn603[Pn-603 = 1] Pn603 --> PressMODE[Press MODE to switch to the normal screen] PressMODE --> ShowStability{Show stability?} ShowStability -- no --> Pn603 ShowStability -- yes --> Pn602[Pn-602 = panel shows the inertia ratio] Pn602 --> End([end]) </pre> Normal face change monitoring and inertia ratio Set the inertia identification related parameters Begin habitual identification Save the inertia ratio	<pre> graph TD Start([start]) --> Pn003[Pn-003 = 0160] Pn003 --> SetSpeed[Set the speed / acceleration time] SetSpeed --> Pn011[Pn-011 = 1] Pn011 --> ServoEnabling[Servo enabling] ServoEnabling --> Pn603[Pn-603 = 1] Pn603 --> PressMODE[Press MODE to switch to the normal screen] PressMODE --> PressUPDOWN[Press the UP to click forward Press DOWN to reverse click] PressUPDOWN --> ShowStability{Show stability?} ShowStability -- no --> PressUPDOWN ShowStability -- yes --> Pn602[Pn-602 = panel shows the inertia ratio] Pn602 --> Pn011_0[Pn-011 = 0] Pn011_0 --> End([end]) </pre> Normal face change monitoring and inertia ratio Set the inertia identification related parameters Point movement mode Begin habitual identification Save the inertia ratio Exit point motion mode
Maximum speed	Pn-605	Pn-007
Acceleration and deceleration time	Pn-606	Pn-008/ Pn-009
Motor stroke	Pn-607	Determined by key time
Interval time	Pn-608	Time interval between two keystrokes
Applicable situation	Running stroke shorter occasions	No limitation on the operating stroke

6.1.2. Online inertia identification

When parameter Pn-686 is set to a non-zero value, the servo system turns on the online inertia recognition function. After it is turned on, the servo continuously estimates and calculates the load inertia ratio according to the command of the upper control system, and automatically saves the load inertia ratio to the parameter Pn-602 every 30 minutes.

There are three modes of online inertia recognition.

Parameter Pn686 = 1, which is suitable for applications where the actual load inertia ratio will hardly change, such as machine tools, wood carving machines, etc.

Parameter Pn686 = 2, which is suitable for applications where the actual load inertia ratio changes slowly.

parameter Pn686 = 3 is applicable to the case where the actual load inertia ratio will change rapidly, such as the handling robot, etc.

The online inertia identification process is as follows.

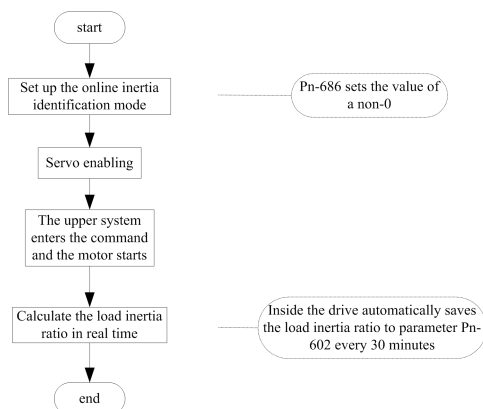


Figure 6-2 Online inertia identification

6.2. Automatic gain adjustment

Automatic gain adjustment refers to the use of rigid grade parameters (Pn-601), the servo driver automatically calculates a set of matching gain-related parameters to meet the rapidity and stability requirements, instead of manual gain tedious repeated debugging process.

In particular, before using the automatic gain mode, you need to correctly obtain the load inertia ratio.

The rigidity grade (Pn-601) ranges from 0 to 31. The rigidity corresponding to the 0 level is the weakest and the gain is the smallest ; the 31 level corresponds to the strongest rigidity and the largest gain.

According to different load types, the rigid level setting reference is as follows :

Rigidity rating	Load type
Level 4~8	Big machinery
Level 8-15	Applications with low rigidity such as belt pulleys
Level 15-20	Ball screw, direct connection and other high rigidity applications

There are two types of auto gain modes, auto gain mode 1 is suitable for most of the applications, and auto gain mode 2 is used only when high speed of positioning is required.

Parameter Pn-600 = 1, auto gain mode 1, focus on stability.

Parameter Pn-600 = 2, auto gain mode 2, focus on positioning.

6.2.1. Auto Gain Mode 1

The first set of gain-related parameters are automatically updated and saved to the corresponding function codes according to the rigidity level.

Parameter	Name
Pn-610	Position loop gain
Pn-611	Speed loop gain
Pn-612	Integral time constant of velocity loop
Pn-502	Torque command filtering time constant

6.2.2. Automatic gain mode 2

The first set of gain-related parameters is automatically updated and saved to the corresponding function code according to the rigidity level.

The second group of gain-related parameters is automatically updated and saved to the corresponding function code according to the rigidity level, and the position loop gain is one rigidity level higher than the first group. At the same time, the integral time constant of the speed loop is fixed to 1000.0 ms, that is, the speed loop integral is invalid, and the speed loop adopts proportional control.

The speed feedforward related parameters are set to fixed values.

The gain switching function is automatically turned on, using the 10 th gain switching condition, and the relevant parameter values of the gain switching are fixed.

Parameter	Name	Description
Pn-610	Position loop gain	
Pn-611	Velocity loop gain	
Pn-612	Speed loop integration time constant	
Pn-502	Torque command filter time constant	
Pn-620	2nd position loop gain	One level of rigidity higher than Pn-610
Pn-621	2nd speed loop gain	
Pn-622	2nd speed loop integration time constant	Fixed to 1000.0ms, i.e. invalid speed loop integration
Pn-637	Speed feedforward gain	Fixed to 30.0%
Pn-638	Speed feedforward filtering time constant	Fixed to 0.5ms
Pn-651	Gain switching condition	Fixed to 0x000A
Pn-652	Gain switching threshold	50
Pn-654	Gain switching hysteresis loop	30
Pn-655	Gain switching time	3.0ms
Pn-656	Gain switching delay time	5.0ms

6.3. Manual gain adjustment

When the automatic gain adjustment does not achieve the desired effect, the manual gain mode ($Pn-600 = 0$) can be used to fine tune the gain and optimize the effect through more detailed adjustment.

The servo system is composed of three control loops, with the position loop, speed loop and current loop in order from outside to inside.

The more inward the loop is, the higher the requirement is accordingly. The Default current loop gain of the servo system has ensured sufficient responsiveness without adjustment, only the position and velocity loop gains, i.e. other auxiliary gains, need to be adjusted.

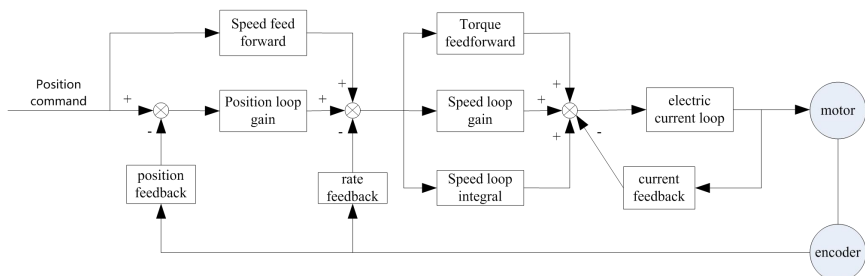
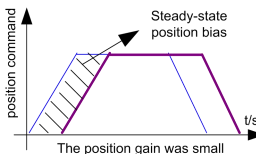
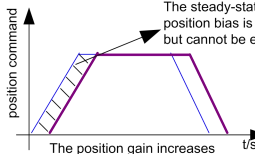
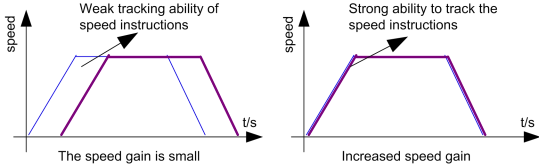
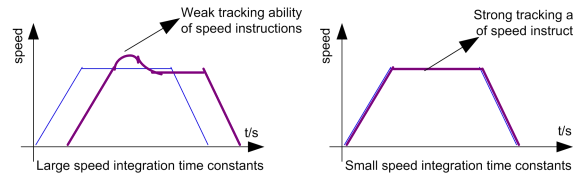
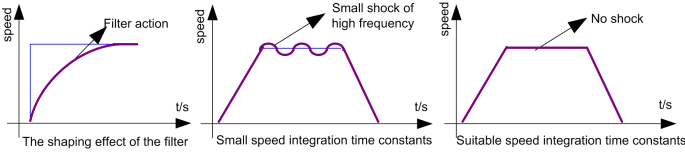


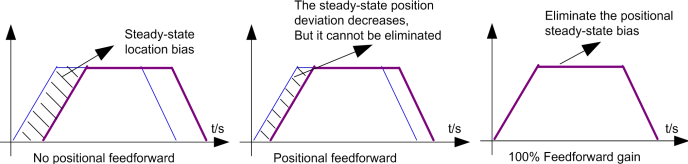
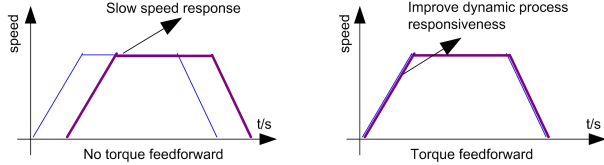
Figure 6-3 Servo control loop

The manual gain adjustment method is as follows :

Table 6-2 Gain related parameters description

Parameter	Name	Adjustment description
Pn-610	Position loop gain	●Parametric action : The greater the ability to follow the position command, the smaller the steady-state deviation, too large a setting is likely to lead to system vibration. After adjusting this parameter, the velocity loop parameter should be adjusted accordingly, and the principle is that the position loop bandwidth is not greater than 1/4 of the velocity loop bandwidth. Under the premise that the inertia ratio Pn-602 is set accurately, the frequency of the position loop following the speed command is the value set by Pn-610.   ●Adjustment method : Increasing the position gain helps to reduce the position deviation. In theory, increasing the position gain without limit can eliminate the position deviation, but in practice, too much position gain can lead to vibration or damage of mechanical equipment, and if you need to eliminate the steady-state position deviation, you can use velocity feedforward (Pn-637).

Pn-611	Speed loop gain	●Parametric action : The larger the response, the faster the response, the stronger the following effect on the speed command, too large may lead to vibration.  ●Adjustment method : In practice, increase this parameter as much as possible, so that the system does not occur vibration to meet the process requirements. In position mode, this parameter is adjusted together with the position gain Pn-610, generally the position gain increases, the speed gain increases, and the position loop gain is greater than 3~4 times the speed loop gain.
Pn-612	Speed loop integration time constant	●Parametric action : Eliminate the steady-state deviation of the speed loop, the smaller the integral effect is stronger, too large lead to loop hysteresis, causing mechanical vibration.  ●Adjustment method : Co-regulation with speed gain, the relationship between the two satisfies : $40 < Pn-610 \cdot Pn-611 < 1000$
Pn-502	Torque command filtering time constant	●Parametric action : The torque filtering effect is to smooth the current and make the machinery run smoothly and softly, the larger the filtering effect is the more obvious but the delay is also bigger.  ●Adjustment method : When setting, note that the torque filter cutoff frequency should be at least four times greater than the speed loop bandwidth, otherwise the delayed link introduced by the filter will lead to untimely torque current response causing oscillation.

Pn-637 Pn-638	Speed feedforward gain and filtering	●Parametric action : The speed feedforward can reduce or eliminate the steady-state position deviation. When the system inputs a fixed frequency position command, the servo motor also runs stably at a certain speed. At this time, the steady-state deviation can be linearly reduced by adjusting the speed feedforward value. When the feedforward gain is 100 %, the steady-state position deviation is 0.  ●Adjustment method : The feedforward filtering time constant and feedforward gain work together to reduce or eliminate the noise caused by too much feedforward gain, too much may lead to lagging feedforward action and cause oscillation after positioning is completed, Default 0.5ms.
Pn-639 Pn-640	Torque feedforward gain and filtering	●Parametric action : Torque feedforward is the amount of advance adjustment made in response to speed command changes, which can increase the following capability of the speed loop and reduce the positioning  ●Adjustment method : The time constant of feed-forward filtering generally cooperates with feed-forward gain, which can reduce or eliminate the noise caused by excessive feed-forward gain. Excessive feed-forward gain may lead to feed-forward lag and cause oscillation after positioning.
Pn-645	Decoupling feedforward gain	●Parametric action : Decoupling feedforward helps to improve the response ability of current loop and compensate the disturbance of motor back EMF to current loop. ●Adjustment method : regulating range 0~100%.

6.4. Filter parameter adjustment

The position command has multiple filters, as shown in the following figure:

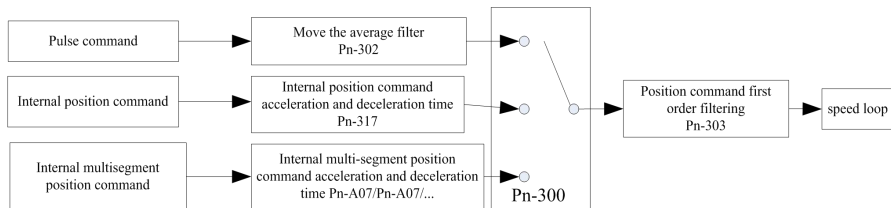
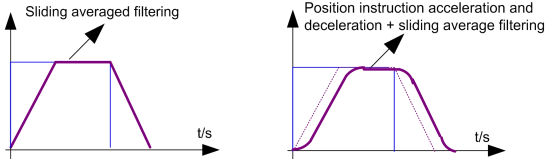
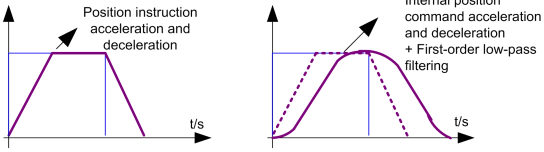
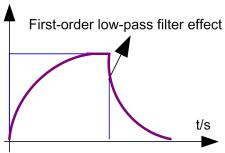


Figure 6-4 Filter

Several filter roles and diagrams are shown below.

Table 6-3 Filter parameter description

Parameter	Name	Adjustment description
Pn-302	Position command moving average filter constant	 For smoothing the command pulse issued by the upper control system, the moving average filtering can change the step command into the harmonic acceleration and deceleration command, and change the slope acceleration and deceleration command into the 'S' type smoothing command.
Pn-317 Pn-A07 ...	Internal/ multi-stage position command acceleration/ deceleration time	 Internal/multi-stage position acceleration/deceleration times can be smoothed with step commands, and "near-S" command curves can be achieved with first-order low-pass filtering.
Pn-303	Position command first-order filter time constant	 Use with sliding filter, or alone.

6.5. Vibration Suppression

Due to the elasticity and backlash of the mechanical components, the mechanical system has a certain resonance frequency. When the servo gain increases, resonance may occur in the mechanical resonance frequency accessories, resulting in the gain cannot continue to increase. There are two ways to suppress mechanical resonance.

Torque command filtering :

By setting the first-order filtering time constant (Pn-502) of the torque command, the torque command is attenuated in the high frequency band above the cut-off frequency to suppress mechanical resonance.

Wave trap :

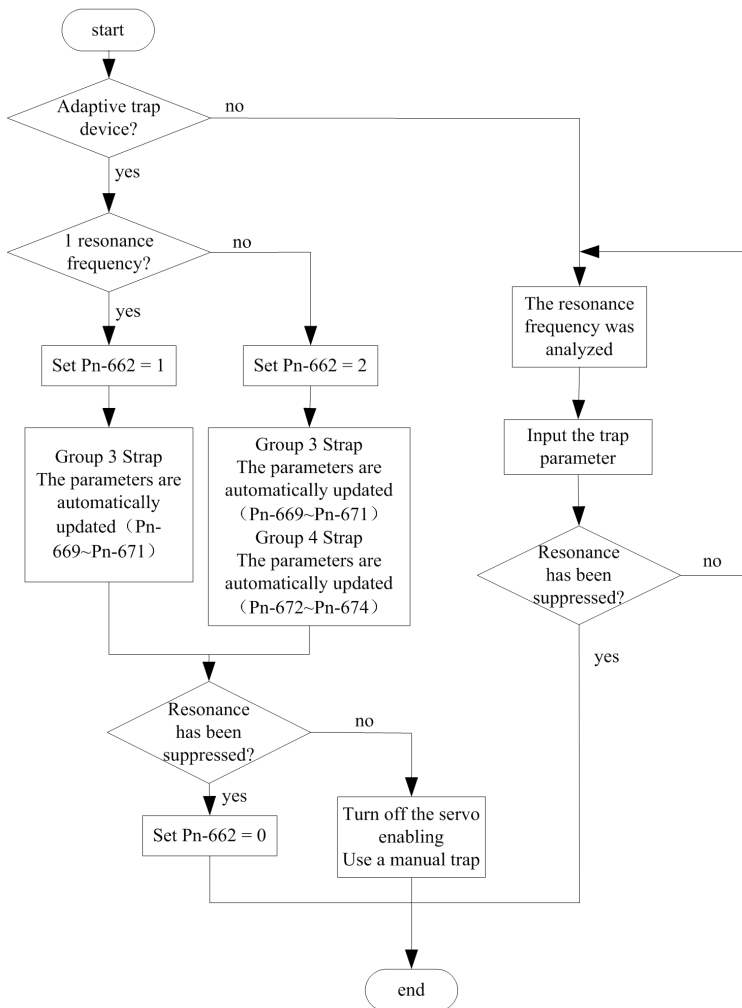


Figure 6-5 Notch function

The mechanical resonance is suppressed by reducing the gain at a specific frequency. The vibration can be effectively suppressed by correctly setting the notch filter parameters, and the servo gain can be continuously increased.

The servo drive contains four sets of notch filters, each containing three parameters (notch frequency / width class / depth class).

The parameter Pn-662 = 0, four sets of notch filter are manual notch filter.

The parameter Pn-662 = 1, the first and second groups of notch filters are manual notch filters, and the third group of notch filters are adaptive notch filters.

The parameter Pn-662 = 2, the first and second groups of notch filters are manual notch filters, and the third and fourth groups of notch filters are adaptive notch filters.

If resonance occurs requiring a notch filter, use an adaptive notch filter first. Adaptive notch is invalid or ineffective, then try to use manual notch.

Adaptive Notch Debugging Steps :

- ①Pn-662 (adaptive notch mode selection) is set to 1 or 2 according to the number of resonance points. When resonance occurs, the Pn-662 is set to 1 to open an adaptive notch filter. After the gain is adjusted, if a new resonance occurs, the Pn-662 is set to 2 to start two adaptive notch filters.
- ②When the servo is running, the 3rd or 4th set of trap parameters are automatically updated and the response function code is automatically stored every 30min.
- ③If the resonance is suppressed, indicating that the adaptive notch filter to achieve results, waiting for a period of time after the servo stable operation, Pn-662 is set to 0, the adaptive notch filter parameters are fixed to the last update value.
- ④If the vibration can not be eliminated for a long time, please close the servo enable in time.
- ⑤When an adaptive notch filter is used, if the servo enable is turned off within 30 minutes, the notch filter parameters will not be stored in the corresponding function code.

Manual wave trap commissioning procedure :

- ①Analyze the mechanical resonance frequency. The resonance frequency can be obtained through the software debugging platform of the servo, or set Pn-662 to 3. When the servo is running, the system automatically tests the resonance frequency and saves the test results in parameter dn-036.
- ②The measured resonance frequency is input into the selected notch filter parameters, and the width level and depth level of the notch filter are input.
- ③If the resonance is suppressed, indicating that the notch effect, you can continue to adjust the gain, if the gain increases after the emergence of new resonance, repeat the above steps.
- ④If the vibration cannot be eliminated for a long time, please turn off the servo enable in time.

Chapter 7 Parameter Description

7.1. Parameter and attribute definition

The parameters were divided into 16 groups, with PC / PD / PE was the reserved parameter group for product function module expansion. Each set of parameters contains 100 parameters, occupying the 2 parameter codes and communication addresses if it is the 32-bit data.

Description of parameter groups		Description of the function code symbol	
Pn-0XX	Basic parameter groups	R	User cannot modify read-only parameters.
Pn-1XX	Basic control parameter groups	H	Hexadecimal parameter, to be changed for each digit.
Pn-2XX	Input and output parameter groups	■	This parameter cannot be changed under Servo On status.
Pn-3XX	Position control parameter groups	▲	Parameter values will not be saved after power-off.
Pn-4XX	Speed control parameter groups	●	After the parameter is modified, it will not be valid until powered on again.
Pn-5XX	Torque control parameter groups	☆	Relevant with the drive's model.
Pn-6XX	Gain adjustment parameter groups	※	This function needs configuring DI/DO. Refer to DI/DO functional configuration table.
Pn-7XX	Full-closed loop parameter groups	P	Position control
Pn-8XX	Fault and protection parameter groups	S	Speed control
Pn-9XX	Communication parameter groups	T	Torque control
Pn-AXX	Multi-segment position parameter group		
Pn-BXX	Multi-segment speed and torque parameter group		
Pn-CXX	Reserved parameter groups		
Pn-DXX	Reserved parameter groups		
Pn-EXX	Reserved parameter groups		
dn-0XX	Monitor parameter groups		

7.2. List of parameters

7.2.1. Basic parameter groups

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-000	Program version	100.04	1		R	P S T
Pn-001	Motor code	1000~4599	1	☆	●	P S T
Pn-002	Reserved					
Pn-003	Panel shows state	Units+Tens : normal monitoring parameters 00~99, Set the serial number of the normal monitoring parameters (The dn group parameter) Hundreds : panel display 0 : Show the status of the drive 1 : Display the normal monitoring parameters Thousands: reserved	1	0000	H	P S T
Pn-004	Parameter management	Units : EEPROM setting 0:Modify EPROM after parameter setting 1:The power off is not saved after the parameter is set and restoring after powering on again Tens : fault parameter clearing 0 : No action 1 : Clearing Default En parameter(zero clearing after finishing) Hundreds : Restore to factory parameter 0: No action 1: Restore to Default. (zero clearing after finishing 0) Thousands : Parameter copy 0 : No action 1 : Copy the native parameters to another drive (Automatic cleaning 0)	1	0000	H/■	P S T
Pn-005	Reserved					
Pn-006	Reserved					
Pn-007	JOG speed value	0.0~6000.0	0.1RPM	200.0		S
Pn-008	JOG acceleration time	0~60000	1ms	200		S
Pn-009	JOG deceleration time	0~60000	1ms	200		S
Pn-010	JOG mode run time torque limit	0~300	1%	300		S
Pn-011	JOG control mode enable	0 : JOG not valid 1 : Enter JOG operation mode		0		S
Pn-012	Custom macro parameters	0 : No action 1 : Custom macro parameters1 2 : Custom macro parameters2 3 : Custom macro parameters3 4 : Custom macro parameters4 5 : Custom macro parameters5 6 : Custom macro parameters6		0	■	P S T
Pn-013	Factory parameter	Factory code			H	P S T

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-014	Reserved					
Pn-015	Reserved					
Pn-016	Absolute position system selection	0 : Incremental position system 1 : Absolute position system		0	■/●	P
Pn-020	Bus voltage correction factor	Setting range:0.950~1.050	0.001	1.000		PST

7.2.2.Basic control parameter groups

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-100	Control mode	0 : Position control 1 : Speed control 2 : Torque control 3 : Speed / position mode ※ 4 : Torque / position mode ※ 5 : Torque / speed mode ※ 6 : CANopen control mode	1	0	■	P S T
Pn-101	Rotation direction selection	0: Forward CCW, Reverse CW 1: Forward CW, Reverse CCW	1	0	■	P S T
Pn-102	Reserved					
Pn-103	Servo OFF stop mode	0 : Free stop, after stopping into free running state . 1 :Zero speed stop, after stopping into the free running state . 2 : Stopping the motor by DB (Dynamic Brake). 3 : Stop the motor with the DB (Dynamic Brake) and then release the DB.		0	■/●	P S T
Pn-104	Overtravel (POT/NOT) stop mode selection	0 : Free stop, after stopping to enter the operation free state . 1 :Zero speed stop, enter servo lock state after stop. 2 :Zero speed stop, after stopping into the free running state . 3 : Set the value of Pn-516 as the maximum torque deceleration stop, and enter the servo lock state after stopping . 4 : Set the value of Pn-516 as the maximum torque deceleration stop, and enter the free running state after stopping.		0	●	P S T
Pn-105	Emergency (EMG) stop mode select	0 : Free stop, after stopping into free running state . 1 :Zero speed stop, after stopping into the free running state . 2 : Set the value of Pn-516 as the maximum torque deceleration stop, and enter the free running state after stopping.		0	●	P S T
Pn-106	Alarm (Error) stop mode select	0:Free stop, keep free running state 1:Stopping the motor by DB (Dynamic Brake) 2:Stop the motor with the DB (dynamic brake) and then release the DB		0	●	P S T

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-107	Reserved					
Pn-108	Electromagnetic brake opening delay time	0~500	1ms	0		P S T
Pn-109	Stationary state, servo OFF delay time	0~500	1ms	0		P S T
Pn-110	Rotating state, electromagnetic brake closing speed threshold	0~1000	1RPM	50		P S T
Pn-111	Rotation state, electromagnetic brake closing delay time	0~1000	1ms	500		P S T
Pn-112 ~ Pn-118	Reserved					
Pn-119	Braking resistor loading rate	0.0~100.0	0.1	10.0		P S T

7.2.3.Input and output parameter groups

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-200	DI terminal source control switch	0 : Input point status is controlled by external hardware terminals 1 : Input point status is controlled by the auxiliary input(Pn-202) Note : bit0~bit15 corresponds to DI1~DI16	1	0000	H/■	P S T
Pn-201	DO terminal source control switch	0 : Output point status controlled by internal program 1 : Output point status is controlled by the auxiliary output(Pn-203) Note: bit0~bit9 corresponds to DO1~DO10	1	0000	H/■	P S T
Pn-202	DI terminal auxiliary input	0 : Invalid input, contact off 1 : Input active, contact on Note: bit0~bit15 corresponds to DI1~DI16	1	0000	H	P S T
Pn-203	DO terminal auxiliary output	0 : Invalid output, contact off 1 : Output active, contact on Note: bit0~bit9 corresponds to DO1~DO10	1	0000	H	P S T
Pn-204	DI terminal auxiliary input management	0 : Auxiliary input status is not saved after power failure 1 : Auxiliary input status saving after power failure Note: bit0~bit15 corresponds to DI1~DI16	1	0000	H	P S T
Pn-205	DO terminal auxiliary output management	0 : Auxiliary output status is not saved after power failure 1 : Auxiliary output status saving after power failure Note: bit0~bit9 corresponds to DI1~DI10	1	0000	H	P S T
Pn-206	Reserved					
Pn-207	Reserved					
Pn-208	DI1~DI8 low speed terminal filtering time	1~32	2ms	5		P S T

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-209	DI7 high-speed terminal filtering grade	0~6	10us	1	•	P S T
Pn-210	DI8 high-speed terminal filtering grade	0~6	10us	1	•	P S T
Pn-211	DI1terminal function planning	Units/Tens : Input function configuration: see DI function assignment table Hundreds : Input Logic 0 : Positive Logic 1 : Negative Logic Thousand:Timer and counter trigger bits 0 : Do not trigger timers and counters 1 : Trigger start timer 1 2 : Trigger start timer 2 3 : Trigger start timer 3 4 : Trigger start timer 4 5 : Trigger off timer 1 6 : Trigger off timer 2 7 : Trigger off timer 3 8 : Trigger off timer 4 9 : Trigger start 16bit counter A : Trigger start 32bit counter B : Trigger off 16bit counter C : Trigger off 32bit counter Note : It only takes effect when the timer configuration or the startup method of the counter configuration is set to external triggering.	1	0001	H/■	P S T
Pn-212	DI2terminal function planning			0004		
Pn-213	DI3terminal function planning			0005		
Pn-214	DI4terminal function planning			0003		
Pn-215	DI5terminal function planning			0013		
Pn-216	DI6terminal function planning			0009		
Pn-217	DI7terminal function planning			0017		
Pn-218	DI8terminal function planning			0002		
Pn-219	DI09 function planning (virtual terminal)			0000		
Pn-220	DI10function planning (virtual terminal)			0000		
Pn-221	DI11function planning (virtual terminal)			0000		
Pn-222	DI12function planning (virtual terminal)			0000		
Pn-223	DI13function planning (virtual terminal)			0000		
Pn-224	DI14function planning (virtual terminal)			0000		
Pn-225	DI15function planning (virtual terminal)			0000		
Pn-226	DI16function planning (virtual terminal)			0000		
Pn-227	Reserved					
Pn-228	Reserved					
Pn-229	Reserved					
Pn-230	Reserved					

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-231	DO1 terminal function planning	Units/Tens: Output function configuration: see DO function assignment table Hundreds: Input Logic 0 : Positive Logic 1 : Negative Logic Thousand:Timer and counter trigger bit 0 : Timer and counter are not triggered 1 : Triggers start timer 1 2 : Triggers start timer 2 3 : Triggers start timer 3 4 : Triggers start timer 4 5 : Triggers off timer 1 6 : Triggers off timer 2 7 : Triggers off timer 3 8 : Triggers off timer 4 9 : Trigger start 16 bit counter A : Trigger start 32 bit counter B : Triggers close of 16 bit counter C : Triggers close of 32 bit counter Note: This takes effect only when the startup mode of timer configuration or counter configuration is set to external trigger	1	0001	H/■	P S T
Pn-232	DO2terminal function planning			0005		
Pn-233	DO3terminal function planning			0008		
Pn-234	DO4terminal function planning			0003		
Pn-235	DO5terminal function planning			0002		
Pn-236	DO6 function planning (virtual terminal)			0000		
Pn-237	DO7 function planning (virtual terminal)			0000		
Pn-238	DO8 function planning (virtual terminal)			0000		
Pn-239	DO9 function planning (virtual terminal)			0000		
Pn-240	DO10 function planning (virtual terminal)			0000		
Pn-241	DO11 Function Planning (Virtual Terminal)			0000		
Pn-242	DO12 Function Planning (Virtual Terminal)			0000		
Pn-243	DO13 Function Planning (Virtual Terminal)			0000		
Pn-244	DO14 Function Planning (Virtual Terminal)			0000		
Pn-245	DO15 Function Planning (Virtual Terminal)			0000		
Pn-246	DO16 Function Planning (Virtual Terminal)			0000		
Pn-247 ~ Pn-249	Reserved					
Pn-250	Analog channel function configuration	Units : speed analog command polarity 0 : Bipolar 1 : Positive unipolar 2 : Negative unipolar Tens: torque analog command polarity 0 : Bipolar 1 : Positive unipolar 2 : Negative unipolar Hundreds:Reserved Thousand:Reserved	1	0000	H	S T

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-251	Speed analog command ratio	10~3000	1RPM/V	300		S
Pn-252	Speed analog command filtering time	0~50	2ms	1		S
Pn-253	Speed analog command zero offset	-300~300	6.03mv	0		S
Pn-254	Speed analog command dead zone	0~5000	1mv	0		S
Pn-255	Reserved					
Pn-256	Torque analog command ratio	1~300	1%/V	30		T
Pn-257	Torque analog command filtering time	0~50	2ms	1		T
Pn-258	Torque analog command zero offset	-300~300	6.03mv	0		T
Pn-259	Torque analog command dead zone	0~5000	1mv	0		T
Pn-260	DO6 Plan the virtual connection function	Units/Tens: Enter the terminal number 0: indicates that the output terminal has no internal virtual connection 1 to 16: indicates DI1 to DI16 Hundreds: virtual output-input connection 0: The virtual output is directly connected to the input 1: The virtual output rising edge is connected to the input 2: The virtual output falling edge is connected to the input 3: Virtual output rising edge or falling edge is connected with input 4: Virtual output is inversely connected to input Thousands: virtual output - input internal connection 0: invalid 1: Effective Note: When the virtual connection function is valid, the DI terminal of the virtual connection is automatically configured as the auxiliary input control	1	0000	H	P S T
Pn-261	DO7 Plan the virtual connection function			0000		
Pn-262	DO8 Plan the virtual connection function			0000		
Pn-263	DO9 Plan the virtual connection function			0000		
Pn-264	DO10 Plan the virtual connection function			0000		
Pn-265	DO11 Plan the virtual connection function			0000		
Pn-266	DO12 Plan the virtual connection function			0000		
Pn-267	DO13 Plan the virtual connection function			0000		
Pn-268	DO14 Plan the virtual connection function			0000		
Pn-269	DO15 Plan the virtual connection function			0000		
Pn-270	DO16 Plan the virtual connection function			0000		

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-271	DO terminal logic and 1 configuration	Units/Tens: indicates the number of the first output terminal 0: The serial number is not defined 1 to 16: correspond to DO1 to DO16 Hundreds/thousands: number of the second output terminal 0 : The serial number is not defined 1 to 16: correspond to DO1 to DO16 Note: 1. When the DO function is configured as AND1, AND2, AND3, AND4, OR1, OR2, OR3, OR4, XOR1, XOR, XOR3, XOR4, XNOR1, XNOR2, XNOR3, the corresponding logical Pn parameter needs to be configured 2. Logical calculation is performed between the first output terminal and the second output terminal. You need to configure the serial numbers of the two DO terminals before using them	1	0000	H	P S T
Pn-272	DO terminal logic and 2 configuration			0000		
Pn-273	DO terminal logic and 3 configuration			0000		
Pn-274	DO terminal logic and 4 configuration			0000		
Pn-275	DO terminal logic or 1 configuration			0000		
Pn-276	DO terminal logic or 2 configuration			0000		
Pn-277	DODO terminal logic or 3 configuration			0000		
Pn-278	DO terminal logic or 4 configuration			0000		
Pn-279	DO terminal logic XOR 1 configuration			0000		
Pn-280	DO terminal logic XOR 2 configuration			0000		
Pn-281	DO terminal logic XOR 3 configuration			0000		
Pn-282	DO terminal logic XOR 4 configuration			0000		
Pn-283	DO terminal logic is configured with or 1			0000		
Pn-284	DO terminal logic is configured with or 2			0000		
Pn-285	DO terminal logic is configured with or 3			0000		
Pn-286	DO terminal logic is configured with or 4			0000		
Pn-287	DO terminal logic is not configured in 1	Units/Tens: indicates the number of the output terminal 0: The serial number is not defined 1 to 16: correspond to DO1 to DO16 Hundreds bit/thousands bit: reserved Note: 1. When the DO function is configured as NOT1, NOT2, NOT3, or NOT4, you need to configure the corresponding logical Pn parameter 2. Perform logical non-calculation on the output terminal. Before using the output terminal, configure the DO terminal number		0000		
Pn-288	DO terminal logic is not configured in 2			0000		
Pn-289	DO terminal logic is not configured in 3			0000		
Pn-290	DO terminal logic is not configured in 4			0000		

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-291	Timer 1 Configuration	Units/Tens: Set value of the timer 0: indicates unlimited duration 1 to 99: Set duration Hundreds: Timer time base 0:1ms 1:10 ms 2:100 ms 3 : 1000ms Thousands: timer startup mode 0: disable 1: Manually start 2: external high level trigger 3: external rising edge trigger Note: Before use, set the DO terminal to TIM1, TIM2, TIM3, or TIM4	1	0000	H/■	P S T
Pn-292	Timer 2 Configuration			0000		
Pn-293	Timer 3 Configuration			0000		
Pn-294	Timer 4 Configuration			0000		
Pn-295	16 bit counter configuration	Units/Tens : The number of the input terminal or output terminal to be counted Value range: 0 ~ 16 Hundreds : Indicates the counting mode of the counter 0: indicates the number of input terminal positions 1: Reset the input terminal count 2: Input terminal rising edge count 3: Input terminal falling edge count 4: indicates the number of output terminals 5: output terminal reset count 6: count the rising edge of the output terminal 7: counts the falling edge of the output terminal Thousands : Starting mode of the counter 0: disable 1: Manually start incremental counting 2: Manually start the decrement count 3: external high level triggers incremental counting 4: external high level trigger decrement count 5: external rising edge triggers increasing count 6: external rising edge triggers decrement count Note: Before use, configure the DO terminal as CNT16 or CNT32	1	0000	H/■	P S T
Pn-296	32 bit counter configuration			0000		
Pn-297	16 bit counter set value	-32768~32767	1	0		P S T
Pn-298 Pn-299	32 bit counter set value	-2147483648~2147483647	1	0		P S T

7.2.4.Position control parameter groups

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-300	Position command source and pulse pattern	Units:Position command source 0 : Pulse command 1 : Internal digital quantity giving (set via Pn-314) 2 : Internal multi-steps position command Tens:Pulse command form 0: pulse + direction 1: Forward/reverse pulse 2: Orthogonal pulse Hundreds:Pulse command input terminal selection 0 : Low-speed input terminal 1 : High-speed input terminal Thousands:Minimum pulse width of external pulse Low-speed terminal 0 : 600ns(830K) 1 : 2.4us(208K) 2 : 4.8us(104K) 3 : 9.6us(52K) 4 : No filter High-speed terminal 0 : 150ns(3.33M) 1 : 600ns(0.83M) 2 : 1.2us(416K) 3 : 2.4us(208K) 4 : No filter		0000	H/■/●	P
Pn-301	Reserved					
Pn-302	Position command moving average filter constant	0~128.0	0.1ms	0.0	■	P
Pn-303	Position command first-order filtering time constant	0~3000.0	0.1ms	0.0	■	P
Pn-304	Electronic gear-numerator 1	1~536870911(17 factory value is 131072, 23 factory value is 8388608)	1PP	131072		P
Pn-306	Electronic gear-numerator 2	1~536870911(17 factory value is 131072, 23 factory value is 8388608)	1PP	131072		P
Pn-308	Electronic gear-numerator 3	1~536870911(17 factory value is 131072, 23 factory value is 8388608)	1PP	131072		P
Pn-310	Electronic gear-numerator4	1~536870911(17 factory value is 131072, 23 factory value is 8388608)	1PP	131072		P
Pn-312	Electronic gear-denominator	1~2147483647	1PP	10000		P
Pn-314	Internal position command setting value	The position command is valid when the source is 1	1PP	10000		P
Pn-316	Internal position command runs at maximum speed	The position command is valid when the source is 1	0.1RPM	1000		P

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-317	Internal position command acceleration and deceleration time	The position command is valid when the source is 1	1ms	100		P
Pn-318	Internal Position Command Form	Units: command type 0 : relative displacement 1 : absolute displacement Tens:Continuous position trigger 0 : After PTRG is triggered, the position command is received. 1 : After the PTRG is triggered, the position command is not received before the end of the previous position. Hundreds:Reserved Thousand:Reserved		0000	H/●	P
Pn-319	Reserved					
Pn-320	Electronic gear ratio switching conditions	0 : Real-time switching 1 : Position command equal to 0 and switched after 2ms		0		P
Pn-321	Position deviation counter clearing settings	0 : ON state continuous clearance 1 : Clear it once when the OFF becomes an ON 2 : OFF status continuously cleared 3 : Clear it once when the ON becomes an OFF Note : This function requires the DI terminal to be configured with position deviation clearing (function number 18).		0		P
Pn-322	Encoder frequency division output pulse number	Single-phase output pulse number of motor rotating one circle 16~16384	1PP	2500	●	P
Pn-323						
Pn-324	Encoder frequency output configuration	Units:Output pulse phase 0 : A pulse phase ahead of B pulse 90° 1 : B pulse phase ahead of A pulse 90° Tens:Z pulse widening n corresponds to 10 widening levels Hundreds:Output source selection 0 : Motor encoder 1 : Full closed loop encoder 2 : pulse command Thousand:Reserved		0000	H/●	P
Pn-325	Reserved					
Pn-326	Complete in position output condition settings	0 : Absolute position deviation less than Pn-327 1 : When the absolute value of position deviation is less than Pn-327 and the filtered position command is 0 2 : When the absolute value of position deviation is less than Pn-327 and the position command input is 0		0		P
Pn-327	Complete in position confirmation range	17 bit maximum 131072, the factory value of 1000	1PP	1000		P
Pn-328		23 maximum value of 8388608, the factory value of 64000				
Pn-329	Nearing position to confirmation range	17 bit maximum 131072, the factory value of 1000	1PP	1000		P
Pn-330		23 maximum value of 8388608, the factory value of 64000				

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-331	Time to complete filtering	0~100	2ms	10		P
Pn-332	Location approach filter detection time	0~100	2ms	10		P
Pn-333	Reserved					
Pn-334	Reserved					
Pn-335	Reserved					
Pn-336	Get home control	Units:Reference point selection 0 : Forward looking for POT signal as reference point 1 : Reverse find the NOT signal as the reference point 2 : Forward looking for REF rising edge as reference point 3 : Reverse finds the REF rising edge as the reference point 4 : Forward looking for REF falling edge as reference point 5 : Reverse looking for REF falling edge as a reference point 6 : Forward looking for Z pulse as reference point 7 : Reversal find Z pulse as reference point 8 : The current position is the reference point Tens:regression model 0 : Find the Z-pulse backward as the origin 1 : Forward the Z pulse as the origin 2 : Direct to the reference point as the origin 3 : Look back for the reference point as the origin 4 : Forward the reference point as the origin 5 : The current position is the origin Hundreds:The limit treatment method 0 : warning 1 : Automatic reverse Thousand: starting mode 0 : Close the homing function 1 : First enable automatic homing after servo power on 2 : homing triggered by GHOM function of DI terminals(function number 30)		0122	H/■	P
Pn-337	First high-speed get home speed	0.1~2000.0	0.1RPM	100.0		P
Pn-338	Second low speed get home speed	0.1~500.0	0.1RPM	20.0		P
Pn-339	Get home acceleration and deceleration time	Acceleration and deceleration of first and second speeds during origin search	1ms	100		P
Pn-340	Home position offset	-2147483648~2147483647	1PP	0		P
Pn-341						
Pn-342	Deviation clearing selection afterget home	0 : clear 1 : no clear		0		P
Pn-343	Get home failure alarm time	0~65535(0 indicates a shield failure alarm)	1s	0		P

7.2.5.Speed control parameter groups

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-400	Speed command source	0 : External analog input (V_REF) 1 : Internal number given (set by Pn-401) 2 : Internal multi-steps speed commands		0	■	S
Pn-401	Internal speed command setting value	Effective when the speed instruction source is 1	0.1RPM	100.0		S
Pn-402	Speed command acceleration time	Valid when the speed command source is 0 or 1	1ms	100		S
Pn-403	Speed command deceleration time	Valid when the speed command source is 0 or 1	1ms	100		S
Pn-404	Speed command S-type smoothing time	0~1000	1ms	50		S
Pn-405	Reserved					
Pn-406	Reserved					
Pn-407	Reserved					
Pn-408	Speed feedback mean filtering setting	0 : No mean filtering 1 : 2-point moving average filtering 2 : 4-point moving average filtering 3 : 8-point moving average filtering 4 : 16-point moving average filtering		0		P S T
Pn-409	Speed feedback first-order filtering time	0.0~50.0	0.1ms	0.0		P S T
Pn-410	Speed command first-order filtering time	0.0~500.0	0.1ms	0.0		S
Pn-411	Maximum speed limit	0~Max.speed	0.1RPM			P S T
Pn-412 ~ Pn-415	Reserved					
Pn-416	Selection of zero speed clamping function	0 : When the zero-speed clamping signal (ZSP) input is ON and the speed command is below Pn-417, the forced speed command is 0 and the motor stops running. 1 : When the zero-speed clamping signal (ZSP) input is ON, the forced speed command is 0, and the motor stops running.		0		S
Pn-417	Zero-speed clamping threshold	0.0~200.0	0.1RPM	10.0		S
Pn-418	Speed consistency signal output threshold	0.0~100.0	0.1RPM	10.0		S
Pn-419	Reserved					
Pn-420	Zero-speed signal detection threshold value	0.0~200.0	0.1RPM	10.0		P S T
Pn-421	Speed reaches signal detection threshold	0.0~6000.0	0.1RPM	3000.0		S
Pn-422	Zero velocity signal detection time	0~100	2ms	5		P S T
Pn-423	Speed to signal detection time	0~100	2ms	5		P S T

7.2.6.Torque control parameter groups

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-500	Torque command source	0 : External analog input(T_REF) 1 : Internal number given (set by Pn-501) 2 : Internal multi-steps torque		0	■	T
Pn-501	Internal torque command setting value	The set value when the torque command source is 0	1%	100		T
Pn-502	Torque command first-order filtering time constant	0~30.00	0.01ms	0.00		T
Pn-503	Torque feedback first-order filtering time constant	0~10.00	0.01ms	0.00		T
Pn-504	Selection of torque limit source	0 : Internal positive and negative torque limit 1 : External positive and negative torque limit (selected by the TCCW / TCW terminal) 2 : Analog input T_REF torque limit 3 : Min[Analog input T_REF, external positive and negative torque limit (TCCW / TCW terminal selection)] 4 : Min[Analog input T_REF, internal positive and negative torque limit]		0	■	P S T
Pn-505	Internal positive torque limit	0~300	1%	300		P S T
Pn-506	Internal negative torque limit	-300~0	1%	-300		P S T
Pn-507	External positive torque limit	0~300	1%	300		P S T
Pn-508	External negative torque limit	-300~0	1%	-300		P S T
Pn-509	Source of torque mode speed limit	0 : Internal positive and negative speed limit (restricted by Pn-511, Pn-512) 1 : Analog input V_REF for speed limit 2 : Select the first or second speed limit input via the DI (ICMD4) function The DI is invalid, and Pn-511 is used as the positive-reversal speed limit value The DI is valid, and Pn-512 serves as the positive-reversal velocity limit value		0	■	T
Pn-510	Torque mode speed limit window time	0~30		10		T
Pn-511	Torque control forward speed limit value	0.0~6000.0	0.1RPM	3000.0		T
Pn-512	Reverse speed limit value of torque control	0.0~6000.0	0.1RPM	3000.0		T
Pn-513	Torque reaches the reference value	0~300	1%	0		P S T
Pn-514	Torque reaching effective value	0~300	1%	20		P S T

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-515	Torque reaches invalid value	0~300	1%	10		P S T
Pn-516	Torque limit during emergency stop (EMG)	0~300	1%	300		P S T
Pn-517	Torque limitation when main circuit voltage drops	0~100	1%	50		P S T
Pn-518	Torque limit release time when main circuit voltage drops	0~1000	1ms	100		P S T
Pn-519	Torque arrival detection time	0~100	2ms	5		P S T
Pn-520	Torque limit check out time	0~100	2ms	5		P S

7.2.7.Gain adjustment parameter groups

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-600	Self-adjusting mode setting	0 : Manual gain mode 1 : Automatic gain mode 1, stability-focused mode 2 : Automatic gain mode 2, positioning-focused mode		0		P S
Pn-601	Rigidity level setting	0~31: The higher the rigidity level, the faster the response		13		P S T
Pn-602	load inertia ratio	1.0~120.0		3.0		P S T
Pn-603	Offline inertia identification enabling	0 : Invalid offline inertia identification 1 : Offline inertia identification enabling		0		P S
Pn-604	Offline inertia identification mode	0 : Positive and inverse triangular wave mode 1 : JOG mode		0		P S
Pn-605	Offline inertia identification maximum speed	Maximum speed instruction of the inertia recognition process (Positive and reverse triangular wave mode)	0.1RPM	500.0		P S
Pn-606	Offline inertia identification acceleration and deceleration time	Time to accelerate inertia identification process to maximum speed (Positive and negative triangular wave mode)	ms	125		P S
Pn-607	Offline inertia identifies the motor stroke	The number of turns of motor rotation in the process of single inertia identification (Positive and negative triangular wave mode)	0.1R	0.5		P S
Pn-608	Offline inertia identification waiting time	The interval time of two consecutive inertia identification processes (Positive and negative triangular wave mode)	ms	800		P S
Pn-609	Reserved					
Pn-610	Position loop gain	The greater the faster the response, too large will cause vibration. Coordinated adjustment of position mode and position loop gain.	0.1	100.0		P
Pn-611	Speed loop gain	The greater the faster the response, too large will cause vibration . Coordinated adjustment of position mode and position loop gain.	0.1Hz	87.0		P S T

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-612	The speed loop integral time constant	The smaller the integral, the stronger the speed tracking ability, with the speed loop gain and position loop gain.	0.1ms	15.0		P S T
Pn-613	The speed loop differential time constant					
Pn-620	2nd position loop gain	The greater the faster the response, too large will cause vibration . Coordinated adjustment of position mode and position loop gain.	0.1	68.0		P
Pn-621	2nd speed loop gain	The greater the faster the response, too large will cause vibration. Coordinated adjustment of position mode and position loop gain.	0.1Hz	45.0		P S T
Pn-622	2nd speed loop integral time constant	The smaller the integral, the stronger the speed tracking ability, with the speed loop gain and position loop gain.	0.1ms	15.0		P S T
Pn-623 ~ Pn-635	Reserved					
Pn-636	Speed feedforward selection	0 : Internal speed feedforward 1 : Analog channel AI1 as a speed feed forward source		0		P
Pn-637	Speed feedforward gain	0.0~100.0	0.1%	0.0		P
Pn-638	Speed feedforward smoothing filter constant	0.10~100.00	0.01ms	0.5		P
Pn-639	Recurrent feedforward selection	0 : Internal torque feedforward 1 : Analog channel AI2 as torque feed forward source		0		P S
Pn-640	Torque feed forward gain	0.0~300.0%	0.1%	0.0		P S
Pn-641	Torque feedforward smoothing filter constant	0.1~100.00	0.01ms	0.20		P S
Pn-642	First torque filtering time constant	0.00~20.00	0.01ms	0.00		P S T
Pn-643	The second torque filtering frequency	50~5000	Hz	5000		P S T
Pn-644	Second torque filter Q value	50.00~100.00		50.00		P S T
Pn-645	Decoupling feedforward gain	0.0~100.0%	0.1%	0.0		P S T
Pn-646 ~ Pn-650	Reserved					

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-651	Gain switching conditions	Units:Gain switching conditions 0 : The first gain was fixed, and the DI terminals were switched by P / PI The GAIN terminal is OFF-PI control The GAIN terminal is under the ON-P control 1 : DI terminal gain switch The GAIN terminal OFF-first gain The GAIN terminal ON-2nd gain 2 : Torque command 3 : Speed command 4 : Speed acceleration command 5 : Speed command high and low speed threshold 6 : Position deviation 7 : Position command 8 : Positioning completed 9 : Motor feedback speed A : Position command + motor feedback speed is available Tens:Reserved Hundreds:Reserved Thousand:Reserved		0000		P S T
Pn-652 Pn-653	Gain switching threshold	Level of gain switching		50		P S T
Pn-654	Gain switching hysteresis loop			30		P S T
Pn-655	Gain switching time	Smooth time, only effective for position gain	0.1ms	3.0		P
Pn-656	Gain switching delay time	Delay time for switching from second gain to first gain	0.1ms	5.0		P S T
Pn-657 ~ Pn-661	Reserved					
Pn-662	Adaptive trap setting	0 : Manual mode, trap parameters for groups 3 and 4 are no longer automatically updated 1 : Group 3 trap parameter adaptive setting 2 : Adaptive setting of trap filter parameters for groups 3 and 4 3 : Test the resonance frequency only (monitoring parameter group) 4 :Restore group 3 and 4 trap parameters to factory values		0		P S
Pn-663	First trap frequency	50~5000	1Hz	5000		P S
Pn-664	The 1st trap width grade	0~20		2		P S
Pn-665	1st trap depth level	0~99		10		P S
Pn-666	Second trap device frequency	50~5000	1Hz	5000		P S
Pn-667	2nd trap width grade	0~20		2		P S
Pn-668	Second trap filter depth level	0~99		10		P S

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-669	Third trap frequency	50~5000	1Hz	5000		P S
Pn-670	3rd trap width grade	0~20		2		P S
Pn-671	third trap depth level	0~99		10		P S
Pn-672	4th trap frequency	50~5000	1Hz	5000		P S
Pn-673	4th trap filter width grade	0~20		2		P S
Pn-674	4th trap filter depth level	0~99		10		P S
Pn-675	End-jitter suppression setting	0 : Manually set the end jitter suppression parameter 1 : Automatic setting of first end jitter suppression parameters 2 : Automatic setting of second end jitter suppression parameters		0		P
Pn-676	First end jitter frequency	1.0~200.0	0.1HZ	1.0		P
Pn-677	First end jitter suppression level	0~10		0		P
Pn-678	The first end jitter position deviation judgment threshold	0~500	1%	100		P
Pn-679	Reserved					
Pn-680	Reserved					
Pn-681	Second end jitter frequency	1.0~200.0	0.1HZ	1.0		P
Pn-682	Second end jitter suppression level	0~10		0		P
Pn-683	Second end jitter position deviation judgment threshold	0~500	1%	100		P
Pn-686	Online inertia identification mode	0 : Online inertia recognition is turned off 1 : Online inertia recognition is on, basically unchanged 2 : Online inertia recognition is turned on, slowly changing 3 : Online inertia recognition is turned on and changes quickly		0		P S
Pn-687	Load disturbance compensation gain	0.0~100.0	1%	0.0		P S T
Pn-688	Load disturbance observer filter time	1.0~100.0	0.1ms	5.0		P S T
Pn-691	Vertical axis torque compensation	0~100.0	0.1%	0		P S T
Pn-692	Forward friction torque compensation	0~100.0	0.1%	0		P S T
Pn-693	Negative friction torque compensation	0~100.0	0.1%	0		P S T

7.2.8.Full closed-loop parameter groups

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-700	Full closed-loop function control switch	Units:Full closed-loop function switch 0 : Full closed loop function off, position feedback from motor encoder 1 : Full closed-loop function on, position feedback from external encoder Tens:External encoder rotation direction 0 : Motor CCW rotation, external full closed-loop encoder forward count 1 : Motor CCW rotation, external all-loop encoder reverse count Hundreds:Reserved Thousand:Reserved		0000	H/■/●	P
Pn-701	Reserved					
Pn-702	Full closed-loop encoder resolution	Feedback pulse number of motor rotating one circle full closed loop encoder. (After A / B pulse double frequency)	1PP	10000		P
Pn-703						
Pn-704	Excessive threshold of position deviation in full closed loop control	0 ~ 1073741824	1PP	50000		P
Pn-705		External encoder unit, when set to 0, the servo drive does not perform full closed loop position deviation is too large fault detection, while always clearing the full closed loop position deviation.				
Pn-706	Full closed-loop position deviation clearance method.	0 : Full closed loop position deviation is not cleared. 1~100 : Servo alarm or servo motor rotation circles small Pn-706, the full closed loop position deviation is greater than Pn-704, clear the full closed loop position deviation ; servo alarm or full closed-loop position deviation is less than Pn-704, when the number of motor rotation is equal to Pn-706, clear the full closed-loop position deviation, motor turns and position deviation from 0 to recount.		0		P
Pn-707 ~ Pn-718	Reserved					
Pn-719	Full closed-loop encoder disconnection detection function	0 : Effective, disconnection detection function on 1 : Invalid, disconnection detection function turned off		0		P

7.2.9.Fault and protection parameter groups

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-800	reset	Units:Fault reset 0 : no-operation 1 : Current fault reset Tens:Soft restart 0 : no-operation 1 : Drive reset Note : This function is equivalent to the drive power-down and power-on, mainly used in the system can not be powered off, but the drive needs to restart the occasion, please ensure that the system security operation. Hundreds:CPLD reset 0: no operation is performed 1: Reset the CPLD Thousand:Reserved		0000	H/■	P S T
Pn-801	Main power supply input is lack of phase protection selection	0 : Fault of enable phase protection (Only three-phase 380V power supply) 1 : Shield phase protection fault (common bus mode, the need to set the parameter to 1)		0		P S T
Pn-802	Motor overload protection gain	50~300	1%	100		P S T
Pn-803	Motor overload shielding enabling	0 : Enabling motor overload detection 1 : Overload alarm and fault detection of shielded motor		0		P S T
Pn-804	Motor stall over temperature protection enabling	0 :Enabling motor stall over temperature protection detection 1 : Blocking over-temperature protection detection of shielded motor		0		P S T
Pn-805	Over temperature protection time window of motor stall	10~65535		200		P S T
Pn-806	Runaway protection function enable	0 : The shielding is set to 0 when the motor is in vertical or towed application, disable this function;. 1 : Enable, turn on the runaway protection function.		0		P S T
Pn-807	Excessive threshold of position deviation	The 17-bit maximum value is 13107200, and the 23-bit maximum value is 838860800.		1310720		P S T
Pn-809	Overspeed fault threshold	Multiple of maximum speed of motor	0.1	1.2		P S T
Pn-810	Encoder multi-steps turn overflow fault shielding selection	0 : not shielded 1 : shielded		0		P S T
Pn-811	Reserved					
Pn-812	Reserved					
Pn-813	Reserved					

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-814	Absolute position to zero	0 : no-operation 1 : Return the absolute position of the current encoder to zero Note: To ensure the safety of the system, first set the Pn-815 to 0x0814, allowing the absolute position to zero.		0		P
Pn-815	Absolute position zeroing enable	When set to 0x0814, the absolute position is allowed to zero, and other values have no effect.		0000	H	P
Pn-816	Reserved					
Pn-817	Soft limit setting	0 : Unable soft limit 1 : Enable soft limit immediately after power on 2 : Enable soft limit after origin return is complete		0		P S T
Pn-818	Positive soft limit	-2147483648~2147483647	1PP	2147483647		P S T
Pn-819						
Pn-820	Negative soft limit	-2147483648~2147483647	1PP	-2147483647		P S T
Pn-821						
Pn-850	Last fault record					
Pn-851	Historical failures record 1				R	P S T
Pn-852	Historical failures record 2				R	P S T
Pn-853	Historical failures record 3				R	P S T
Pn-854	Historical failures record 4				R	P S T
Pn-855	Historical failures record 5				R	P S T
Pn-856	Historical failures record 6				R	P S T
Pn-857	Historical failures record 7				R	P S T
Pn-858	Historical failures record 8				R	P S T
Pn-859	Historical failures record 9				R	P S T
Pn-860	Rotational speed at the last fault time		1RPM		R	P S T
Pn-861	Current at the last time of failure		0.01A		R	P S T
Pn-862	Output voltage at the last failure time		1V		R	P S T
Pn-863	DC voltage at the last time of failure		0.1V		R	P S T
Pn-864	Radiator / IGBT temperature at the last fault time		0.1℃		R	P S T

7.2.10.Communication parameter groups

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-900	Servo axis address	Servo communication slave address, also suitable for RS485 communication, CAN communication		1	•	P S T
Pn-901	RS485 communication configuration	Units:baud rate 0 : 2400bps 1 : 4800bps 2 : 9600bps 3 : 19200bps 4 : 38400bps 5 : 57600bps 6 : 115200bps Tens: Modbus data format 0 : 1-8-1No check(Modbus,RTU) 1 : 1-8-1Even check(Modbus,RTU) 2 : 1-8-1Odd check(Modbus,RTU) 3 : 1-8-2No check(Modbus,RTU) 4 : 1-8-1No check(Modbus,ASCII) 5 : 1-8-1Even check(Modbus,ASCII) 6 : 1-8-1Odd check(Modbus,ASCII) 7 : 1-8-2No check(Modbus,ASCII) 8 : 1-7-1No check(Modbus,ASCII) 9 : 1-7-1Even check(Modbus,ASCII) A : 1-7-1Odd check(Modbus,ASCII) B : 1-7-2No check(Modbus,ASCII) Hundreds: Reserved Thousands: Reserved		0003	H	P S T
Pn-902	Local response delay	0~5000	1ms	0		P S T
Pn-903	Communication failure judging time	0~2000	1ms	1000		P S T
Pn-904	Communication error handling	0 : Warning and continue the execution 1 : Warning and slow down shutdown		0		P S T
Pn-905	Reserved					
Pn-906	Reserved					
Pn-907	Reserved					
Pn-908	Reserved					
Pn-909	Reserved					
Pn-910	CANopen data format	Units:baud rate 0 : 20Kbps 1 : 50Kbps 2 : 100Kbps 3 : 125Kbps 4 : 250Kbps 5 : 500Kbps 6 : 800Kbps 7 : 1Mbps Tens:Reserved Hundreds:Reserved Thousands: Reserved		0005	H/•	P S T

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-911	CANopen synchronization setting	Units:Synchronization time correction Range : 0 ~ F, Unit : us Tens:Synchronous determination of dead time Range : 0 ~ F, Unit : us Hundreds: Target value preposition Range : 0 ~ F, Unit : us Thousands:error range Range : 0 ~ F, Unit : us		0000	H	P S T
Pn-912	CANopen protocol setting	Units:Action when communication error 0 : Motor continues to run when communication error occurs 1 : servo off occurred during the communication error Tens:PDO abnormal alarm is automatically cleared 0 : Can not automatically clear 1 : Automatically clear PDO communication when an alarm occurs Hundreds:Reserved Thousands:Reserved		0000	H	P S T
Pn-913	CANopen parameter save setting	PDO parameter save setting 0 : PDO parameters are not saved to EEPROM when they are written 1 : When PDO parameters are written, they are saved to EEPROM		0		P S T
Pn-914	CANopen quick stop settings	Units: Trigger the servo on process and quick stop support Tens: Whether to trigger quick stop when the drive alarms Hundreds: Parameter loading settings Thousands:Reserved		0000	H	P S T
Pn-915	Reserved					
Pn-916	Reserved					
Pn-917	Reserved					
Pn-918	USB monitoring settings	Units:USB low-speed monitoring sampling interval (unit ms)0~5 Tens:USB monitor mode settings 0 : Close the surveillance 1 : Low speed monitoring 2 : High-speed monitoring, with a sampling frequency of 8k 3 : High-speed monitoring, with a sampling frequency of 16k Hundreds:Reserved Thousands:Reserved		0000	H	P S T
Pn-919	Reserved					
Pn-920	Reserved					

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-921	Mapping parameter1	Mapping parameters access address		0		P S T
Pn-922	Mapping parameter2					
Pn-923	Mapping parameter3					
Pn-924	Mapping parameter4					
Pn-925	Mapping parameter5					
Pn-926	Mapping parameter6					
Pn-927	Mapping parameter7					
Pn-928	Mapping parameter8					
Pn-929	Mapping parameter9					
Pn-930	Mapping parameter10					
Pn-931	Mapping the parameter target setting1	Map the parameter target address		0000	H/●	P S T
Pn-932	Mapping the parameter target setting2					
Pn-933	Mapping the parameter target setting3					
Pn-934	Mapping the parameter target setting4					
Pn-935	Mapping the parameter target setting5					
Pn-936	Mapping the parameter target setting6					
Pn-937	Mapping the parameter target setting7					
Pn-938	Mapping the parameter target setting8					
Pn-939	Mapping the parameter target setting9					
Pn-940	Mapping the parameter target setting10					

7.2.11.Multi-segment position parameter groups

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-A00	Multiple-segment displacement control mode	0 : DI switch runs 1 : Loop execution 2 : Stop after a single run		0	■	P
Pn-A01	Number of multiple-segment displacement end-point steps	1~16		16	■	P
Pn-A02	Multiple-segment displacement instruction type	0 : Relative displacement command 1 : Absolute displacement command		0	■	P
Pn-A03	Reserved					
Pn-A04	Paragraph 1 displacement	-2147483648~2147483647	1PP	10000		P
Pn-A05						
Pn-A06	Paragraph 1 displacement running maximum speed	0.1~6000.0	0.1RPM	200.0		P
Pn-A07	Paragraph 1 displacement, acceleration and deceleration time	0~60000	1ms	200		P
Pn-A08	Paragraph 1 displacement completion waiting time	0~3600000	1ms	1000		P
Pn-A09						
Pn-A10	Paragraph 2 displacement	-2147483648~2147483647	1PP	10000		P
Pn-A11						
Pn-A12	Paragraph 2 displacement running maximum speed	0.1~6000.0	0.1RPM	200.0		P
Pn-A13	Paragraph 2 displacement, acceleration and deceleration time	0~60000	1ms	200		P
Pn-A14	Paragraph 2 displacement completion waiting time	0~3600000	1ms	1000		P
Pn-A15						
Pn-A16	Paragraph 3 displacement	-2147483648~2147483647	1PP	10000		P
Pn-A17						
Pn-A18	Paragraph 3 displacement running maximum speed	0.1~6000.0	0.1RPM	200.0		P
Pn-A19	Paragraph 3 displacement, acceleration and deceleration time	0~60000	1ms	200		P
Pn-A20	Paragraph 3 displacement completion waiting time	0~3600000	1ms	1000		P
Pn-A21						
Pn-A22	Paragraph 4 displacement	-2147483648~2147483647	1PP	10000		P
Pn-A23						
Pn-A24	Paragraph 4 displacement running maximum speed	0.1~6000.0	0.1RPM	200.0		P
Pn-A25	Paragraph 4 displacement, acceleration and deceleration time	0~60000	1ms	200		P
Pn-A26	Paragraph 4 displacement completion waiting time	0~3600000	1ms	1000		P
Pn-A27						
Pn-A28	Paragraph 5 displacement	-2147483648~2147483647	1PP	10000		P
Pn-A29						
Pn-A30	Paragraph 5 displacement running maximum speed	0.1~6000.0	0.1RPM	200.0		P
Pn-A31	Paragraph 5 displacement, acceleration and deceleration time	0~60000	1ms	200		P
Pn-A32	Paragraph 5 displacement completion waiting time	0~3600000	1ms	1000		P
Pn-A33						

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-A34	Paragraph 6 displacement	-2147483648~2147483647	1PP	10000		P
Pn-A35						
Pn-A36	Paragraph 6 displacement running maximum speed	0.1~6000.0	0.1RPM	200.0		P
Pn-A37	Paragraph 6 displacement, acceleration and deceleration time	0~60000	1ms	200		P
Pn-A38	Paragraph 6 displacement completion waiting time	0~3600000	1ms	1000		P
Pn-A39						
Pn-A40	Paragraph 7 displacement	-2147483648~2147483647	1PP	10000		P
Pn-A41						
Pn-A42	Paragraph 7 displacement running maximum speed	0.1~6000.0	0.1RPM	200.0		P
Pn-A43	Paragraph 7 displacement, acceleration and deceleration time	0~60000	1ms	200		P
Pn-A44	Paragraph 7 displacement completion waiting time	0~3600000	1ms	1000		P
Pn-A45						
Pn-A46	Paragraph 8 displacement	-2147483648~2147483647	1PP	10000		P
Pn-A47						
Pn-A48	Paragraph 8 displacement running maximum speed	0.1~6000.0	0.1RPM	200.0		P
Pn-A49	Paragraph 8 displacement, acceleration and deceleration time	0~60000	1ms	200		P
Pn-A50	Paragraph 8 displacement completion waiting time	0~3600000	1ms	1000		P
Pn-A51						
Pn-A52	Paragraph 9 displacement	-2147483648~2147483647	1PP	10000		P
Pn-A53						
Pn-A54	Paragraph 9 displacement running maximum speed	0.1~6000.0	0.1RPM	200.0		P
Pn-A55	Paragraph 9 displacement, acceleration and deceleration time	0~60000	1ms	200		P
Pn-A56	Paragraph 9 displacement completion waiting time	0~3600000	1ms	1000		P
Pn-A57						
Pn-A58	Paragraph 10 displacement	-2147483648~2147483647	1PP	10000		P
Pn-A59						
Pn-A60	Paragraph 10 displacement running maximum speed	0.1~6000.0	0.1RPM	200.0		P
Pn-A61	Paragraph 10 displacement, acceleration and deceleration time	0~60000	1ms	200		P
Pn-A62	Paragraph 10 displacement completion waiting time	0~3600000	1ms	1000		P
Pn-A63						
Pn-A64	Paragraph 11 displacement	-2147483648~2147483647	1PP	10000		P
Pn-A65						
Pn-A66	Paragraph 11 displacement running maximum speed	0.1~6000.0	0.1RPM	200.0		P
Pn-A67	Paragraph 11 displacement, acceleration and deceleration time	0~60000	1ms	200		P
Pn-A68	Paragraph 11 displacement completion waiting time	0~3600000	1ms	1000		P
Pn-A69						
Pn-A70	Paragraph 12 displacement	-2147483648~2147483647	1PP	10000		P
Pn-A71						
Pn-A72	Paragraph 12 displacement running maximum speed	0.1~6000.0	0.1RPM	200.0		P

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-A73	Paragraph 12 displacement, acceleration and deceleration time	0~60000	1ms	200		P
Pn-A74	Paragraph 12 displacement completion waiting time	0~3600000	1ms	1000		P
Pn-A75						
Pn-A76	Paragraph 13 displacement	-2147483648~2147483647	1PP	10000		P
Pn-A77						
Pn-678	Paragraph 13 displacement running maximum speed	0.1~6000.0	0.1RPM	200.0		P
Pn-A79	Paragraph 13 displacement, acceleration and deceleration time	0~60000	1ms	200		P
Pn-A80	Paragraph 13 displacement completion waiting time	0~3600000	1ms	1000		P
Pn-A81						
Pn-A82	Paragraph 14 displacement	-2147483648~2147483647	1PP	10000		P
Pn-A83						
Pn-A84	Paragraph 14 displacement running maximum speed	0.1~6000.0	0.1RPM	200.0		P
Pn-A85	Paragraph 14 displacement, acceleration and deceleration time	0~60000	1ms	200		P
Pn-A86	Paragraph 14 displacement completion waiting time	0~3600000	1ms	1000		P
Pn-A87						
Pn-A88	Paragraph 15 displacement	-2147483648~2147483647	1PP	10000		P
Pn-A89						
Pn-A90	Paragraph 15 displacement running maximum speed	0.1~6000.0	0.1RPM	200.0		P
Pn-A91	Paragraph 15 displacement, acceleration and deceleration time	0~60000	1ms	200		P
Pn-A92	Paragraph 15 displacement completion waiting time	0~3600000	1ms	1000		P
Pn-A93						
Pn-A94	Paragraph 16 displacement	-2147483648~2147483647	1PP	10000		P
Pn-A95						
Pn-A96	Paragraph 16 displacement running maximum speed	0.1~6000.0	0.1RPM	200.0		P
Pn-A97	Paragraph 16 displacement, acceleration and deceleration time	0~60000	1ms	200		P
Pn-A98	Paragraph 16 displacement completion waiting time	0~3600000	1ms	1000		P
Pn-A99						

7.2.12. Multi-steps speed and torque parameter groups

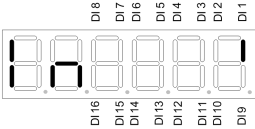
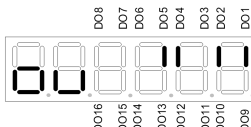
Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-B00	Multi-steps speed control mode	0 : DI switch runs 1 : Loop execution 2 : Stop after a single run		0	■	S
Pn-B01	Number of multiple-speed end steps	1~16		16	■	S
Pn-B02	Multi-steps speed running time unit	0 : Seconds 1 : Minutes effective only for loop execution and single run		0	■	S
Pn-B03	Reserved					
Pn-B04	Reserved					
Pn-B05	Reserved					
Pn-B06	Paragraph 1 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B07	Paragraph 1 speed acceleration time	0~65535	1ms	20		S
Pn-B08	Paragraph 1 speed deceleration time	0~65535	1ms	20		S
Pn-B09	Paragraph 1 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B10	Paragraph 2 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B11	Paragraph 2 speed acceleration time	0~65535	1ms	20		S
Pn-B12	Paragraph 2 speed deceleration time	0~65535	1ms	20		S
Pn-B13	Paragraph 2 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B14	Paragraph 3 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B15	Paragraph 3 speed acceleration time	0~65535	1ms	20		S
Pn-B16	Paragraph 3 speed deceleration time	0~65535	1ms	20		S
Pn-B17	Paragraph 3 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B18	Paragraph 4 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B19	Paragraph 4 speed acceleration time	0~65535	1ms	20		S
Pn-B20	Paragraph 4 speed deceleration time	0~65535	1ms	20		S
Pn-B21	Paragraph 4 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B22	Paragraph 5 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B23	Paragraph 5 speed acceleration time	0~65535	1ms	20		S
Pn-B24	Paragraph 5 speed deceleration time	0~65535	1ms	20		S
Pn-B25	Paragraph 5 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B26	Paragraph 6 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B27	Paragraph 6 speed acceleration time	0~65535	1ms	20		S
Pn-B28	Paragraph 6 speed deceleration time	0~65535	1ms	20		S
Pn-B29	Paragraph 6 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B30	Paragraph 7 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B31	Paragraph 7 speed acceleration time	0~65535	1ms	20		S
Pn-B32	Paragraph 7 speed deceleration time	0~65535	1ms	20		S
Pn-B33	Paragraph 7 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B34	Paragraph 8 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B35	Paragraph 8 speed acceleration time	0~65535	1ms	20		S
Pn-B36	Paragraph 8 speed deceleration time	0~65535	1ms	20		S
Pn-B37	Paragraph 8 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B38	Paragraph 9 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B39	Paragraph 9 speed acceleration time	0~65535	1ms	20		S
Pn-B40	Paragraph 9 speed deceleration time	0~65535	1ms	20		S
Pn-B41	Paragraph 9 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B42	Paragraph 10 speed command	-6000.0~6000.0	0.1RPM	0.0		S

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-B43	Paragraph 8 speed acceleration time	0~65535	1ms	20		S
Pn-B44	Paragraph 8 speed deceleration time	0~65535	1ms	20		S
Pn-B45	Paragraph 8 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B46	Paragraph 9 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B47						
Pn-B48	Paragraph 9 speed acceleration time	0~65535	1ms	20		S
Pn-B49	Paragraph 9 speed deceleration time	0~65535	1ms	20		S
Pn-B50	Paragraph 9 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B51	Paragraph 10 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B52						
Pn-B53	Paragraph 10 speed acceleration time	0~65535	1ms	20		S
Pn-B54	Paragraph 10 speed deceleration time	0~65535	1ms	20		S
Pn-B55	Paragraph 10 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B56	Paragraph 11 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B57						
Pn-B58	Paragraph 11 speed acceleration time	0~65535	1ms	20		S
Pn-B59	Paragraph 11 speed deceleration time	0~65535	1ms	20		S
Pn-B60	Paragraph 11 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B61	Paragraph 12 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B62						
Pn-B63	Paragraph 12 speed acceleration time	0~65535	1ms	20		S
Pn-B64	Paragraph 12 speed deceleration time	0~65535	1ms	20		S
Pn-B65	Paragraph 12 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B66	Paragraph 13 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B67						
Pn-B68	Paragraph 13 speed acceleration time	0~65535	1ms	20		S
Pn-B69	Paragraph 13 speed deceleration time	0~65535	1ms	20		S
Pn-B70	Paragraph 13 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B71	Paragraph 14 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B72						
Pn-B73	Paragraph 14 speed acceleration time	0~65535	1ms	20		S
Pn-B74	Paragraph 14 speed deceleration time	0~65535	1ms	20		S
Pn-B75	Paragraph 14 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B76	Paragraph 15 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B77						
Pn-B78	Paragraph 15 speed acceleration time	0~65535	1ms	20		S
Pn-B79	Paragraph 15 speed deceleration time	0~65535	1ms	20		S
Pn-B80	Paragraph 15 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B81	Paragraph 16 speed command	-6000.0~6000.0	0.1RPM	0.0		S
Pn-B82						
Pn-B83	Paragraph 16 speed acceleration time	0~65535	1ms	20		S
Pn-B84	Paragraph 16 speed deceleration time	0~65535	1ms	20		S
Pn-B85	Paragraph 16 command running time	0~6553.5	0.1S/Min	5.0		S
Pn-B86	Reserved					
Pn-B87	Reserved					

Function Code	Name	Description	Resolution	Default	Property	Mode
Pn-B88	The first torque value	-300~300	1%	100		T
Pn-B89	The first torque value establishment time	0~65535	1ms	50		T
Pn-B90	The second torque value	-300~300	1%	200		T
Pn-B91	The second torque value establishment time	0~65535	1ms	50		T
Pn-B92	The third torque value	-300~300	1%	50		T
Pn-B93	The third torque value establishment time	0~65535	1ms	50		T
Pn-B94	The fourth torque value	-300~300	1%	50		T
Pn-B95	The fourth torque value establishment time	0~65535	1ms	50		T
Pn-B96	Reserved					
Pn-B97	Reserved					
Pn-B98	Reserved					
Pn-B99	Reserved					

7.2.13. Monitoring parameter groups

Function Code	Name	Description	Resolution	Default	Property	Mode
dn-000	Motor speed	-6000~6000	1RPM		R	P S T
dn-001	Speed command	-6000~6000	1RPM		R	P S T
dn-002	Speed analog command	-12.00~12.00	0.01V		R	S
dn-003	Torque command	-300~300	1%		R	T
dn-004	Torque analog command	-12.00~12.00	0.01V		R	T
dn-005	Average torque	Effective torque within 60S	1%		R	P S T
dn-006	peak torque	Peak torque within 60S	1%		R	P S T
dn-007	External pulse input accumulation value (front of gear ratio)	-2147483648~2147483647	1PP		R	P
dn-009	External pulse input accumulation value (after gear ratio)	-2147483648~2147483647	1PP		R	P
dn-011	Encoder feedback pulse cumulative value	-2147483648~2147483647	1PP		R	P
dn-013	Number of position-deviation pulses	Command unit	1PP		R	P
dn-015	System absolute position (PUU value)	-2147483648~2147483647	1PUU		R	P
dn-017	Absolute value encoder circles		1R		R	P S T
dn-018	Absolute Value encoder location		1PP		R	P S T
dn-020	System absolute position (pulse value)	-2147483648~2147483647	1PP		R	P
dn-022	Effective value of motor current		0.01A		R	P S T
dn-023	bus voltage	Main Circuit BUS Voltage	0.1V		R	P S T

Function Code	Name	Description	Resolution	Default	Property	Mode
dn-024	Power module temperature	-50.0~120.0	0.1℃		R	P S T
dn-025	DI state		As shown in the left figure : ON ="Light up the vertical line" OFF ="Extinguished vertical line"		R	P S T
dn-026	DO state				R	P S T
dn-027	Warning code				R	P S T
dn-028	Fault code				R	P S T
dn-029	Servo running time	Total power-on time of servo from factory to present	0.01H		R	P S T
dn-030	Reserved					
dn-031	Reserved					
dn-032	Average load rate		1%		R	P S T
dn-033	Brake resistance load rate	Load rate in 10S cycles	1%		R	P S T
dn-034	Encoder battery fault state	0-No fault 1-Encoder battery failure	0.1V		R	P S T
dn-035	Cumulative number of frames of communication errors				R	P S T
dn-036	Resonance frequency 1	50~5000	1Hz		R	P S T
dn-037	Resonance frequency 2	50~5000	1Hz		R	P S T
dn-038	Resonance frequency 3	50~5000	1Hz		R	P S T
dn-039	Reserved					
dn-040	Full closed-loop encoder feedback pulse cumulative value	-1073741824~1073741824	1PP		R	P
dn-042	Full closed-loop position deviation counter	-1073741824~1073741824	1PP		R	P
dn-044	Full closed-loop encoder position	Single-ring position relative to Z-pulse of full closed-loop encoder	1PP		R	P
dn-046	Timer 1 counts values	Value range: 0 ~ 65535		0	▲/R	P S T
dn-047	Timer 2 counts values	Note:		0		
dn-048	Timer 3 counts values	1. When the timer count value is greater than or equal to the timer set value, the corresponding DO configuration terminal is set. Otherwise, the terminal is cleared		0		
dn-049	Timer 4 counts values	2. When the calculated value reaches the set value (the set value is greater than 0), the calculated value will continue to accumulate with the timer set time base until the maximum value.		0		

Function Code	Name	Description	Resolution	Default	Property	Mode
		3. When the set value of the timer is 0, the count value will be accumulated in a cycle based on the set time when the timer is enabled. 4. When the timer is off, the count value is cleared to zero and the corresponding DO terminal is reset.				
dn-050	Zero partial voltage of U phase		0.01V		R	P S T
dn-051	Zero partial voltage of V phase		0.01V		R	P S T
dn-052	The 16 bit counter counts the value	Note: 1. When the counter value is greater than or equal to the set value of the counter, the corresponding configuration DO terminal is set. Otherwise, zero is cleared 2. When the count value reaches the set value, the count value remains unchanged 3. When the counter is closed, the count value is cleared to zero and the corresponding configured DO terminal is reset		0	▲/R	P S T
dn-053 dn-054	The 32 bit counter counts the value			0		
dn-055	Reserved					
dn-056	Actual position loop gain		0.1		R	P
dn-057	Actual speed loop gain		0.1Hz		R	P S T
dn-058	Actual velocity loop integral time constant		0.1ms		R	P S T
dn-059	Load inertia		Kg·cm ²		R	P S T
dn-060	Load inertia ratio		0.1		R	P S T
dn-061	Reserved					
dn-062	Overload ratio		1%		R	P S T
dn-063 ~ dn-069	Reserved					
dn-070	Number of encoder communication count errors				R	P S T
dn-071	Encoder error state				R/H	P S T
dn-072	Encoder warning status				R/H	P S T
dn-073 ~ dn-079	Reserved					
dn-080	Actual position of motor				R	P S T
dn-081 dn-082	External pulses measure the frequency		0.1KHz		R	P
dn-085	Motor output voltage		V		R	P S T

Chapter 8 Communication Function

8.1. Overview of Modbus protocol

The Modbus protocol is a general-purpose protocol applied to industrial controllers. The protocol has become an industrial common standard due to its use convenience, and is widely used in the integration of main controllers and slave devices. Devices of the different brands can be connected to industrial network via this protocol.

Modbus defines three modes of transmission, they are the ASCII, RTU and TCP, for which CD100 servo drive supports ASCII and RTU.

8.2. Interface and transmission mode

CD100 uses RS485 as the Modbus physical interface, and a host controls one or more (up to 127) servo drives.

Terminal marking	Application	Function
RS+	Data receiving and sending terminal (+)	When connecting to PC/PLC with RS485 communication interface, please connect to (+) signal
RS-	Data receiving and sending terminal (-)	When connecting to PC/PLC with RS485 communication interface, please connect to (-) signal

It uses the transmission mode of asynchronous serial and half-duplex, by which, only one side (the host and the slave) can send data at the same time, while the other can only receive data.

8.3. Data structure

1) baud rate (set by Pn-901 Units bits)

2400bps、4800bps、9600bps、19200bps、38400bps、57600bps、115200bps、

2) Data transmission format (set by Pn-901 ten bits)

- 1-bit start bit -8-bit data bit -1-bit stop bit-no parity , (Modbus,RTU)
- 1-bit start bit -8-bit data bit -1-bit stop bit-even parity, (Modbus,RTU)
- 1-bit start bit -8-bit data bit -1-bit stop bit-odd parity, (Modbus,RTU)
- 1-bit start bit -8-bit data bit -2-bit stop bit-no parity, (Modbus,RTU)
- 1-bit start bit -8-bit data bit -1-bit stop bit-no parity, (Modbus,ASCII)
- 1-bit start bit -8-bit data bit -1-bit stop bit-even parity, (Modbus,ASCII)
- 1-bit start bit -8-bit data bit -1-bit stop bit-odd parity, (Modbus,ASCII)
- 1-bit start bit -8-bit data bit -2-bit stop bit-no parity, (Modbus,ASCII)
- 1-bit start bit -7-bit data bit -1-bit stop bit-no parity, (Modbus,ASCII)
- 1-bit start bit -7-bit data bit -1-bit stop bit-even parity, (Modbus,ASCII)
- 1-bit start bit -7-bit data bit -1-bit stop bit-odd parity, (Modbus,ASCII)
- 1-bit start bit -7-bit data bit -2-bit stop bit-no parity, (Modbus,ASCII)

3) Slave address (set by Pn-900)

1 ~ 127, support up to 127 servo drive network.

4) Communication rule

The start interval between data frames is greater than the 3.5-byte transmission period (standard), but the minimum interval is not less than 0.5ms.

8.4. Servo drive parameter configuration

The Modbus function codes supported by the servo drive are as follows :

0x03 : Read the value of the servo drive parameters ;

0x06 : Write the data to the servo drive parameters (16bit) ;

0x10 : Write multi-steps data to the servo drive parameters (16bit or 32bit);

Parameter access address :

Parameter identifier	RAM access address	EEPROM access address
Pn-000 ~ Pn-099	0x2000 ~ 0x2063	0xE000 ~ 0xE063
Pn-100 ~ Pn-199	0x2064 ~ 0x20C7	0xE064 ~ 0xE0C7
Pn-200 ~ Pn-299	0x20C8 ~ 0x212B	0xE0C8 ~ 0xE12B
Pn-300 ~ Pn-399	0x212C ~ 0x218F	0xE12C ~ 0xE18F
Pn-400 ~ Pn-499	0x2190 ~ 0x21F3	0xE190 ~ 0xE1F3
Pn-500 ~ Pn-599	0x21F4 ~ 0x2257	0xE1F4 ~ 0xE257
Pn-600 ~ Pn-699	0x2258 ~ 0x22BB	0xE258 ~ 0xE2BB
Pn-700 ~ Pn-799	0x22BC ~ 0x231F	0xE2BC ~ 0xE31F
Pn-800 ~ Pn-899	0x2320 ~ 0x2383	0xE320 ~ 0xE383
Pn-900 ~ Pn-999	0x2384 ~ 0x23E7	0xE384 ~ 0xE3E7
Pn-A00 ~ Pn-A99	0x23E8 ~ 0x244B	0xE3E8 ~ 0xE44B
Pn-B00 ~ Pn-B99	0x244C ~ 0x24AF	0xE44C ~ 0xE4AF
Pn-C00 ~ Pn-C99	0x24B0 ~ 0x2513	0xE4B0 ~ 0xE513
Pn-D00 ~ Pn-D99	0x2514 ~ 0x2577	0xE514 ~ 0xE577
Pn-E00 ~ Pn-E99	0x2578 ~ 0x25DB	0xE578 ~ 0xE5DB
dn-000 ~ dn-099	0x25DC ~ 0x263F	0xE5DC ~ 0xE63F

Pay special attention to the use of communication:

1. When the parameters for power-down save are not required, the parameter value can be written to the RAM area, and the date required to be stored for a long time or power-down saved data can be written to the EEPROM. However, the frequent writing of EEPROM will reduce the EEPROM life.

2. All 32-bit parameters occupy two consecutive register addresses. Do not access high 16-bit or low 16-bit parameters separately, otherwise it may cause data errors.

3. The maximum number of registers accessed by 03H and 10H function codes is 10.

4. When multi-steps discontinuous parameters need to be accessed at the same time, the mapping parameters can be used. This method only needs one frame of data to complete the access (refer to the Pn-921 ~ Pn-940 parameter description).

5. For ease of operation, the target address of the mapping parameter does not need to be converted to the actual physical address, and is directly set to the parameter code number. For example, to map the access parameter dn-000, set the mapping destination address to 0x0F00 instead of 0x25DC.

8.5. Application example

1. Read actual motor speed and driver speed commands (03H function code, read dn-000 and dn-001 parameters)

RTU mode :

Request (main station)		Response (slave station)	
Slave address	01	Slave address	01
Function code	03	Function code	03
High-order address of starting	25	Data length	04
Low-order address of starting	DC	Data high-order of 1st register	03
High-order number of registers	00	Data low-order of 1st register	E8
Low-order number of registers	02	Data high-order of 2st register	03
CRC low-order	0E	Data low-order of 2st register	E8
CRC high-order	FD	CRC low-order	7A
		CRC high-order	FD

ASCII mode :

Request (main station)		Response (slave station)	
Start character	3A	Start character	3A
Slave address	30	Slave address	30
	31		31
Function code	30	Function code	30
	33		33
High-order address of starting	32	Data length	30
	35		34
Low-order address of starting	44	Data high-order of 1st register	30
	43		33
High-order number of registers	30	Data low-order of 1st register	45
	30		38
Low-order number of registers	30	Data high-order of 2st register	30
	32		33
LRC	46	Data low-order of 2st register	45
	39		38
End character 0	0D	LRC	32
End character 1	0A		32
		End character 0	0D
		End character 1	0A

2. Set position command source and pulse shape (06H function code, write Pn-300 parameters)

RTU mode :

Request (main station)		Response (slave station)	
Slave address	01	Slave address	01
Function code	06	Function code	06
High-order number of registers	21	High-order number of registers	21
Low-order number of registers	2C	Low-order number of registers	2C
Data high-order	00	Data high-order	00
Data low-order	01	Data low-order	01
CRC low-order	83	CRC low-order	83
CRC high-order	FF	CRC high-order	FF

ASCII mode :

Request (main station)		Response (slave station)	
Start character	3A	Start character	3A
Slave address	30	Slave address	30
	31		31
Function code	30	Function code	30
	36		36
High-order number of registers	32	High-order number of registers	32
	31		31
Low-order number of registers	32	Low-order number of registers	32
	43		43
Data high-order	30	Data high-order	30
	30		30
Data low-order	30	Data low-order	30
	31		31
LRC	41	LRC	41
	42		42
End character 0	0D	End character 0	0D
End character 1	0A	End character 1	0A

3. Set the first-order low-pass filter constant and electron gear molecule 1 (10H function code, write Pn-303, Pn-304 and Pn-305 parameters)

RTU mode :

Request (main station)		Response (slave station)	
Slave address	01	Slave address	01
Function code	10	Function code	10
High-order address of starting	21	High-order address of starting	21
Low-order address of starting	2F	Low-order address of starting	2F
High-order number of registers	00	High-order number of registers	00
Low-order number of registers	03	Low-order number of registers	03

Number of data	06	CRC low-order	D1
1st data high-order	00	CRC high-order	38
1st data low-order	64		
2st data high-order	00		
2st data low-order	00		
3st data high-order	00		
3st data low-order	02		
CRC low-order	88		
CRC high-order	EF		

ASCII mode :

Request (main station)		Response (slave station)	
Start character	3A	Start character	3A
Slave address	30	Slave address	30
	31		31
Function code	31	Function code	31
	30		30
High-order address of starting	32	High-order address of starting	32
	31		31
Low-order address of starting	32	Low-order address of starting	32
	46		46
High-order number of registers	30	High-order number of registers	30
	30		30
Low-order number of registers	30	Low-order number of registers	30
	33		33
Number of data	30	LRC	39
	36		43
1st data high-order	30	End character 0	0D
	30	End character 1	0A
1st data low-order	36		
	34		
2st data high-order	30		
	30		
2st data low-order	30		
	30		
3st data high-order	30		
	30		
3st data low-order	30		
	32		
LRC	33		
	30		
End character 0	0D		
End character 1	0A		

Chapter 9 Servo motor and accessories

9.1. Servo cable naming rule

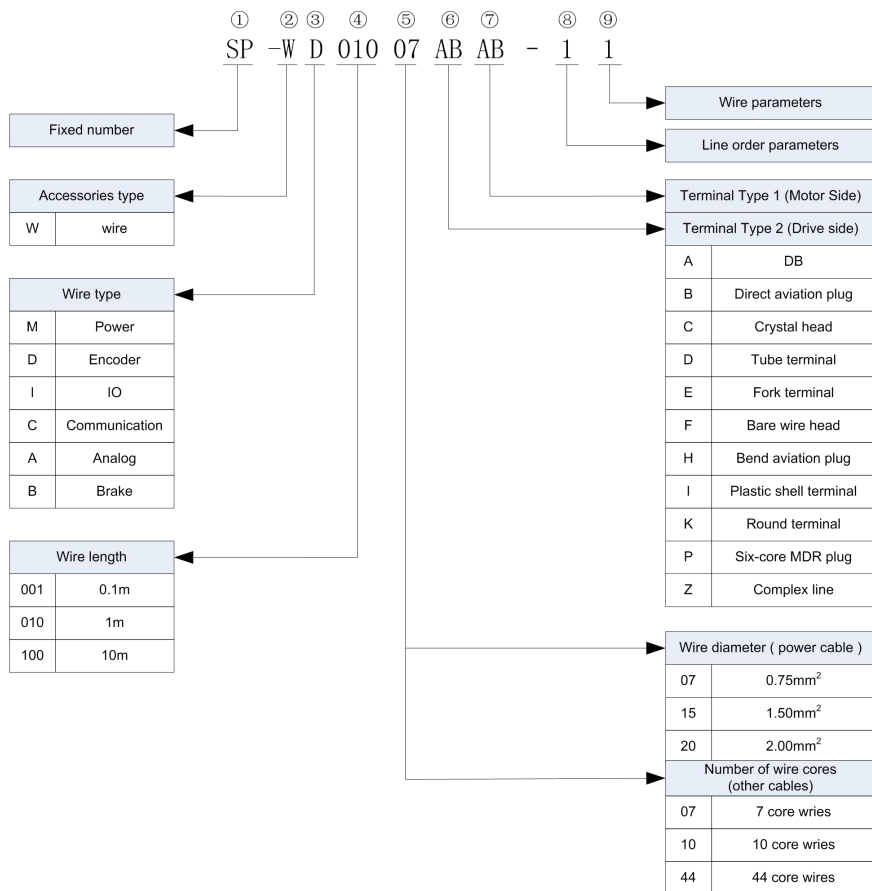


Figure 9-1 Servo cable naming rule

Note:

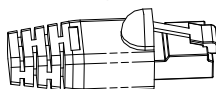
⑧Represents the line sequence parameters, for the distinction between the motor power cable sequence, CD100 series adapter over 110 flange (using aviation plug) power cable fixed selection 1

⑨ Represents wire parameters, 1=ordinary wire, 2=high flexible wire

9.2. Terminal Resistance

Model : SP-R135-02

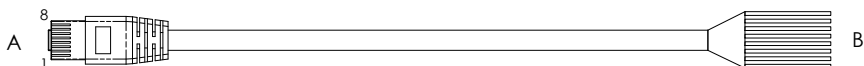
Note: Electrical matching for RS485 or CANopen communications.



9.3. RS485 communication and CANopen communication wire

Model : SP-WCXXX08CCFA-01

Note : RS485 or CANopen communication between the driver and the host device such as PLC.
2m has standing stock, other lengths need to be customized.

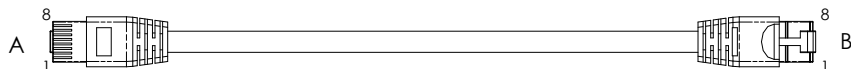


Pin number	Name	Illustrate
8	CANL	1-2wrings 3-6wrings 4-5wrings 7-8wrings
7	CANH	
6	GND	
5	RS-	
4	RS+	
3	NC	
2	NC	
1	NC	
outer casing		PE shielding layer

Model : SP-WCXXX08ACA-01

Note : It is used to drive multi-steps machine parallel connection to realize RS485 or CANopen communication networking.

15cm / 20cm / 30cm has standing stock, other lengths need to be customized.



9.4. Standard Motor and Cable Reference Table

Table 9-1 Servo motor code and cable comparison table

Power	Motor model	CODE ^①	Power cable ^②	Encoder cable ^{③③}
T1R8	CM10-A60TR6430B3□S3	1010	SP-WMXXX05DAIB-01	SP-WDXXX05PAID-01
T3R0	CM10-A80T02430B3□M3	1020		
T4R5	CM10-A80T04025B3□M3	1030		
T5R5	CM10-A130T05025Q3□M3	1140	SP-WMXXX07DCHB-11	SP-WDXXX05PAHC-01
T7R5	CM10-A130T07725Q3□M3	1150		
F4R0	CM10-A130F06025Q3□M3	1513		
F6R5	CM10-A130F15015Q3□M3	1510	SP-WMXXX15DBHA-11	SP-WDXXX05PAHC-01
F8R5	CM10-A180F19015Q3□L3	1520		
F12R	CM10-A180F21520Q3□L3	1530		

Note① : The motor code is used to set the parameter Pn-001.

Note② : Power cable and encoder cable ' XXX ' for the cable length, please order according to demand. 3m / 5m / 10m standing stock.

Note③ : When the encoder is a multi-steps turn absolute value application, select 7-wire (with battery box) for the cable. When the encoder is a single-turn or incremental absolute value.

The motors in the above table are standard motors for each drive, and the other adaptation motors are shown in Table 9-2 servo motor technical parameters.

9.5. Servo motor technical specifications

Table 9-2 Technical parameters of servo motor

Motor model	CM10-			
	A60TR6430B	A60T01330B	A80T02430B	A80T04025B
Voltage(V)	AC220			
Motor base size	60	60	80	80
Rated output power(kW)	0.2	0.4	0.75	1.0
Rated current(A)	1.2	2.8	3.0	4.4
Rated torque(N·m)	0.64	1.27	2.39	4.0
Maximum torque(N·m)	1.92	3.81	7.17	12.0
Rated speed(r/min)	3000	3000	3000	2500
Maximum speed(r/min)	5000	5000	3500	3000
Rotor inertia(Kg·m ²)	0.175×10 ⁻⁴	0.29×10 ⁻⁴	1.40×10 ⁻⁴	2.7×10 ⁻⁴
Torque constant(N·m/A)	0.53	0.45	0.8	0.91
Back electromotive force (V/1000r/min)	31	29.7	48	55
Line resistance(Ω)	6.18	2.35	2.2	1.6
Line inductance(mH)	29.3	14	15.7	11.8
Number of pole-pairs	4	4	4	4
Insulation grade	F	F	F	F
Protection level	IP65	IP65	IP65	IP65
Motor body length L(mm)	115(147)	140(171)	151(191)	191(231)

Note : Motor body length, in parentheses is the length with brake function.

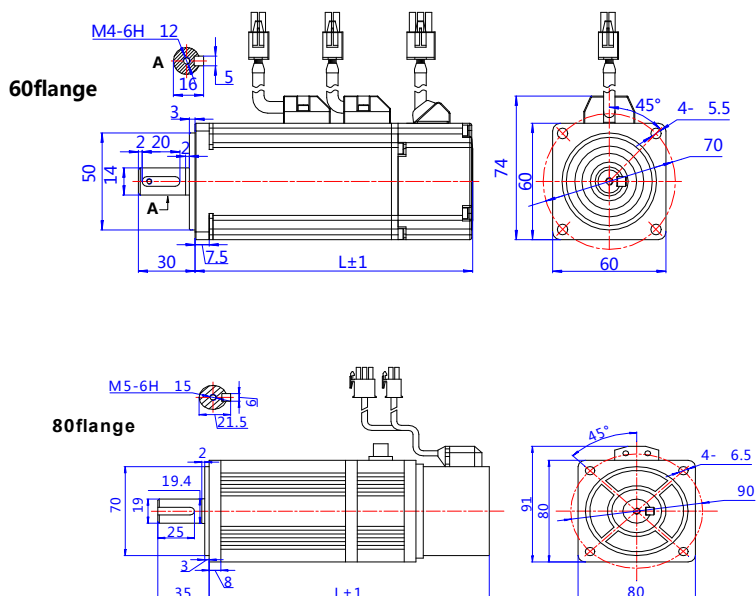
Motor model	CM10-			
	A130T05025	A130T060025	A130T07725	CM10-A130T10015
Voltage(V)	AC220			
Motor base size	130	130	130	130
Rated output power(kW)	1.3	1.5	2.0	1.5
Rated current(A)	5.0	6.0	7.5	6
Rated torque(N·m)	5.0	6.0	7.7	10
Maximum torque(N·m)	12.5	15.0	18.7	25
Rated speed(r/min)	2500	2500	2500	1500
Maximum speed(r/min)	3000	3000	3000	2000
Rotor inertia(Kg·m ²)	10.7×10 ⁻⁴	12.9×10 ⁻⁴	14.1×10 ⁻⁴	18.8×10 ⁻⁴
Torque constant(N·m/A)	1.0	1.0	1.03	1.67
Back electromotive force (V/1000r/min)	65	65	68	108
Line resistance(Ω)	1.6	1.3	1.2	1.85
Line inductance(mH)	8.0	6.2	5.8	9.9
Number of pole-pairs	4	4	4	4
Insulation grade	F	F	F	F
Protection level	IP65	IP65	IP65	IP65
Motor body length L(mm)	171(224)	179(224)	192(229)	209 (265)

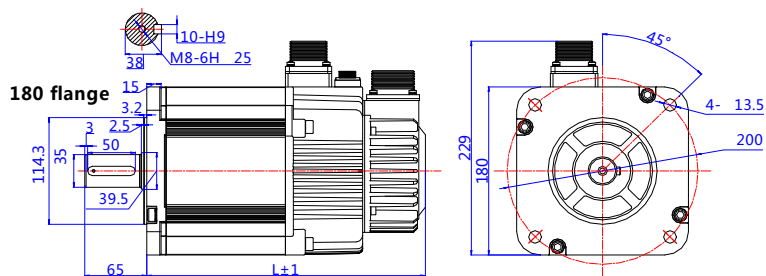
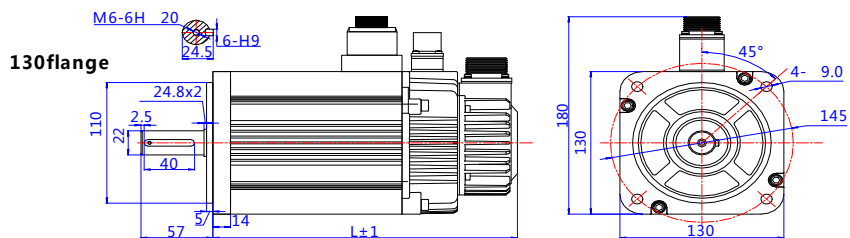
Note : Motor body length, in parentheses is the length with brake function.

Motor model	CM10-			
	CM10-A130F06025	CM10-A130F15015	CM10-A180F19015	CM10-A180F21520
Voltage(V)	380V			
Motor base size	130	130	180	180
Rated output power(kW)	1.5	2.3	3	4.5
Rated current(A)	4.0	5.0	7.5	9.5
Rated torque(N·m)	6.0	15.0	19	21.5
Maximum torque(N·m)	15.0	30.0	57	64.5
Rated speed(r/min)	2500	1500	1500	2000
Maximum speed(r/min)	3000	2000	1800	2150
Rotor inertia(Kg·m ²)	12.9	25.5	63.5	72.7
Torque constant(N·m/A)	1.5	3.0	2.53	2.26
Back electromotive force (V/1000r/min)	108	180	166	140
Line resistance(Ω)	3.1	1.2	1.33	0.84
Line inductance(mH)	17.1	8.3	14.2	8.4
Number of pole-pairs	4	4	4	4
Insulation grade	F	F	F	F
Protection level	IP65	IP65	IP65	IP65
Motor body length L(mm)	179 (224)	231 (282)	205 (252)	215 (262)

Note : Motor body length, in parentheses is the length with brake function.

9.6. Servo motor size





Appendix A Warnings and Faults

Warnings :

When a warning occurs in the servo driver, the warning code is displayed as an AL prefix, which indicates that the driver detects an exception. It is necessary to check and exclude the cause of the warning, otherwise it will affect the continuous normal use of the servo system. The reference solution is as follows.

Warning code	Name	Possible Causes	Resolutions
AL-001	Emergency stop	External emergency stop input	Reset emergency button to ensure system security
AL-002	Overload warning	Overload factor setting is too small	Adjust the overload coefficient
AL-003	Forward limit	Mechanical touch to the forward limit switch	Check the machinery to confirm the forward of the limit switch
AL-004	Reverse limit	Mechanical touch reverse limit switch	Check the machinery to confirm the position of the limit switch
AL-007	Software forward limit	The position command exceeds the set value of Pn-818	Adjust the target position of the system or Appropriately modify the value of Pn-818
AL-008	Software reverse limit	The position command exceeds the set value of the Pn-820	Adjust the target position of the system or Appropriately modify the value of Pn-820
AL-010	Absolute position loss	Battery voltage is lower than 2.8 V, or the drive control power supply is invalid in the case of replacement batteries, or battery-powered line contact.	Check the battery wiring, replace the battery after the origin reset process.
AL-011	Absolute position encoder Low battery voltage	Battery voltage below 3.1 V	Replace battery when drive control power is active
AL-012	Absolute position encoder Multi-steps turn data overflow	The number of turns of the motor exceeds the range of origin 32768 to 32767	Reset the origin process
AL-015	High internal temperature of encoder	Encoder temperature exceeding 100 °C	Increase motor heat dissipation
AL-020	Absolute position counter overflow	Absolute feedback position calculation overflow	The electronic gear ratio is properly set according to the actual use and the total stroke of the absolute position to avoid position calculation overflow.

Faults :

When the servo driver fails to alarm, the fault code is shown as the Er prefix, and the power will stop running at this time. The reference solution is shown in the following table.

Fault code	Name	Possible Causes	Resolutions
Er-001	Hardware overcurrent	External short circuit, IPM module damage	Check the motor phase deficiency or short circuit, whether the parameter setting is reasonable.
Er-003	Overload	Motor stall or excessive load	Check the machinery or select high-power drives
Er-004	Temperature detection of the sampling channel fault	Temperature sensor is not connected	Check the temperature sensor terminal is reliable connection
Er-005	U-phase current sampling channel fault	U phase current detection circuit fault	Check the reliable connection between the control board wiring and the main circuit
Er-006	V-phase current sampling channel fault	V-phase current detection circuit fault	Check the reliable connection between the control board wiring and the main circuit
Er-008	Regenerative brake overload	Excessive braking power	Increase external braking resistance
Er-009	Busbar undervoltage	Main circuit power supply voltage is too low	Correct use of voltage source or series converter
Er-010	Busbar overvoltage	Main circuit power supply voltage is too high or frequent braking	Increase external braking resistance
Er-011	Motor overspeed	Driver power line sequence error	Check drive power cord sequence
Er-012	Temperature	High ambient temperature or fan failure	Check and replace cooling fan
Er-013	Frequent EEPROM write failure	Frequent write operations on EEPROM	Write the parameters that need to be saved once, no need Save write RAM.
Er-014	EEPROM access fault	EEPROM chip damage	Replace EEPROM chip
Er-015	High pulse input frequency	Pulse command input frequency is higher than the rated input frequency	Correctly set pulse input frequency
Er-016	Position tracking deviation	Excessive deviation between target position and actual position	Check whether the motor power line is connected, maximum position deviation.
Er-017	Encoder not connected	Encoder connection line disconnection	Check or replace the encoder connection
Er-020	Internal data anomaly of encoder	Driver and encoder communication interfered or encoder corruption	Check the encoder for damage
Er-022	Electronic gear ratio is set over the limit	Molecular to denominator ratio of electronic gear ratio exceeds range	Correct setting of the molecule and denominator of the electronic gear ratio (1 / 50~ Encoder resolution / 50)
Er-023	Electromagnetic interference	Switch circuit disturbed	Check whether the drive secondary circuit wiring is reasonable, driver Reliable connection to the earth.
Er-028	Main circuit power input phase missing	Main circuit power open	Check the main circuit power input
Er-029	Control circuit power supply 24V is abnormal	The drive 24V power failure or the internal wiring is not reliably connected	Check driver wiring or internal 24V circuits
Er-030	System power supply of 15V is abnormal	15V power failure inside the driver	Check the drive wiring or internal 15V circuit
Er-031	QC test tag exception1	Single-board test failed	Contact the manufacturer

Fault code	Name	Possible Causes	Resolutions
Er-032	QC test tag exception2	The test before aging did not pass	Contact the manufacturer
Er-033	QC test tag exception3	12 hour aging test failed	Contact the manufacturer
Er-034	QC test tag exception4	failed the test after aging	Contact the manufacturer
Er-035	Software overcurrent	Load short circuit, or imbalance leads to excessive current	1. Check the grid and load operating conditions 2. Contact the manufacturer
Er-036	Full closed-loop encoder disconnection	Full closed-loop encoder connector disconnected	Check the full closed-loop connection line
Er-037	Full closed loop position deviation is too large	The fully closed-loop connector loosens or falls off, or the Pn-704 setting is too small.	Check the full closed-loop connector and adjust appropriately The value of the Pn-704
Er-039	Driver code error	Driver ID code error	Contact the manufacturer
Er-040	Motor code error	Motor code does not match drive	Correctly set motor code
Er-045	DI function repeated allocation	When the DI function is assigned, the same function is repeatedly assigned to multi-steps DI terminals.	Check whether the parameters Pn-211 ~ Pn-226 set the same non-zero DI function.
Er-070	Absolute position mode motor does not match	Incremental motor opens absolute position function	Check if the motor is incremental. To use the absolute position function, use an absolute value motor.

Appendix B DI function configuration

S/N	Symbol	Function	Functional description			Mode
0	NULL	No function	Input status has no effect on the system			
1	SON	Servo enabled	OFF	Servo is not enabled and the motor is not energized		P S T
			ON	Servo enabled, motor power on		
2	EMG	Emergency stop	OFF	The servo works properly		P S T
			ON	Servo shuts down, and motor power is failed		
3	ERCLR	Fault reset	Clear the fault on the rising edge			P S T
			Note : Before reset, please confirm that the fault has been eliminated.			
4	POT	Positive overtravel	OFF	Forward rotation disabled		P S T
			ON	Forward rotation allowed		
5	NOT	Negative overtravel	OFF	Reverse rotation disabled		P S T
			ON	Reverse rotation allowed		
6	TCCW	External forward torque limiting	OFF	CCW is not limited by Pn507 in forward		P S T
			ON	CCW is limited by Pn507 in forward		
			Note: The forward torque is limited by Pn505 regardless of whether TCCW is valid or not			
7	TCW	External reverse torque limiting	OFF	CW reverse is not limited by Pn508		P S T
			ON	CW reverse is limited by Pn508		
			Note: The reverse torque is limited by Pn506 regardless of whether TCW is valid or not			
8	NULL	Reserved				
9	ZCLMP	Zero speed clamp	OFF	Invalid		P S T
			ON	Valid		
			Note : Reference Pn-416 and Pn-417 function description			
10	CMOD1	Control mode switching	CMOD	Pn-100	Control mode	P S T
11	CMOD2		OFF	3	Position control	
			ON		Speed control	
12	CMOD3		OFF	4	Position control	
			ON		Torque control	
13	CMOD3		OFF	5	Speed control	
			ON		Torque control	
13	GAIN	Gains switching	OFF	The speed loop uses the PI control / Group 1 gain		P S T
			ON	The speed loop uses the P control / Group 2 gain		
			Note : Reference Pn-651 function description			
14	NULL	Reserved				
15	GEAR1	Electronic gear ratio selection	GEAR2	GEAR1	Electronic gear ratio molecules	P
			OFF	OFF	Electronic gear ratio molecules1	
			OFF	ON	Electronic gear ratio molecules2	
			ON	OFF	Electronic gear ratio molecules3	
16	GEAR2		ON	ON	Electronic gear ratio molecules4	

S/N	Symbol	Function	Functional description					Mode
17	INHP	Position pulse input disabled	OFF	External position pulse command is valid				P
			ON	Ignore external position pulse command				
18	PECLR	Position deviation clearing	OFF	Invalid				P
			ON	Clearing position deviation counter				
			Note : Reference Pn-321 function description					
19	ICMD1	Internal multi-steps command switching 1		ICMD4	ICMD3	ICMD2	ICMD1	P S T
20	ICMD2	Internal multi-steps command switching 2	Paragraph 1	0	0	0	0	
21	ICMD3	Internal multi-steps command switching 3	Paragraph n	---				
22	ICMD4	Internal multi-steps command switching 4	Paragraph 16	1	1	1	1	
23	NULL	Reserved						
24	NULL	Reserved						
25	PTRG	Internal position command triggered	Internal position command trigger when rising edge					P
			Note : Valid only for internal position command control					
26	CINV	Command inverse	OFF	Normal command				P S T
			ON	Current invert command				
			Note : External pulse instruction mode is invalid					
27	JOGP	Positive jog	OFF	The motor stops turning				P S T
			ON	The motor turns forward				
28	JOGN	Negative jog	OFF	The motor stops turning				P S T
			ON	The motor turns Reverse				
29	REF	Get home reference signal	Note : Refer to Pn-336 parameter for detailed description					P
30	GHOM	Get home trigger signal	OFF	Invalid				P
			ON	Returning home enabling				
			Note : Refer to Pn-336 parameter for detailed description					

Appendix C DO function configuration

S/N	Symbol	Function	Functional description		mode
0	NULL	No functional	OFF state		
1	RDY	Servo is ready	OFF	Servo failure	P S T
			ON	Servo ready to respond to SON signal	
2	RUN	Servo running	OFF	Servo motor is not energized	P S T
			ON	Servo motor is energized and runs	
3	BRK	Electromagnetic brake	OFF	Electromagnetic brake enabled	P S T
			ON	Electromagnetic brake released	
4	ALM	Servo warning	OFF	Servo unit warning state	P S T
			ON	Normal state of servo unit	
5	ERR	Servo failure	OFF	Fault state of servo unit	P S T
			ON	Normal state of servo unit	
6	COIN	Complete in position	OFF	Positioning not completed	P
			ON	Positioning completed	
			Reference Pn-327 function description		
7	NEAR	Near in position	OFF	Position deviation counter greater than the set value	P
			ON	Position deviation counter less than the set value	
			Reference Pn-329 function description		
8	ZSP	Zero speed	OFF	Motor speed higher than set value	P S T
			ON	Motor speed below set value	
			Reference Pn-420 function description		
9	SPA	Speed achieve	OFF	The actual speed of the motor is less than the set value	P S T
			ON	The actual speed of the motor is greater than the set value	
			Reference Pn-421 function description		
10	SPL	Speed limiting	OFF	Motor speed is failed to reach the limit	T
			ON	Motor speed reaches the limit	
11	TQA	Torque achieve	OFF	Motor actual torque is less than the set value	P S T
			ON	Motor actual torque is greater than the set value	
			Reference Pn-513 / Pn-514 / Pn-515 function description		
12	TQL	Torque limiting	OFF	Motor torque is failed to reach the limited value	P S
			ON	Motor torque reaches the limited value	
13	HOM	Home completed	OFF	no completed	P S T
			ON	Returning home position completed	
17	AND1	Logic and 1	OFF	The two terminals configured in Pn271 perform logic and calculation, but the result is false	P S T
			ON	The two terminals configured in Pn271 perform logic and calculation, and the result is true	
18	AND2	Logic and 2	OFF	The two terminals configured for Pn272 perform logic and calculation, but the result is false	P S T
			ON	The two terminals configured with Pn272 perform logic and calculation, and the result is true	
19	AND3	Logic and 3	OFF	The two terminals configured with Pn273 perform logic and calculation, but the result is false	P S T
			ON	The two terminals configured with Pn273 perform logic and calculation, and the result is true	

S/N	Symbol	Function	Functional description		mode
20	AND4	Logic and 4	OFF	The two terminals configured for Pn274 perform logic and calculation, but the result is false	P S T
			ON	The two terminals configured for Pn274 perform logic and calculation, and the result is true	
21	OR1	Logic or 1	OFF	The two terminals configured on the Pn275 perform logic or calculation, but the result is false	P S T
			ON	The two terminals configured on the Pn275 perform logic or calculation, and the result is true	
22	OR2	Logic or 2	OFF	The two terminals configured on Pn276 perform logic or calculation, but the result is false	P S T
			ON	The two terminals configured on Pn276 perform logic or calculation, and the result is true	
23	OR3	Logic or 3	OFF	The two terminals configured on Pn277 perform logic or calculation , but the result is false	P S T
			ON	The two terminals configured on Pn277 perform logic or calculation, and the result is true	
24	OR4	Logic or 4	OFF	The two terminals configured on Pn278 perform logic or calculation, but the result is false	P S T
			ON	The two terminals configured on Pn278 perform logic or calculation, and the result is true	
25	XOR1	Logic XOR 1	OFF	The logical XOR operation on the two terminals configured on Pn279 ,but the result is false	P S T
			ON	The logical XOR operation on the two terminals configured on Pn279 , and the result is true	
26	XOR2	Logic XOR 2	OFF	The logical XOR operation on the two terminals configured on Pn280 ,but the result is false	P S T
			ON	The logical XOR operation on the two terminals configured on Pn280, and the result is true	
27	XOR3	Logic XOR 3	OFF	The logical XOR operation on the two terminals configured on Pn281 ,but the result is false	P S T
			ON	The logical XOR operation on the two terminals configured on Pn281, and the result is true	
28	XOR4	Logic XOR 4	OFF	The logical XOR operation on the two terminals configured on Pn282 ,but the result is false	P S T
			ON	The logical XOR operation on the two terminals configured on Pn282, and the result is true	
29	XNOR1	Logic same or 1	OFF	The logical same or operation on the two terminals configured in Pn283 , but the result was false	P S T
			ON	The logical same or operation on the two terminals configured in Pn283, and the result is true	
30	XNOR2	Logic same or 2	OFF	The logical same or operation on the two terminals configured in Pn284 , but the result was false	P S T
			ON	The logical same or operation on the two terminals configured in Pn284, and the result is true	
31	XNOR3	Logic same or 3	OFF	The logical same or operation on the two terminals configured in Pn285 , but the result was false	P S T
			ON	The logical same or operation on the two terminals configured in Pn285, and the result is true	

S/N	Symbol	Function	Functional description		mode
32	XNOR4	Logic same or 4	OFF	The logical same or operation on the two terminals configured in Pn286 , but the result was false	P S T
			ON	The logical same or operation on the two terminals configured in Pn286, and the result is true	
33	NOT1	Logic non 1	OFF	The logical inverse operation is performed on the two terminals configured on Pn287, but the result is false	P S T
			ON	The logical inverse operation is performed on the two terminals configured on Pn287, and the result is true	
34	NOT2	Logic non 2	OFF	The logical inverse operation is performed on the two terminals configured on Pn288, but the result is false	P S T
			ON	The logical inverse operation is performed on the two terminals configured on Pn288, and the result is true	
35	NOT3	Logic non 3	OFF	The logical inverse operation is performed on the two terminals configured on Pn289, but the result is false	P S T
			ON	The logical inverse operation is performed on the two terminals configured on Pn289, and the result is true	
36	NOT4	Logic non 4	OFF	The logical inverse operation is performed on the two terminals configured on Pn290, but the result is false	P S T
			ON	The logical inverse operation is performed on the two terminals configured on Pn290, and the result is true	
37	TIM1	Timer 1	OFF	Timer count value not reaching set value	P S T
			ON	Timer count value reaches the set value	
			Refer to Pn-291/dn-046 function description		
38	TIM2	Timer2	OFF	Timer count value not reaching set value	P S T
			ON	Timer count value reaches the set value	
			Refer to Pn-292/dn-047 function description		
39	TIM3	Timer3	OFF	Timer count value not reaching set value	P S T
			ON	Timer count value reaches the set value	
			Refer to Pn-293/dn-048 function description		
40	TIM4	Timer4	OFF	Timer count value not reaching set value	P S T
			ON	Timer count value reaches the set value	
			Refer to Pn-294/dn-049 function description		
41	CNT16	16bit counter	OFF	The counter value not reaching set value	P S T
			ON	The counter value reaches set value	
			Refer to dn-052/Pn-295/Pn-297 function description		
42	CNT32	32bit counter	OFF	The counter value not reaching set value	P S T
			ON	The counter value reaches set value	
			Refer to dn-053/Pn-296/Pn-298 function description		



Warranty Card

Maintenance Company: _____ User: _____

Model No.: _____ Purchase Date: _____

Invoice No.: _____ Purchased From: _____

Add: Building A, Huichao Industrial Park, Gushu 2nd Rd., Xixiang, Bao'an District, Shenzhen, China.

Post Code: 401336

Tel: 0086-0755-26607756

Service: 0086-0755-26910928

Fax: 0086-0755-26919882

Note:

1. Please keep this card properly, please contact service center with this card and invoice when need maintenance.
2. The warranty period is 18 months.

Shenzhen Simphoenix Electric Technology Co.,Ltd

Cut along the dotted line



Certificate of Approval

This product is approved to delivery
according to the standard

Inspector: _____



Shenzhen Simphoenix Electric Technology Co.,Ltd